

Australian/New Zealand Standard™

Structural design actions

Part 2: Wind actions

AS/NZS 1170.2:2002

This Joint Australian/New Zealand Standard was prepared by Joint Technical Committee BD-006, General Design Requirements and Loading on Structures. It was approved on behalf of the Council of Standards Australia on 29 March 2002 and on behalf of the Council of Standards New Zealand on 28 March 2002. This Standard was published on 4 June 2002.

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Australian/New Zealand Standard™

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Part 2: Wind actions

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AS 1170.2—1989 and NZS 4203:1992 jointly revised, amalgamated and redesignated in part as AS/NZS 1170.2:2002.
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PREFACE

This Standard was prepared by the Joint Standards Australia/Standards New Zealand Committee, BD-006, General Design Requirements and Loading on Structures, to supersede AS 1170.2—1989, *Minimum design loads on structures*, Part 2: *Wind loads*, and in part Part 5 of NZS 4203:1992, *Code of practice for general structural design and design loading for buildings*.

This Standard incorporates Amendment No. 1 (April 2005). The changes required by the Amendment are indicated in the text by a marginal bar and amendment number against the clause, note, table, figure or part thereof affected.

This Standard is published as a joint Standard (as are also AS/NZS 1170.0 and AS/NZS 1170.1) and it is intended that it is suitable for use in New Zealand as well as Australia. However, NZS 4203, *General structural design and design loadings for buildings* remains current in New Zealand until the publication of all parts (including Part 4: Earthquake action) and for a transition period afterwards.

This Standard will be referenced in the Building Code of Australia by way of BCA Amendment 11 to be Published on 1 July 2002, thereby superseding the previous edition, AS 1170.2—1989, which will be withdrawn 12 months from the date of publication of this Edition.

The objective of this Standard is to provide designers of structures with wind actions for use in the design of structures subject to wind action. It provides a detailed procedure for the determination of wind actions on structures, varying from those less sensitive to wind action to those for which dynamic response must be taken into consideration.

This Standard is Part 2 of the AS/NZS 1170 series *Structural design actions*, which comprises the following parts, each of which will have an accompanying Commentary published as a Supplement:

- Part 0: General principles
- Part 1: Permanent, imposed and other actions
- Part 2: Wind actions
- Part 3: Snow and ice actions
- Part 4: Earthquake action

The Commentary to this Standard is AS/NZS 1170.2 Supp 1, *Structural design actions—Wind actions—Commentary* (Supplement to AS/NZS 1170.2:2002).

The wind speeds provided are based on existing data. At the time of drafting, it was considered that there was insufficient evidence to indicate any trend in wind speeds due to climatic change.

This Standard is based on ISO 4354, *Wind actions on structures*. ISO 4354 gives general format and guidance on detail for the drafting of national Standards on wind actions.

This edition differs from the previous editions as follows:

- (a) The format of ISO 4354 has been adopted except that the effects of exposure are applied to the wind speed to give directional site wind speeds before conversion to pressure (see Note to Clause 2.4).
- (b) Importance factors have been replaced with variable annual probability of exceedance, to enable reliability-based design. Values of wind speed are determined using the annual probability of exceedance (see AS/NZS 1170.0).

- (c) It is a joint Standard intended for use in Australia and New Zealand.
- (d) Average roof height is used to calculate wind pressures for rectangular enclosed buildings.
- (e) Actions determined from wind tunnel tests or other methods are not covered through the 'deemed-to-comply' solution given in this Standard, but must be separately established (by a special study; see AS/NZS 1170.0).
- (f) A simplified procedure is not included.
- (g) Wind speeds for permissible stress design are not included (see the Commentary, AS/NZS 1170.2 Supp 1).
- (h) Directional wind speed multipliers have been revised and extended beyond the capital cities in Australia.
- (i) Generally, the clauses have been updated to incorporate the latest research and to improve useability. The following new information has been included:
 - (i) Elevation effect for Tasmania.
 - (ii) Separation zone for crests of steeper slopes.
 - (iii) Combination factor for major structural members.
 - (iv) Parapet reduction factor.
 - (v) Hyperbolic paraboloid roofs.
 - (vi) Methods for open lattice structures and lattice towers (including ancillaries).
 - (vii) Flags and spheres.
- (j) The calculation of dynamic wind response has been simplified and the use of an hourly mean wind speed for dynamic analysis has been replaced with the 3 s gust. When dynamic response is to be determined, a single additional factor is determined.

The Joint Committee has considered exhaustive research and testing information from Australian, New Zealand and overseas sources in the preparation of this Standard. The design wind actions prescribed in this Standard are the minimum for the general cases described.

Particular acknowledgment should be given to Monash University, University of Queensland, James Cook University, Curtin University of Technology, Building Research Association of New Zealand, Bureau of Meteorology (Aust) and National Institute of Water and Atmospheric Research (NZ) for their contributions to the drafting of this Standard.

The terms 'normative' and 'informative' have been used in this Standard to define the application of the appendix to which they apply. A 'normative' appendix is an integral part of a Standard, whereas an 'informative' appendix is only for information and guidance.

Statements expressed in mandatory terms in notes to tables and figures are deemed to be an integral part of this Standard.

Notes to the text contain information and guidance and are not considered to be an integral part of the Standard.

- A1 | Essentially, AS/NZS 1170.2 is independent of the design requirement specified—thus becoming a purely technical document. It may be used to calculate wind speeds, wind pressures and wind forces given *any* annual probability of exceedance. The annual probability of exceedance (P) defines the wind event (in which pressures and forces occur). Thus a major wind storm with an annual probability of exceedance in the range of 1/200 to 1/2500 is used for ultimate limit states, while a much more frequent wind event (say of 1/25) might be used for serviceability limit states.

^{A1} In AS/NZS 1170.2, Table 3.1 gives the regional wind speeds for selected annual probabilities of exceedance and Section 2 sets out the determination of site wind speeds, design wind speeds, design wind pressures and wind actions (forces or loads) that are appropriate for the annual probability specified.

The annual probability of exceedance is not defined in AS/NZS 1170.2, but is set out in AS/NZS 1170.0 [and by further reference for ultimate limit states in the BCA (Table of annual probabilities of exceedance) and in the Guide to the BCA (examples of structures for importance levels)]. AS/NZS 1170.0 gives the annual probability of exceedance for New Zealand structures and for Australian structures not covered by the BCA. Guidance for serviceability events and associated limits is given in an informative Appendix of AS/NZS 1170.0 for loads associated with an appropriate annual probability of exceedance (P) for serviceability.

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STANDARDS AUSTRALIA/STANDARDS NEW ZEALAND

Australian/New Zealand Standard

Structural design actions

Part 2: Wind actions

SECTION 1 GENERAL

1.1 SCOPE

This Standard sets out procedures for determining wind speeds and resulting wind actions to be used in the structural design of structures subjected to wind actions other than those caused by tornadoes.

The Standard covers structures within the following criteria:

- (a) Buildings less than 200 m high.
- (b) Structures with roof spans less than 100 m.
- (c) Structures other than offshore structures, bridges and transmission towers.

NOTES:

1 This Standard is a stand-alone document for structures within the above criteria. It may be used, in general, for all structures but other information may be necessary. Guidance on wind tunnel testing, reliable references and alternative data is given in AS/NZS 1170.2 Supp 1, *Structural design actions—Wind actions—Commentary* (Supplement to AS/NZS 1170.2:2002).

2 Where structures have natural frequencies less than 1 Hz, Section 6 requires dynamic analysis to be carried out (see Section 6).

A1 3 In this document, the words 'this Standard' indicate AS/NZS 1170.2, which is regarded as Part 2 of the AS/NZS 1170 series of Standards (see Preface).

1.2 APPLICATION

This Standard shall be read in conjunction with AS/NZS 1170.0.

This Standard may be used as a means for demonstrating compliance with the Requirements of Part B1 of the Building Code of Australia.

A1 NOTE: Use of methods or information not given in this Standard should be justified by a special study (see AS/NZS 1170.0).

1.3 REFERENCED DOCUMENTS

The following documents are referred to in this Standard:

AS

4040 Methods of testing sheet roof and wall cladding

4040.3 Part 3: Resistance to wind pressures for cyclone regions

AS/NZS

1170 Structural design actions

1170.0 Part 0: General principles

ISO

2394 General principles on reliability for structures

4354 Wind actions on structures

Australian Building Codes Board
Building Code of Australia

1.4 DETERMINATION OF WIND ACTIONS

Values of wind actions (W) for use in design shall be established that are appropriate for the type of structure or structural element, its intended use, design working life and exposure to wind action.

A1 The following wind actions, determined in accordance with this Standard (using the procedures detailed in Section 2 and the values given in the remaining Sections), shall be deemed to comply with the requirements of this Clause:

- (a) W_u determined using a regional wind speed appropriate to the annual probability of exceedance (P) specified for ultimate limit states as given in AS/NZS 1170.0.
- (b) W_s determined using a regional wind speed appropriate to the annual probability of exceedance for the serviceability limit states (see Note 3).

NOTES:

- 1 Information on serviceability conditions and criteria can be found in AS/NZS 1170.0 (see Preface).
- 2 Some design processes require the determination of wind pressure (ultimate or serviceability wind pressure). Such pressures should be calculated for the wind speed associated with the annual probability of exceedance (P) appropriate to the limit state being considered.
- 3 For guidance on Item (b), see AS/NZS 1170.0.

1.5 UNITS

Except where specifically noted, this Standard uses the SI units of kilograms, metres, seconds, pascals, newtons and hertz (kg, m, s, Pa, N, Hz).

1.6 DEFINITIONS

Definitions of the terms used in this Standard are given in Appendix A.

1.7 NOTATION

The notation used in this Standard is given in Appendix B.

SECTION 2 CALCULATION OF WIND ACTIONS

2.1 GENERAL

The procedure for determining wind actions (W) on structures and elements of structures or buildings shall be as follows:

- (a) Determine site wind speeds (see Clause 2.2).
- (b) Determine design wind speed from the site wind speeds (see Clause 2.3).
- (c) Determine design wind pressures and distributed forces (see Clause 2.4).
- (d) Calculate wind actions (see Clause 2.5).

2.2 SITE WIND SPEED

The site wind speeds ($V_{\text{sit},\beta}$) defined for the 8 cardinal directions (β) at the reference height (z) above ground (see Figure 2.1) shall be as follows:

$$V_{\text{sit},\beta} = V_R M_d (M_{z,\text{cat}} M_s M_t) \quad \dots 2.2$$

where

V_R = regional 3 s gust wind speed, in metres per second, for annual probability of exceedance of $1/R$, as given in Section 3

M_d = wind directional multipliers for the 8 cardinal directions (β) as given in Section 3

$M_{z,\text{cat}}$ = terrain/height multiplier, as given in Section 4

M_s = shielding multiplier, as given in Section 4

M_t = topographic multiplier, as given in Section 4

Generally, the wind speed is determined at the average roof height (h). In some cases this varies, as given in the appropriate sections, according to the structure.

Where the orientation of a building is not known, the regional wind speed shall be assumed to act from any cardinal direction (i.e. $M_d = 1.0$ for all directions).

2.3 DESIGN WIND SPEED

The building orthogonal design wind speeds ($V_{\text{des},\theta}$) shall be taken as the maximum cardinal direction site wind speed ($V_{\text{sit},\beta}$) linearly interpolated between cardinal points within a sector ± 45 degrees to the orthogonal direction being considered (see Figures 2.2 and 2.3).

NOTE: That is, $V_{\text{des},\theta}$ equals the maximum value of site wind speed ($V_{\text{sit},\beta}$) in the range $[\beta = \theta \pm 45 \text{ degrees}]$ where β is the cardinal direction clockwise from true North and θ is the angle to the building orthogonal axes.

In cases such as walls and hoardings and lattice towers, where an angle of 45° is considered, $V_{\text{des},\theta}$ shall be the value of $V_{\text{sit},\beta}$ in a sector $\pm 22.5^\circ$ from the 45° direction being considered.

For ultimate limit states design, $V_{\text{des},\theta}$ shall not be less than 30 m/s.

NOTE: A conservative approach is to design the structure using the wind velocity and multipliers for the worst direction. For example, for a building on an escarpment, it may be easily checked whether the $V_R M_d (M_{z,\text{cat}} M_s M_t)$ on the exposed face (towards the escarpment) is the worst case. To simplify design, this value could then be used as the design wind speed for all directions on the building.

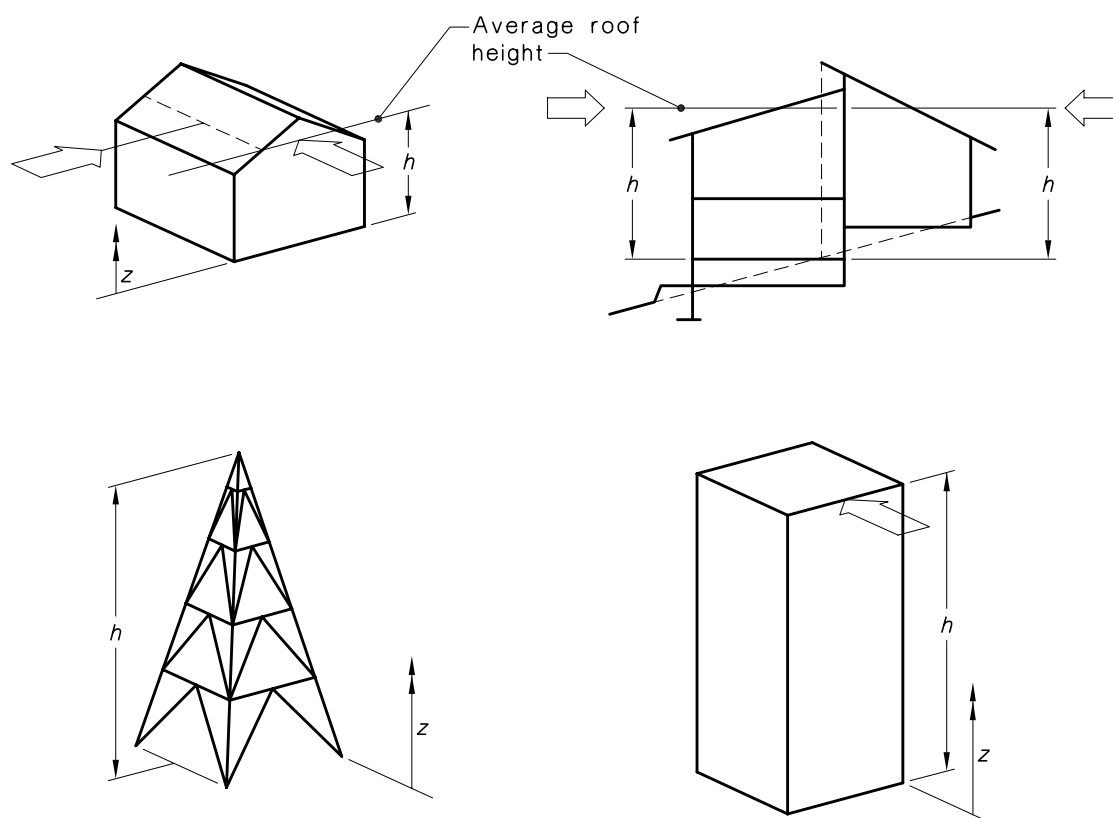


FIGURE 2.1 REFERENCE HEIGHT OF STRUCTURES

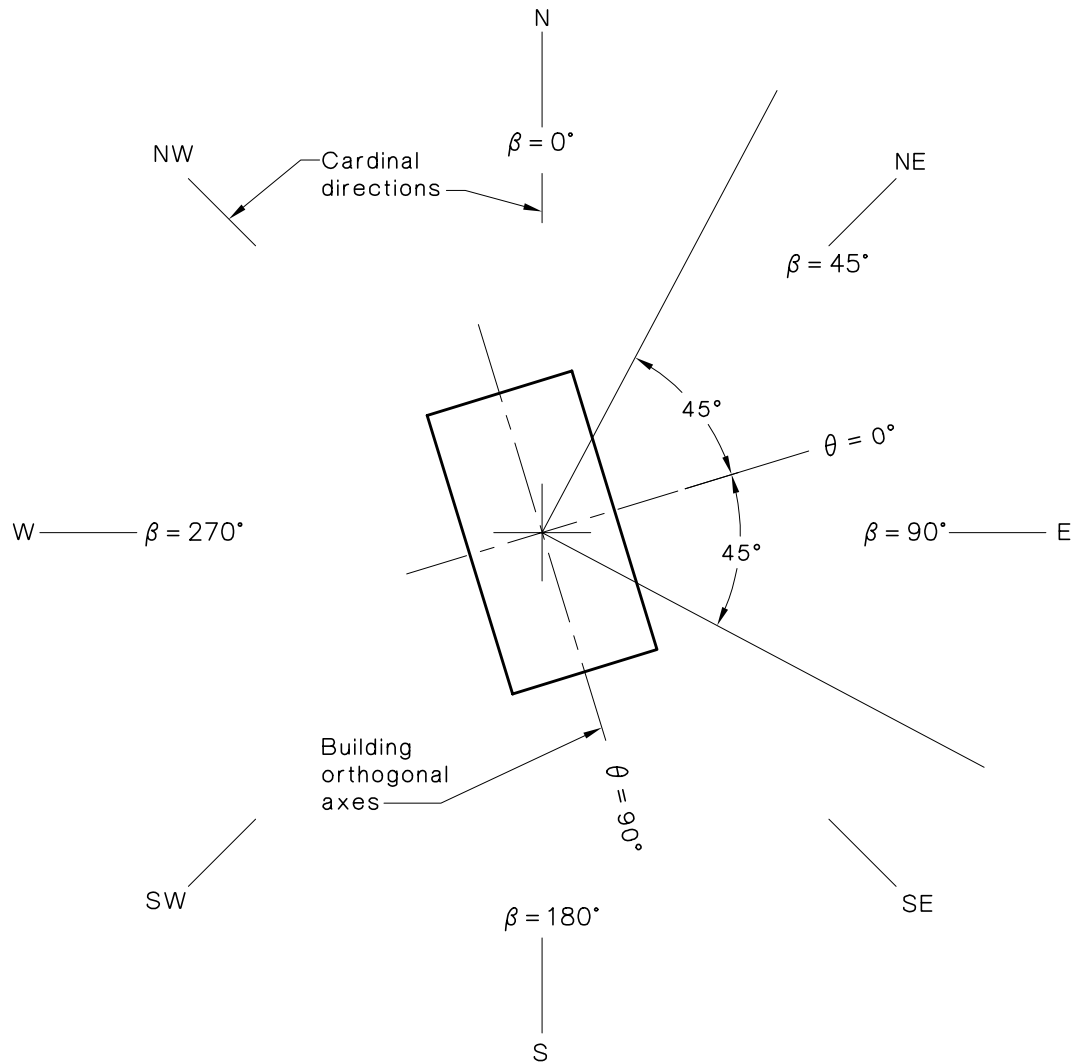
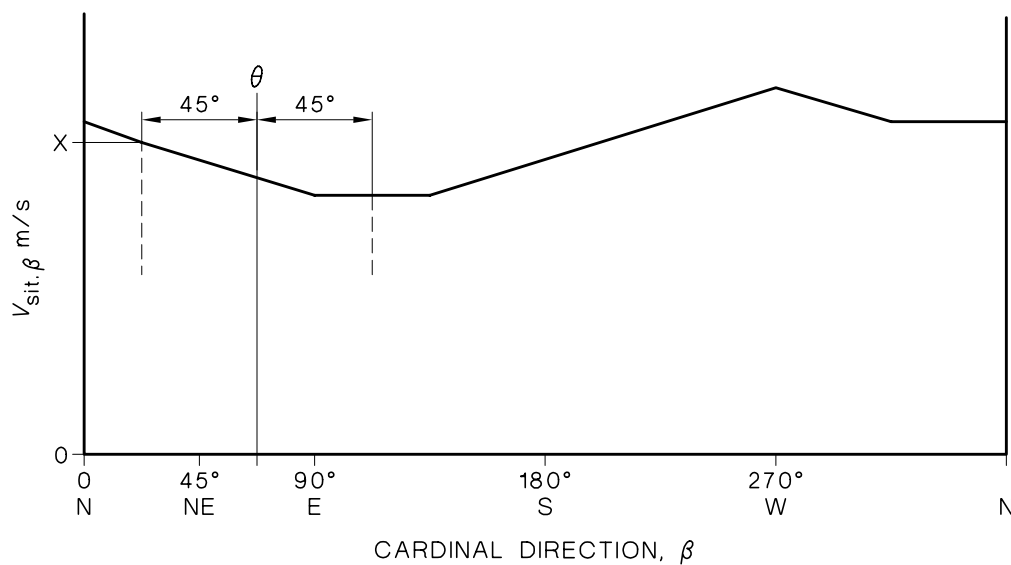


FIGURE 2.2 RELATIONSHIP OF WIND DIRECTIONS AND BUILDING ORTHOGONAL AXES



NOTE: The value of $V_{des,\theta}$ is the maximum of $V_{sit,\beta}$ in the range $\theta \pm 45^\circ$ degrees, which, in the case shown here, is the wind speed X.

FIGURE 2.3 EXAMPLE OF $V_{sit,\beta}$ CONVERSION TO $V_{des,\theta}$

2.4 DESIGN WIND PRESSURE AND DISTRIBUTED FORCES

2.4.1 Design wind pressures

The design wind pressures (p) in pascals, shall be determined for structures and parts of structures as follows:

$$p = (0.5 \rho_{\text{air}}) [V_{\text{des},\theta}]^2 C_{\text{fig}} C_{\text{dyn}} \quad \dots 2.4(1)$$

where

- p = design wind pressure acting normal to a surface, in pascals
 - = p_e , p_i or p_n where the sign is given by the C_p values used to evaluate C_{fig}
 - NOTE: Pressures are taken as positive, indicating pressures above ambient and negative, indicating pressures below ambient.
- ρ_{air} = density of air, which shall be taken as 1.2 kg/m^3
- $V_{\text{des},\theta}$ = building orthogonal design wind speeds (usually, $\theta = 0^\circ, 90^\circ, 180^\circ$, and 270°), as given in Clause 2.3
 - NOTE: For some applications, $V_{\text{des},\theta}$ may be a single value or may be expressed as a function of height (z) e.g., windward walls of tall buildings ($>25\text{m}$).
- C_{fig} = aerodynamic shape factor, as given in Section 5
- C_{dyn} = dynamic response factor, as given in Section 6 (the value is 1.0 except where the structure is wind sensitive, see Section 6)

2.4.2 Design wind distributed forces

The design wind frictional drag force per unit area (f) in pascals, shall be taken for structures and parts of structures as follows:

$$f = (0.5 \rho_{\text{air}}) [V_{\text{des},\theta}]^2 C_{\text{fig}} C_{\text{dyn}} \quad \dots 2.4(2)$$

2.5 WIND ACTIONS

2.5.1 General

- A1 | Wind actions (W_u and W_s) for use in AS/NZS 1170.0 shall be determined as given in Clauses 2.5.2 to 2.5.5 and accelerations as given in Clause 2.5.6.

2.5.2 Directions to be considered

Wind actions shall be derived by considering wind from no fewer than four orthogonal directions aligned to the structure.

2.5.3 Forces on surfaces or structural elements

2.5.3.1 Forces derived from wind pressure

To determine wind actions, the forces (F) in newtons, on surfaces or structural elements, such as a wall or a roof, shall be the vector sum of the forces calculated from the pressures applicable to the assumed areas (A), as follows:

$$F = \sum(p_z A_z) \quad \dots 2.5(1)$$

where

- p_z = design wind pressure, in pascals (normal to the surface), at height z , calculated in Clause 2.4.1
 - NOTE: The sign convention for pressures leads to forces towards the surface for positive pressures and forces away from the surface for negative pressures.
- A_z = a reference area, in square metres, at height z , upon which the pressure at that height (p_z) acts

For enclosed buildings, internal pressures shall be taken to act simultaneously with external pressures including the effects of local pressure factors (K_e). The most severe combinations of internal and external pressures shall be selected for design.

Where variations in the surface pressure with height are considered, the area shall be subdivided so that the specified pressures are taken over appropriate areas (see Clause 4.2 for variation of wind speed with height).

2.5.3.2 Forces derived from frictional drag

To determine wind actions, the forces (F) in newtons, on a building element, such as a wall or a roof, shall be the vector sum of the forces calculated from distributed frictional pressures applicable to the assumed areas, as follows:

$$F = \sum(f_z A_z) \quad \dots 2.5(2)$$

where

f_z = the design frictional distributed force parallel to the surface, calculated in Clause 2.4.2 at height z , in pascals

2.5.3.3 Forces derived from force coefficients

Appendices E and F cover structures for which shape factors are given in the form of force coefficients rather than pressure coefficients. In these cases, to determine wind actions, the forces (F) in newtons, shall be determined as follows:

$$F = (0.5 \rho_{\text{air}}) [V_{\text{des},\theta}]^2 C_{\text{fig}} C_{\text{dyn}} A_{\text{ref}} \quad \dots 2.5(3)$$

where

A_{ref} = as defined in Appendix F, for flags

= $l \times b$ for other structures or elements of structures covered in Appendices E and F

2.5.4 Forces and moments on complete structures

To determine wind actions, the total resultant forces and overturning moments on complete structures shall be taken to be the summation of the effects of the external pressures on all surfaces of the building.

For rectangular enclosed buildings where the ratio d/h or d/b (see Clause 5.4) is greater than 4, the total resultant force on a complete structure shall include the frictional drag calculated in accordance with Clause 5.5.

For dynamic effects, the combination of along-wind and crosswind responses shall be calculated in accordance with Section 6.

2.5.5 Performance of fatigue-sensitive elements

In regions C and D, cladding, its connections and immediate supporting members shall demonstrate performance under the pressure sequences defined in AS 4040.3, based on the ultimate limit state wind pressure on external and internal surfaces, as determined in accordance with this Standard.

2.5.6 Serviceability of wind-sensitive structures

For the purpose of calculating wind actions for serviceability of wind-sensitive chimneys, masts and poles of circular cross-section (as defined in Clause 6.1), deflections shall be calculated in accordance with Section 6.

NOTE: Information on peak acceleration of other wind-sensitive structures is given in Appendix G.

SECTION 3 REGIONAL WIND SPEEDS

3.1 GENERAL

This Section shall be used to calculate gust wind speeds appropriate to the region in which a structure is to be constructed, including wind direction effects.

3.2 REGIONAL WIND SPEEDS (V_R)

A1 | Regional wind speeds (V_R) for all directions based on 3 second gust wind data shall be as given in Table 3.1 for the regions shown in Figure 3.1 where R (average recurrence interval) is the inverse of the annual probability of exceedance of the wind speed (i.e., P for ultimate or serviceability limit states). Refer to AS/NZS 1170.0 for information on values of annual probability of exceedance appropriate for the design of structures.

TABLE 3.1
REGIONAL WIND SPEEDS

Regional wind speed (m/s)	Region				
	Non-cyclonic			Cyclonic	
	A (1 to 7)	W	B	C	D
V_5	32	39	28	F_C 33	F_D 35
V_{10}	34	41	33	F_C 39	F_D 43
V_{20}	37	43	38	F_C 45	F_D 51
A1 V_{25}	37	43	39	F_C 47	F_D 53
V_{50}	39	45	44	F_C 52	F_D 60
V_{100}	41	47	48	F_C 56	F_D 66
V_{200}	43	49	52	F_C 61	F_D 72
V_{500}	45	51	57	F_C 66	F_D 80
V_{1000}	46	53	60	F_C 70	F_D 85
V_{2000}	48	54	63	F_C 73	F_D 90
A1 V_R (see Note)	$67 - 41R^{-0.1}$	$104 - 70R^{-0.045}$	$106 - 92R^{-0.1}$	$F_C \times (122 - 104R^{-0.1})$	$F_D \times (156 - 142R^{-0.1})$

NOTE: The calculated value shall be rounded to the nearest 1 m/s.

3.3 WIND DIRECTION MULTIPLIER (M_d)

3.3.1 Regions A and W

The wind direction multiplier (M_d) for regions A and W shall be as given in Table 3.2.

3.3.2 Regions B, C and D

The wind direction multiplier (M_d) for all directions in regions B, C and D shall be as follows:

- 0.95 for determining the resultant forces and overturning moments on complete buildings and wind actions on major structural elements (members resisting collapse of the whole structure).
- 1.0 for all other cases (including cladding).

3.4 FACTORS FOR REGIONS C AND D (F_C , F_D)

The wind speeds given in Table 3.1 for regions C and D include additional factors (F_C and F_D) which shall be as follows:

- (a) For ultimate limit states wind speeds, $F_D = 1.1$.
- (b) For ultimate limit states wind speeds, $F_C = 1.05$.
- (c) For serviceability limit states wind speeds, F_C and $F_D = 1.0$.

NOTE: The frequency of Category 5 cyclone crossings in Region D in the years 1998 to 2002 is much greater than predicted from the historic data on which the wind speeds in this Standard are based. The factors in this Clause have been introduced to allow for uncertainties in the prediction of ultimate design wind speeds in Regions C and D (tropical cyclones regions) when they are based on recorded wind speeds. The values of these factors may be revised in the future following simulations based on recorded cyclone tracks. Such an analysis would naturally include cyclone activity throughout the northern coast of Australia (i.e., in regions C and D). The effects of long-term climate change may also be included.

TABLE 3.2
WIND DIRECTION MULTIPLIER (M_d)

Cardinal directions	Region A1	Region A2	Region A3	Region A4	Region A5	Region A6	Region A7	Region W
N	0.90	0.80	0.85	0.90	1.00	0.85	0.90	1.0
NE	0.80	0.80	0.80	0.85	0.85	0.95	0.90	0.95
E	0.80	0.80	0.80	0.90	0.80	1.0	0.80	0.80
SE	0.80	0.95	0.80	0.90	0.80	0.95	0.90	0.90
S	0.85	0.90	0.80	0.95	0.85	0.85	0.90	1.0
SW	0.95	0.95	0.85	0.95	0.90	0.95	0.90	1.0
W	1.00	1.00	0.90	0.95	1.00	1.0	1.0	0.90
NW	0.95	0.95	1.00	0.90	0.95	0.95	1.0	0.95
Any direction	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0

A1

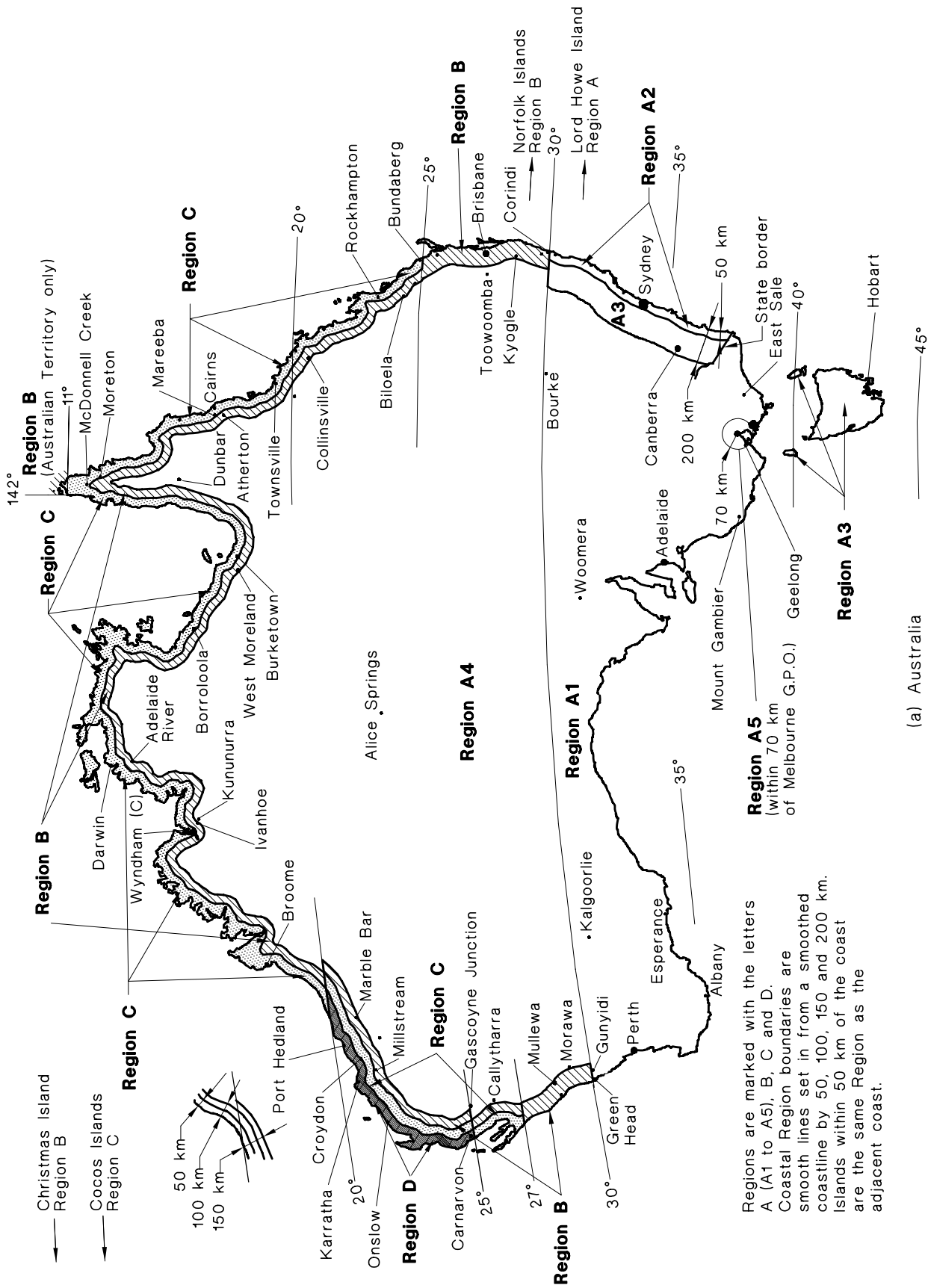
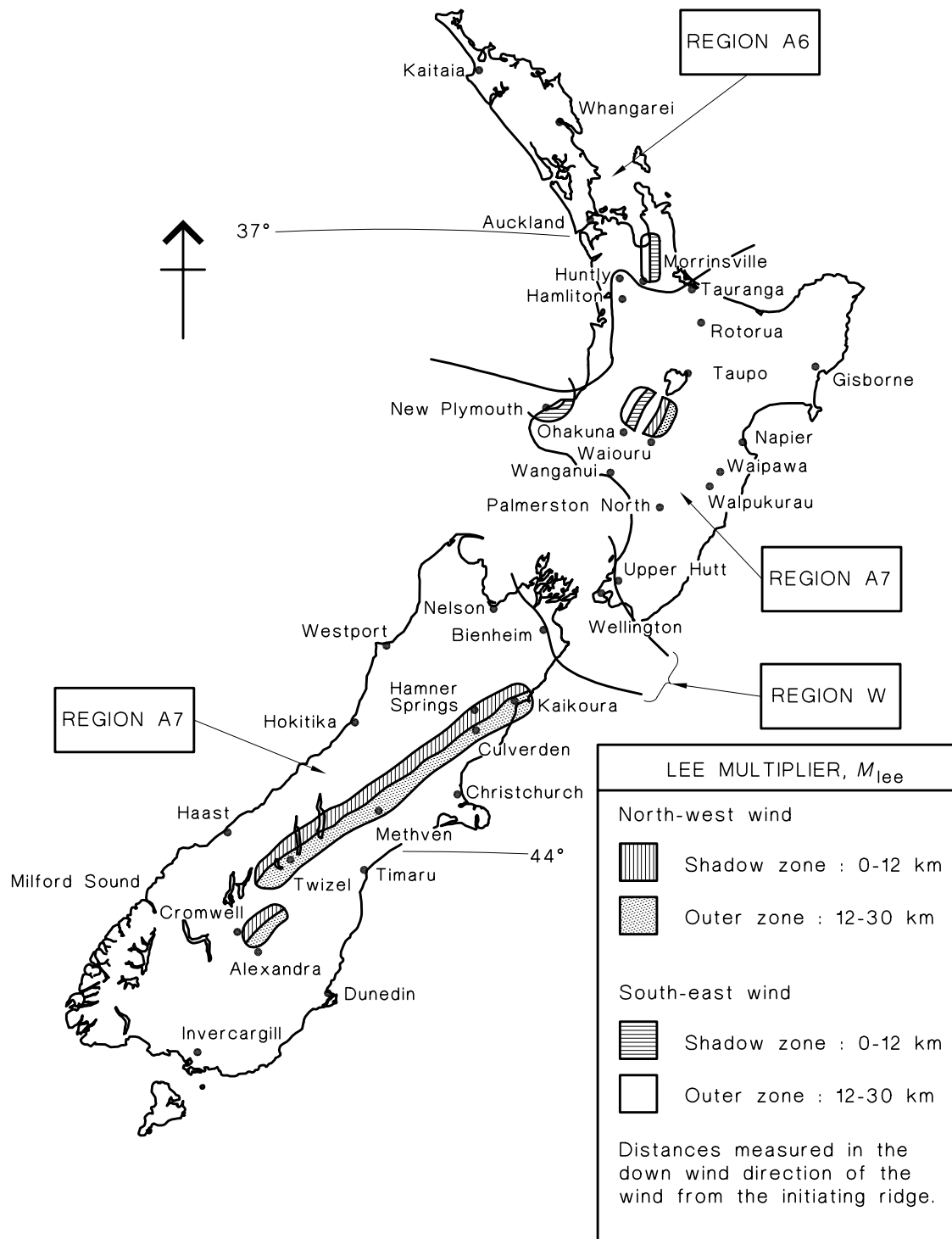


FIGURE 3.1 (in part) WIND REGIONS



(b) New Zealand

FIGURE 3.1 (in part) WIND REGIONS

SECTION 4 SITE EXPOSURE MULTIPLIERS

4.1 GENERAL

This Section shall be used to calculate the exposure multipliers relating to site conditions related to terrain/height ($M_{z,cat}$), shielding (M_s) and topography (M_t).

The design shall take account of known future changes to terrain roughness when assessing terrain category and to buildings providing shielding when assessing shielding.

4.2 TERRAIN/HEIGHT MULTIPLIER ($M_{z,cat}$)

4.2.1 Terrain category definitions

Terrain, over which the approach wind flows towards a structure, shall be assessed on the basis of the following category descriptions:

- (a) *Category 1*—Exposed open terrain with few or no obstructions and water surfaces at serviceability wind speeds.
- (b) *Category 2*—Water surfaces, open terrain, grassland with few, well-scattered obstructions having heights generally from 1.5 m to 10 m.
- (c) *Category 3*—Terrain with numerous closely spaced obstructions 3 m to 5 m high such as areas of suburban housing.
- (d) *Category 4*—Terrain with numerous large, high (10 m to 30 m high) and closely spaced obstructions such as large city centres and well-developed industrial complexes.

Selection of terrain category shall be made with due regard to the permanence of the obstructions that constitute the surface roughness. In particular, vegetation in tropical cyclonic regions shall not be relied upon to maintain surface roughness during wind events.

4.2.2 Determination of terrain/height multiplier ($M_{z,cat}$)

The variation with height (z) of the effect of terrain roughness on wind speed (terrain and structure height multiplier, $M_{z,cat}$) shall be taken from the values for fully developed profiles given in Tables 4.1(A) and 4.1(B). For intermediate values of height and terrain category, use linear interpolation.

TABLE 4.1(A)
TERRAIN/HEIGHT MULTIPLIERS FOR GUST WIND SPEEDS
IN FULLY DEVELOPED TERRAINS—SERVICEABILITY
LIMIT STATE DESIGN—ALL REGIONS AND
ULTIMATE LIMIT STATE—REGIONS A1 TO A7, W AND B

Height (z) m	Terrain/height multiplier ($M_{z,cat}$)			
	Terrain category 1	Terrain category 2	Terrain category 3	Terrain category 4
≤3	0.99	0.91	0.83	0.75
5	1.05	0.91	0.83	0.75
10	1.12	1.00	0.83	0.75
15	1.16	1.05	0.89	0.75
20	1.19	1.08	0.94	0.75
30	1.22	1.12	1.00	0.80
40	1.24	1.16	1.04	0.85
50	1.25	1.18	1.07	0.90
75	1.27	1.22	1.12	0.98
100	1.29	1.24	1.16	1.03
150	1.31	1.27	1.21	1.11
200	1.32	1.29	1.24	1.16
250	1.34	1.31	1.27	1.20
300	1.35	1.32	1.29	1.23
400	1.37	1.35	1.32	1.28
500	1.38	1.37	1.35	1.31

NOTE: For intermediate values of height z and terrain category, use linear interpolation.

TABLE 4.1(B)
TERRAIN/HEIGHT MULTIPLIERS FOR GUST WIND
SPEEDS IN FULLY DEVELOPED TERRAINS—ULTIMATE LIMIT
STATE DESIGN—REGIONS C AND D ONLY

Height (z) m	Multiplier ($M_{z,cat}$)	
	Terrain categories 1 and 2	Terrain categories 3 and 4
≤3	0.90	0.80
5	0.95	0.80
10	1.00	0.89
15	1.07	0.95
20	1.13	1.05
30	1.20	1.15
40	1.25	1.25
50	1.29	1.29
75	1.35	1.35
≥100	1.40	1.40

NOTE: For intermediate values of height z and terrain category, use linear interpolation.

4.2.3 Changes in terrain category

When considering a direction where the wind approaches across ground with changes in terrain category that lie within the averaging distances given in Table 4.2(A) for structure height, the terrain and structure height multiplier ($M_{z,cat}$) shall be taken as the weighted average value over the averaging distance upwind of the structure at height z above ground level (see Figure 4.1(a)).

The weighted average of $M_{z,cat}$ shall be weighted by the length of each terrain upwind of the structure allowing for the lag distance at each terrain category change. An example is given in Figure 4.1(b).

For evaluation at height z , a change in terrain incorporates a lag distance (x_i) given as follows:

$$x_i = z_{0,r} \left[\frac{z}{0.3z_{0,r}} \right]^{1.25}$$

... 4.2

where

- x_i = distance downwind from the start of a new terrain roughness to the position where the developed height of the inner layer equals z (lag distance)
- $z_{0,r}$ = larger of the two roughness lengths at a boundary between roughnesses, as given in Table 4.2(B)
- z = reference height on the structure above the average local ground level

NOTE: Lag distance is not a significant effect for heights less than 15 m.

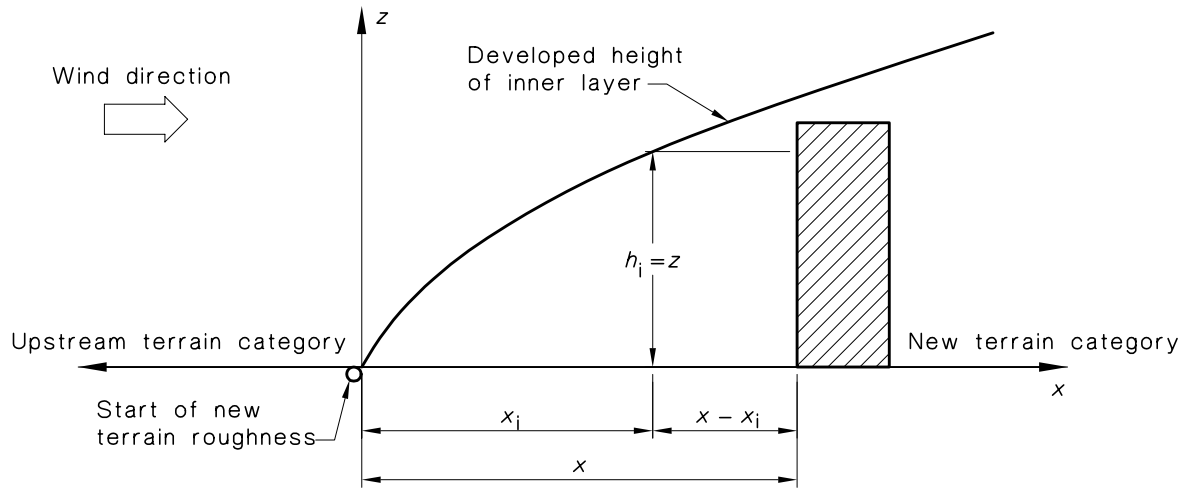
TABLE 4.2(A)
AVERAGING DISTANCE FOR STRUCTURE HEIGHT

Structure height (m)	Averaging distance upwind of structure (m)
$h < 50$	1000
$50 \leq h < 100$	2000
$100 \leq h < 200$	3000

‘Not applicable’

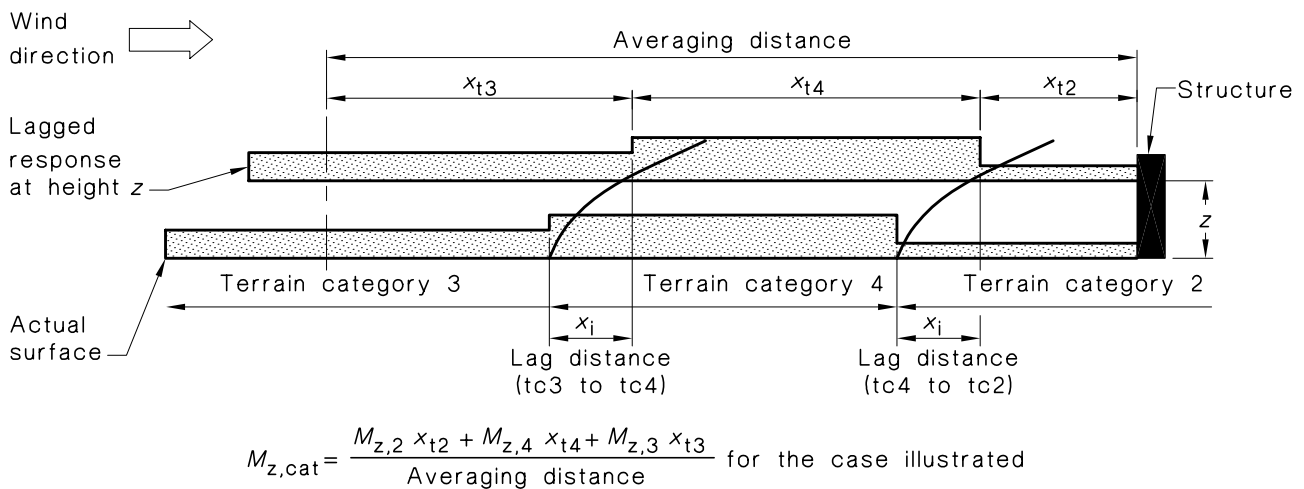
TABLE 4.2(B)
ROUGHNESS LENGTHS FOR TERRAIN CATEGORIES

Terrain category	Roughness length (m)
1	0.002
2	0.02
3	0.2
4	2.0



(a) Notation for changes in terrain category

(a) Notation for changes in terrain category



(b) Example of changes in terrain category

FIGURE 4.1 CHANGES IN TERRAIN CATEGORY

4.3 SHIELDING MULTIPLIER (M_s)

4.3.1 General

The shielding multiplier (M_s) that is appropriate to a particular direction shall be as given in Table 4.3. The shielding multiplier shall be 1.0 where the average upwind ground gradient is greater than 0.2 or where the effects of shielding are not applicable for a particular wind direction or are ignored.

TABLE 4.3
SHIELDING MULTIPLIER (M_s)

Shielding parameter (s)	Shielding multiplier (M_s)
≤ 1.5	0.7
3.0	0.8
6.0	0.9
≥ 12.0	1.0

NOTE: For intermediate values of s , use linear interpolation.

4.3.2 Buildings providing shielding

- A1 Only buildings within a 45° sector of radius $20h$ (symmetrically positioned about the directions being considered) and whose height is greater than or equal to z shall be taken to provide shielding.

4.3.3 Shielding parameter (s)

The shielding parameter (s) in Table 4.3 shall be determined as follows:

$$s = \frac{l_s}{\sqrt{h_s b_s}} \quad \dots 4.3(1)$$

where

$$\begin{aligned} l_s &= \text{average spacing of shielding buildings, given by:} \\ &= h \left(\frac{10}{n_s} + 5 \right) \quad \dots 4.3(2) \end{aligned}$$

h_s = average roof height of shielding buildings

b_s = average breadth of shielding buildings, normal to the wind stream

h = average roof height, above ground, of the structure being shielded

- A1 n_s = number of upwind shielding buildings within a 45° sector of radius $20h$ and with $h_s \geq z$

4.4 TOPOGRAPHIC MULTIPLIER (M_t)

4.4.1 General

The topographic multiplier (M_t) shall be taken as follows:

- (a) For sites in New Zealand and Tasmania over 500 m above sea level:

$$M_t = M_h M_{lee} (1 + 0.00015 E) \quad \dots 4.4(1)$$

where

M_h = hill shape multiplier

M_{lee} = lee (effect) multiplier (taken as 1.0, except in New Zealand lee zones, see Clause 4.4.3)

E = site elevation above mean sea level in meters

(b) Elsewhere, the larger value of the following:

(i) $M_t = M_h$

(ii) $M_t = M_{lee}$

4.4.2 Hill-shape multiplier (M_h)

The hill shape multiplier (M_h) shall be taken as 1.0 except that for the particular cardinal direction in the local topographic zones shown in Figures 4.2 to 4.4, the value shall be as follows:

(a) For $H/(2L_u) < 0.05$, $M_h = 1.0$

(b) For $0.05 \leq H/(2L_u) < 0.45$, (see Figures 4.2 and 4.3)

$$M_h = 1 + \left(\frac{H}{(3.5(z + L_1))} \right) \left(1 - \frac{|x|}{L_2} \right) \quad \dots 4.4(2)$$

(c) For $H/(2L_u) > 0.45$, (see Figure 4.4)

(i) Within the separation zone (see Figure 4.4)

$$M_h = 1 + 0.71 \left[1 - \frac{|x|}{L_2} \right] \quad \dots 4.4(3)$$

(ii) Elsewhere within the local topographic zone (see Figures 4.2 and 4.3), M_h shall be as given in Equation 4.4(2)

where

H = height of the hill, ridge or escarpment

L_u = horizontal distance upwind from the crest of the hill, ridge or escarpment to a level half the height below the crest

x = horizontal distance upwind or downwind of the structure to the crest of the hill, ridge or escarpment

L_1 = length scale, in metres, to determine the vertical variation of M_h , to be taken as the greater of $0.36 L_u$ or $0.4 H$

L_2 = length scale, in metres, to determine the horizontal variation of M_h , to be taken as $4 L_1$ upwind for all types, and downwind for hills and ridges, or $10 L_1$ downwind for escarpments

z = reference height on the structure above the average local ground level

NOTE: Figures 4.2, 4.3 and 4.4 are cross-sections through the structure's site for a particular wind direction.

For the case where x and z are zero, the value of M_h is given in Table 4.4.

Irrespective of the provisions of this Clause, the influence of any peak may be ignored, provided it is distant from the site of the structure by more than 10 times its elevation above sea level.

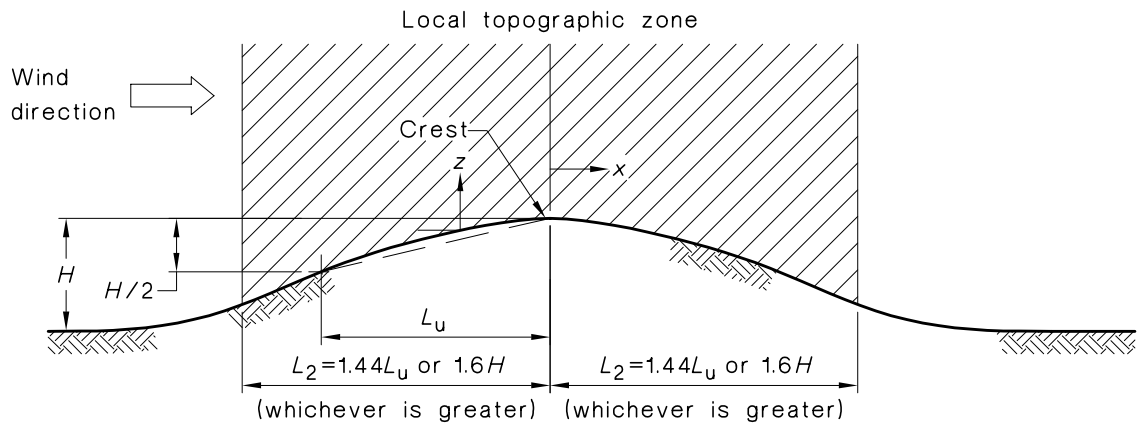
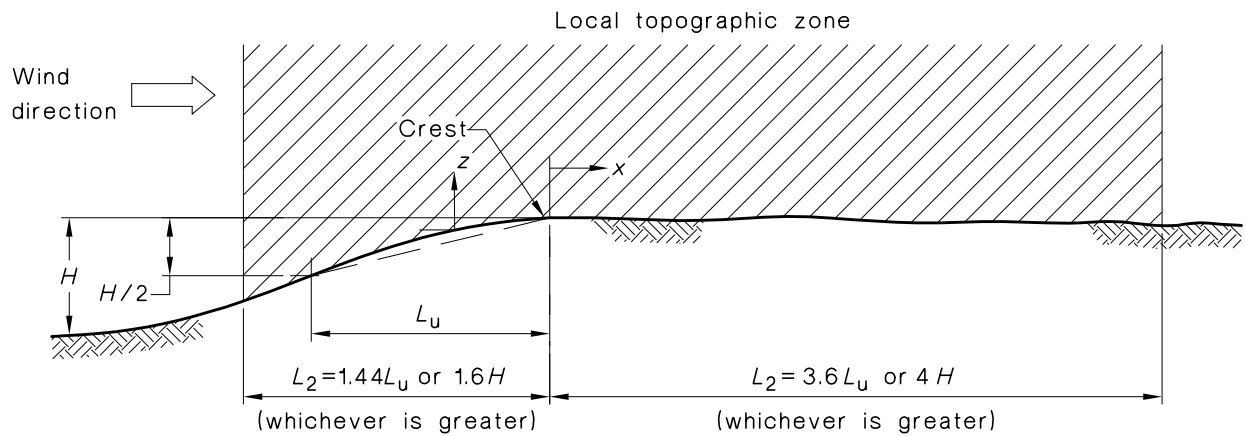


FIGURE 4.2 HILLS AND RIDGES



NOTE: For escarpments, the average downwind slope, measured from the crest to a distance of the greater of $3.6L_u$ or $4H$ shall not exceed 0.05.

FIGURE 4.3 ESCARPMENTS

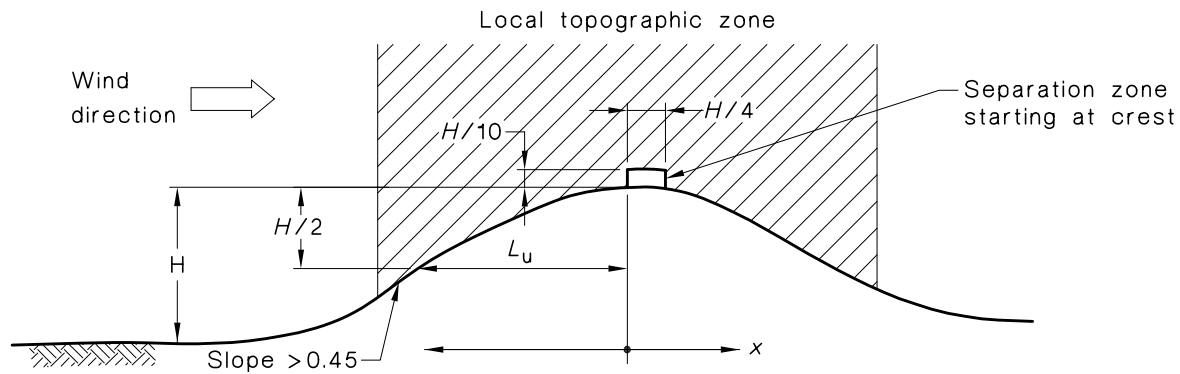


FIGURE 4.4 SEPARATION ZONE FOR SLOPES GREATER THAN 0.45

A1

TABLE 4.4
HILL-SHAPE MULTIPLIER AT CREST
($|x| = 0$), $z = 0$ (FOR GUST WIND SPEEDS)

Upwind slope ($H/2L_u$)	M_h
<0.05	1.0
0.05	1.08
0.10	1.16
0.20	1.32
0.30	1.48
≥ 0.45	1.71

4.4.3 Lee multiplier (M_{lee})

The lee (effect) multiplier (M_{lee}) shall be evaluated for New Zealand sites in the lee zones as shown in Figure 3.1(b). For all other sites, the lee multiplier shall be 1.0. Within the lee zones, the lee multiplier shall apply only to wind from the cardinal directions nominated in Figure 3.1(b).

Each lee zone shall be 30 km in width, measured from the leeward crest of the initiating range, downwind in the direction of the wind nominated. The lee zone comprises a 'shadow lee zone', which extends 12 km from the upwind boundary of the lee zone (crest of the initiating range), and an 'outer lee zone' over the remaining 18 km.

The lee multiplier shall be 1.35 for sites within the shadow lee zone (i.e., within 12 km of the crest of the range). Within the outer lee zone, the lee multiplier shall be determined by linear interpolation with horizontal distance, from the shadow/outer zone boundary (where $M_{lee} = 1.35$), to the downwind lee zone boundary (where $M_{lee} = 1.0$).

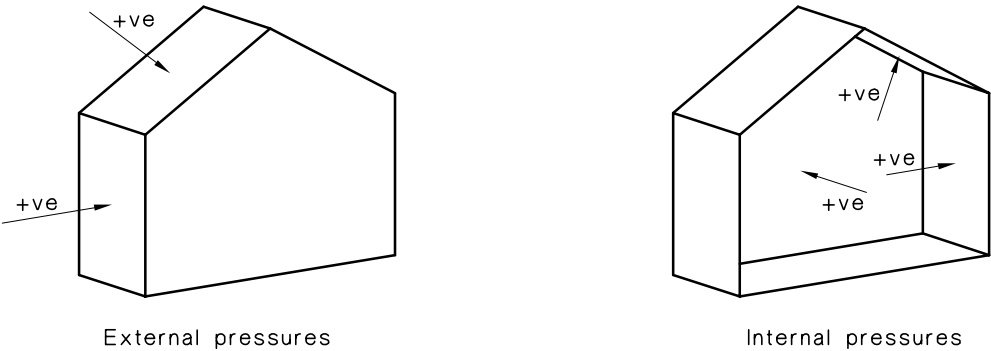
NOTE: No lee zones have been identified in Australia.

SECTION 5 AERODYNAMIC SHAPE FACTOR

5.1 GENERAL

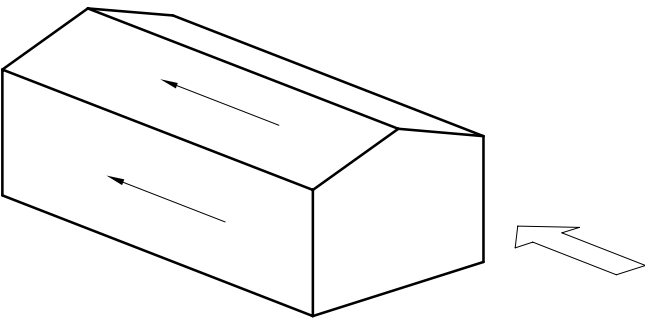
This Section shall be used to calculate the aerodynamic shape factor (C_{fig}) for structures or parts of structures. Values of C_{fig} shall be used in determining the pressures applied to each surface. For calculating pressures, the sign of C_{fig} indicates the direction of the pressure on the surface or element (see Figure 5.1), positive values indicating pressure acting towards the surface and negative values pressure acting away from the surface (less than ambient pressure, i.e., suction). The wind action effects used for design shall be the sum of values determined for different pressure effects such as the combination of internal and external pressure on enclosed buildings.

Clauses 5.3, 5.4 and 5.5 provide values for enclosed rectangular buildings. For the purposes of this Standard, rectangular buildings includes buildings generally made up of rectangular shapes in plan. Methods for particular cases for buildings, free walls, free roofs, exposed members and other structures are given in the appropriate Appendices.



NOTE: C_{fig} is used to give a pressure on one face of the surface under consideration. Positive value of C_{fig} indicates pressure acting towards the surface, negative acting away from the surface.

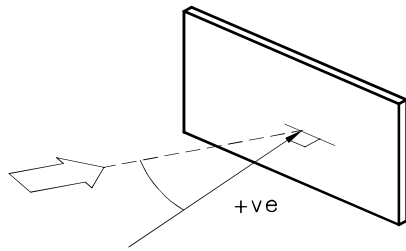
(a) Pressures normal to the surfaces of enclosed buildings



NOTE: C_{fig} is used to give a frictional drag on external surfaces of the structure only. Load per unit area acts parallel to the surface.

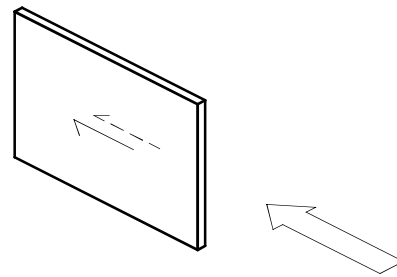
(b) Frictional drag on enclosed buildings

FIGURE 5.1 (in part) SIGN CONVENTIONS FOR C_{fig}



NOTE: C_{fig} is used to give a net pressure normal to the wall derived from face pressures on both upwind and downwind faces. The net pressure always acts normal to the longitudinal axis of the wall.

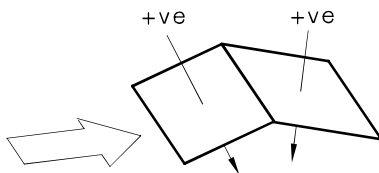
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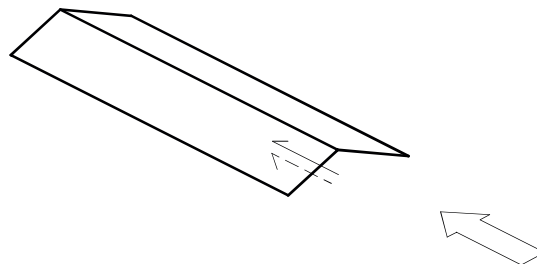
NOTE: C_{fig} is used to give frictional drag on both sides of the wall. Load per unit area acts parallel to both the surfaces of the wall.

(c) Pressure normal to the surfaces of walls and hoardings

(d) Frictional drag on walls and hoardings



NOTE: C_{fig} is used to give net pressure normal to the roof derived from face pressures on both upper and lower surfaces. The net pressure always acts normal to the surface and positive indicates downwards.



NOTE: C_{fig} is used to give the total frictional drag forces derived from face frictional forces on both upper and lower surfaces. Load per unit area acts parallel to both the surfaces of the roof.

(e) Pressure normal to the surfaces freestanding roofs

(f) Frictional drag on freestanding roofs

FIGURE 5.1 (in part) SIGN CONVENTIONS FOR C_{fig}

5.2 EVALUATION OF AERODYNAMIC SHAPE FACTOR

The aerodynamic shape factor (C_{fig}) shall be determined for specific surfaces or parts of surfaces as follows:

- (a) *Enclosed buildings (see this Section 5 and Appendix C)—*

$$C_{fig} = C_{p,e} K_a K_c K_\ell K_p, \text{ for external pressures} \quad \dots 5.2(1)$$

$$C_{fig} = C_{p,i} K_c, \text{ for internal pressures} \quad \dots 5.2(2)$$

$$C_{fig} = C_f K_c, \text{ for frictional drag forces} \quad \dots 5.2(3)$$

- (b) *Circular bins, silos and tanks—see Appendix C.*

- (c) *Freestanding walls, hoardings, canopies and roofs (see Appendix D)*—

$$C_{fig} = C_{p,n} K_a K_\ell K_p, \text{ for pressure normal to surface} \quad \dots 5.2(4)$$

$$C_{fig} = C_f, \text{ for frictional drag forces} \quad \dots 5.2(5)$$

- (d) *Exposed structural members, frames and lattice towers*—see Appendix E.

- (e) *Flags and circular shapes*—see Appendix F.

where

$C_{p,e}$ = external pressure coefficient

K_a = area reduction factor

K_c = combination factor

K_ℓ = local pressure factor

K_p = porous cladding reduction factor

$C_{p,i}$ = internal pressure coefficient

C_f = frictional drag force coefficient

$C_{p,n}$ = net pressure coefficient acting normal to the surface for canopies, freestanding roofs, walls, and the like

5.3 INTERNAL PRESSURE FOR ENCLOSED RECTANGULAR BUILDINGS

5.3.1 General

Aerodynamic shape factors for internal pressure ($C_{p,i}$) shall be determined from Tables 5.1(A) and 5.1(B). Table 5.1(A) shall be used for the design case where openings are shut and the wall permeability dominates. Table 5.1(B) shall be used for the design case where openings are assumed to be open. In all cases, the height at which the wind speed is determined shall be the average roof height (h).

Internal pressure is a function of the relative permeability of the external surfaces of the building. The permeability of a surface shall be calculated by adding areas of opening to leakage on that surface of the building (e.g., vents, gaps in windows).

5.3.2 Openings

Combinations of openings shall be assumed to give internal pressures, which together with external pressures give the most adverse wind actions. Potential openings include doors, windows and vents.

In regions C and D, internal pressure resulting from the dominant opening shall be applied, unless the building envelope (windows, doors and cladding) can be shown to be capable of resisting impact loading equivalent to a 4 kg piece of timber of 100 mm × 50 mm cross-section, projected at 15 m/s at any angle.

5.3.3 Dominant openings

A surface is considered to contain dominant openings if the sum of all openings in that surface exceeds the sum of openings in each of the other surfaces considered one at a time. This does not include normal surface permeability.

NOTE: A dominant opening does not need to be large and can occur as a result of a particular proposed scenario, such as an open air vent, while all other potential openings are shut.

TABLE 5.1(A)
INTERNAL PRESSURE COEFFICIENTS ($C_{p,i}$) FOR BUILDINGS WITH OPEN
INTERIOR PLAN—CASES FOR PERMEABLE WALLS WITHOUT DOMINANT
OPENINGS

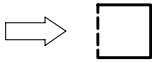
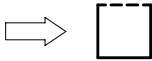
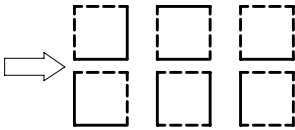
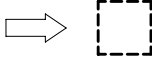
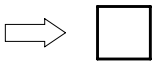
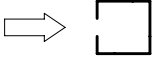
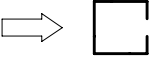
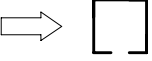
Condition	$C_{p,i}$	Examples showing openings, permeability and wind direction
One wall permeable, other walls impermeable: (a) Windward wall permeable (b) Windward wall impermeable	0.6 -0.3	 
Two or three walls equally permeable, other walls impermeable: (a) Windward wall permeable (b) Windward wall impermeable	-0.1, 0.2 -0.3	
All walls equally permeable	-0.3 or 0.0, whichever is the more severe for combined forces	
A building effectively sealed and having non-opening windows	-0.2 or 0.0, whichever is the more severe for combined forces	

TABLE 5.1(B)
INTERNAL PRESSURE COEFFICIENTS ($C_{p,i}$) FOR BUILDINGS WITH OPEN
INTERIOR PLAN—DOMINANT OPENINGS ON ONE SURFACE

Ratio of dominant opening to total open area (including permeability) of other wall and roof surfaces	Dominant opening on windward wall	Dominant opening on leeward wall	Dominant opening on side wall	Dominant opening on roof
0.5 or less	-0.3, 0.0	-0.3, 0.0	-0.3, 0.0	-0.3, 0.0
1	-0.1, 0.2	-0.3, 0.0	-0.3, 0.0	-0.3, 0.15 $C_{p,e}$
2	0.7 $C_{p,e}$	0.7 $C_{p,e}$	0.7 $C_{p,e}$	0.7 $C_{p,e}$
3	0.85 $C_{p,e}$	0.85 $C_{p,e}$	0.85 $C_{p,e}$	0.85 $C_{p,e}$
6 or more	$C_{p,e}$	$C_{p,e}$	$C_{p,e}$	$C_{p,e}$
				

NOTE: $C_{p,e}$ is the relevant external pressure coefficient at the location of the dominant opening.

5.4 EXTERNAL PRESSURES FOR ENCLOSED RECTANGULAR BUILDINGS

5.4.1 External pressure coefficients ($C_{p,e}$)

The external pressure coefficients ($C_{p,e}$) for surfaces of rectangular enclosed buildings shall be as given in Tables 5.2(A), 5.2(B) and 5.2(C) for walls and 5.3(A), 5.3(B) and 5.3(C) for roofs and for some special roofs in Appendix C. The parameters (e.g. dimensions) referred to in these Tables are set out in Figure 5.2.

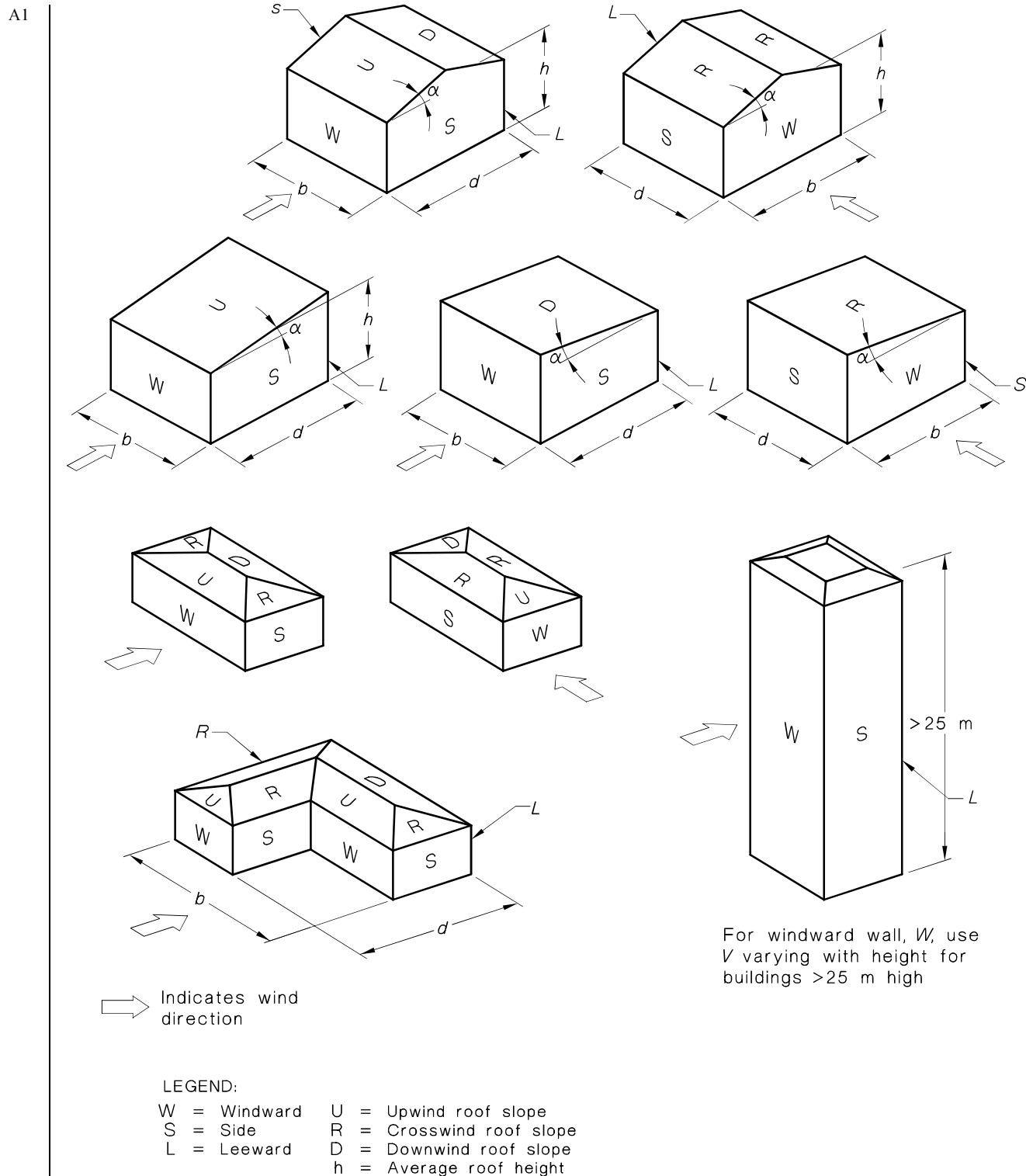


FIGURE 5.2 PARAMETERS FOR RECTANGULAR ENCLOSED BUILDINGS

For leeward walls, side walls and roofs wind speed shall be taken as the value at $z = h$. The reference height (h) shall be taken as the average height of the roof.

Where two values are listed, roofs shall be designed for both values. In these cases, roof surfaces may be subjected to either value due to turbulence. Alternative combinations of external and internal pressures (see also Clause 5.3) shall be considered, to obtain the most severe conditions for design.

A1 For roofs, the following alternative load cases shall be considered:

- (a) When using Table 5.3(A) only, for the appropriate roof type, slope and edge distance—
 - (i) apply the more negative value of $C_{p,e}$ to all pressure zones and surfaces; and
 - (ii) apply the less negative (or most positive) value of $C_{p,e}$ to all pressure zones and surfaces.
- (b) When using both Tables 5.3(B) and 5.3(C), and for the appropriate parameters—
 - (i) apply the more negative value of $C_{p,e}$ from Table 5.3(B) to the upwind slope together with the value from Table 5.3(C) to the downwind slope; and
 - (ii) apply the less negative (or positive) value of $C_{p,e}$ from Table 5.3(B) to the upwind slope together with the value from Table 5.3(C) to the downwind slope.
- (c) When using Table 5.3(C) only, for steeper crosswind slopes on hip roofs, apply the appropriate $C_{p,e}$ value to both slopes.

For the underside of elevated buildings, $C_{p,e}$ shall be taken as 0.8 and -0.6 . For buildings with less elevation above ground than one-third of the height, use linear interpolation between these values and 0.0, according to the ratio of clear unwall height underneath first floor level to the total building height. For the calculation of underside external pressures, wind speed shall be taken as the value at h for all z .

Under-eaves pressures shall be taken as equal to those applied to the adjacent wall surface below the surface under consideration.

TABLE 5.2(A)

**WALLS—EXTERNAL PRESSURE COEFFICIENTS ($C_{p,e}$) FOR
RECTANGULAR ENCLOSED BUILDINGS—WINDWARD WALL (W)**

h	External pressure coefficients ($C_{p,e}$)
> 25.0 m	0.8 (wind speed varies with height)
≤ 25.0 m	For buildings on ground— 0.8, when wind speed varies with height; or 0.7, when wind speed is taken for $z = h$
	For elevated buildings— 0.8 (wind speed taken at h)

A1

TABLE 5.2(B)**WALLS—EXTERNAL PRESSURE COEFFICIENTS ($C_{p,e}$) FOR
RECTANGULAR ENCLOSED BUILDINGS—LEEWARD WALL (L)**

Wind direction θ , degrees (see Figure 2.2)	Roof shape	Roof pitch α , degrees (see Note 1)	d/b (see Note 1)	External pressure coefficients ($C_{p,e}$)
0	Hip or gable	<10	≤ 1 2 ≥ 4	−0.5 −0.3 −0.2
0	Hip or gable	10	All values	−0.3
0	Hip or gable	15		−0.3
0	Hip or gable	20		−0.4
0	Hip or gable	≥ 25	≤ 0.1	−0.75
90	Gable (see Note 2)	All values	≤ 1 2 ≥ 4	−0.5 −0.3 −0.2

NOTES:

- 1 For intermediate values of d/b and α , use linear interpolation.
- 2 For hip roofs use the same values as for $\theta = 0$.

TABLE 5.2(C)**WALLS—EXTERNAL PRESSURE COEFFICIENTS ($C_{p,e}$) FOR
RECTANGULAR ENCLOSED BUILDINGS—SIDE WALLS (S)**

Horizontal distance from windward edge	External pressure coefficients ($C_{p,e}$)
0 to $1h$	−0.65
$1h$ to $2h$	−0.5
$2h$ to $3h$	−0.3
$> 3h$	−0.2

TABLE 5.3(A)**ROOFS—EXTERNAL PRESSURE COEFFICIENTS ($C_{p,e}$) FOR
RECTANGULAR ENCLOSED BUILDINGS—FOR UPWIND SLOPE (U) AND
DOWNWIND SLOPE, (D) FOR $\alpha < 10^\circ$ AND FOR (R) FOR GABLE ROOFS**

Roof type and slope		Horizontal distance from windward edge of roof	External pressure coefficient ($C_{p,e}$)	
Crosswind slopes for gable roofs, (R)	Upwind slope, (U), Downwind slope, (D)		$h/d \leq 0.5$ (see Note 1)	$h/d \geq 1.0$ (see Note 1)
All α	$\alpha < 10^\circ$	0 to $0.5h$	−0.9, −0.4	−1.3, −0.6
		$0.5h$ to $1h$	−0.9, −0.4	−0.7, −0.3
		$1h$ to $2h$	−0.5, 0	(−0.7), (−0.3)
		$2h$ to $3h$	−0.3, 0.1	see Note 2
		$> 3h$	−0.2, 0.2	

NOTES:

- 1 For intermediate values of roof slopes and h/d ratios, use linear interpolation. Interpolation shall only be carried out on values of the same sign.
- 2 The values given in parentheses are provided for interpolation purposes.

TABLE 5.3(B)
ROOFS—EXTERNAL PRESSURE COEFFICIENTS ($C_{p,e}$) FOR
RECTANGULAR ENCLOSED BUILDINGS—UPWIND SLOPE (U) $\alpha \geq 10^\circ$

Roof type and slope	Ratio h/d (see Note)	External pressure coefficients ($C_{p,e}$)						
		Roof pitch, α degrees (see Note)						
		10	15	20	25	30	35	≥ 45
$\alpha \geq 10^\circ$	≤ 0.25	-0.7, -0.3	-0.5, 0.0	-0.3, 0.2	-0.2, 0.3	-0.2, 0.4	0.0, 0.5	$0, 0.8 \sin \alpha$
	0.5	-0.9, -0.4	-0.7, -0.3	-0.4, 0.0	-0.3, 0.2	-0.2, 0.3	-0.2, 0.4	
	≥ 1.0	-1.3, -0.6	-1.0, -0.5	-0.7, -0.3	-0.5, 0.0	-0.3, 0.2	-0.2, 0.3	

NOTE: For intermediate values of roof slopes and h/d ratios, use linear interpolation. Interpolation shall only be carried out on values of the same sign.

TABLE 5.3(C)
ROOFS—EXTERNAL PRESSURE COEFFICIENTS ($C_{p,e}$) FOR
RECTANGULAR ENCLOSED BUILDINGS—DOWNWIND
SLOPE (D), $\alpha \geq 10$ DEGREES AND (R) FOR HIP ROOFS

Roof type and slope		Ratio h/d (see Note)	External pressure coefficients ($C_{p,e}$)			
Crosswind slopes for hip roofs (R)	Downwind slopes (D)		Roof pitch (α), degrees (see Note)			
			10	15	20	≥ 25
For any α	$\alpha \geq 10^\circ$	≤ 0.25	-0.3	-0.5	-0.6	For $b/d < 3$; -0.6
		0.5	-0.5	-0.5	-0.6	For $3 < b/d < 8$; -0.06 (7 + b/d)
		≥ 1.0	-0.7	-0.6	-0.6	For $b/d > 8$; -0.9

NOTE: For intermediate values of roof slopes and h/d ratios, use linear interpolation. Interpolation shall only be carried out on values of the same sign.

5.4.2 Area reduction factor (K_a) for roofs and side walls

For roofs and sidewalls, the area reduction factor (K_a) shall be as given in Table 5.4. For all other cases, K_a shall be taken as 1.0. Tributary area is the area contributing to the force being considered.

TABLE 5.4
AREA REDUCTION FACTOR (K_a)

Tributary area (A), m^2 (see Note)	Area reduction factor (K_a)
≤ 10	1.0
25	0.9
≥ 100	0.8

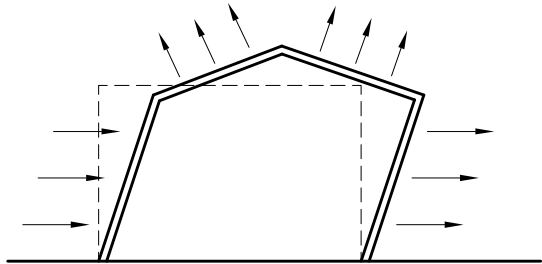
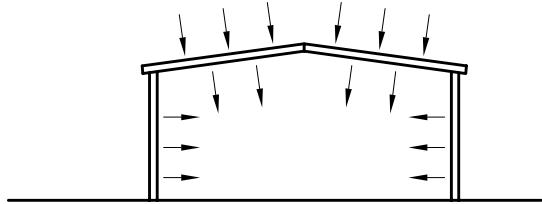
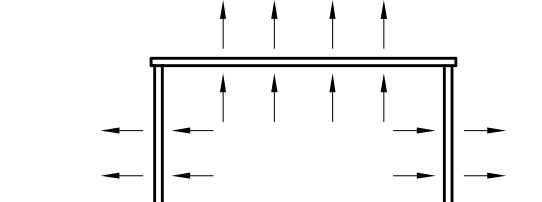
NOTE: For intermediate values of A , use linear interpolation.

5.4.3 Combination factor (K_c)

Where wind pressures acting on two or more surfaces of an enclosed building (e.g., windward wall, upwind roof, side wall, internal pressure, etc.) contribute simultaneously to a structural action effect (e.g. member force) on a major structural element, the combination factor (K_c) given in Table 5.5 may be applied to the combined forces calculated for the critical external and internal surfaces. This factor shall not be applied to cladding or immediate supporting structure such as purlins.

For all surfaces, K_c shall not be less than $0.8/K_a$ (see Clause 5.4.2).

TABLE 5.5
ACTION COMBINATION FACTORS FOR WIND PRESSURE
CONTRIBUTING FROM TWO OR MORE BUILDING SURFACES TO
EFFECTS ON MAJOR STRUCTURAL ELEMENTS

Design case	Combination factor (K_c)	Example diagrams
(a) Where wind action from any single surface contributes 75 percent or more to an action effect	1.0	—
(b) Pressures from windward and leeward walls in combination with positive or negative roof pressures	0.8	
(c) Positive pressures on roofs in combination with negative internal pressures (from a wall opening)	0.8	
(d) Negative pressures on roofs or walls in combination with positive internal pressures	0.95	
(e) All other cases	1.0	—

NOTE: The action combination factors less than 1.0 can be justified because wind pressures are highly fluctuating and do not occur simultaneously on all building surfaces.

5.4.4 Local pressure factor (K_ℓ) for cladding

The local pressure factor (K_ℓ) shall be taken as 1.0 in all cases except when determining the wind forces applied to cladding, their fixings, the members that directly support the cladding, and the immediate fixings of these members. In these cases K_ℓ shall be taken either as 1.0 or from Table 5.6 for the area and locations indicated, whichever gives the most adverse effect when combined with the external and internal pressures. Where more than one case applies, the largest value of K_ℓ from Table 5.6 shall be used.

Where the cladding or the supporting member extends beyond the zone a given in Table 5.6, a value of $K_\ell = 1.0$ shall apply to wind force contributions imposed from beyond that zone.

The value of dimension a is the minimum of $0.2b$ or $0.2d$ or the height (h) as shown in Figure 5.3.

Where interaction is possible, external pressures shall be taken to act simultaneously with internal pressures given in Clause 5.3 and with the under eaves pressures given in Clause 5.4.1 and the resultant forces shall be added. Design cases for negative pressures in Table 5.6 are alternative cases and shall not be applied simultaneously.

For rectangular buildings, the negative limit on the product $K_\ell C_{p,e}$ shall be -2.0 in all cases.

For flat or near flat roofs (slope not greater than 10°) with parapets, values of K_ℓ for areas RA1 and RA2 in the lee of the parapet may be modified by multiplying the values from Table 5.6 by the parapet reduction factor, K_r , given in Table 5.7.

TABLE 5.6
LOCAL PRESSURE FACTOR (K_ℓ)

Design case	Figure 5.3 reference number	h (m)	Area, A	Proximity to edge	K_ℓ
Positive pressures					
Windward wall	WA1	All	$A \leq 0.25a^2$	Anywhere	1.25
All other areas	—	All	—	—	1.0
Negative pressures					
Roof edges	RA1	All	$A \leq a^2$	$< a$	1.5
	RA2	All	$A \leq 0.25a^2$	$< 0.5a$	2.0
Hips and ridges of roofs with pitch $\geq 10^\circ$	RA3	All	$A \leq a^2$	$< a$	1.5
	RA4	All	$A \leq 0.25a^2$	$< 0.5a$	2.0
Side walls near windward wall edges	SA1	≤ 25	$A \leq a^2$	$< a$	1.5
	SA2	≤ 25	$A \leq 0.25a^2$	$< 0.5a$	2.0
	SA3	> 25	$A \leq 0.25a^2$	$> a$	1.5
	SA4	> 25	$A \leq a^2$	$< a$	2.0
	SA5	> 25	$A \leq 0.25a^2$	$< 0.5a$	3.0
All other areas	—	All	—	—	1.0

NOTES:

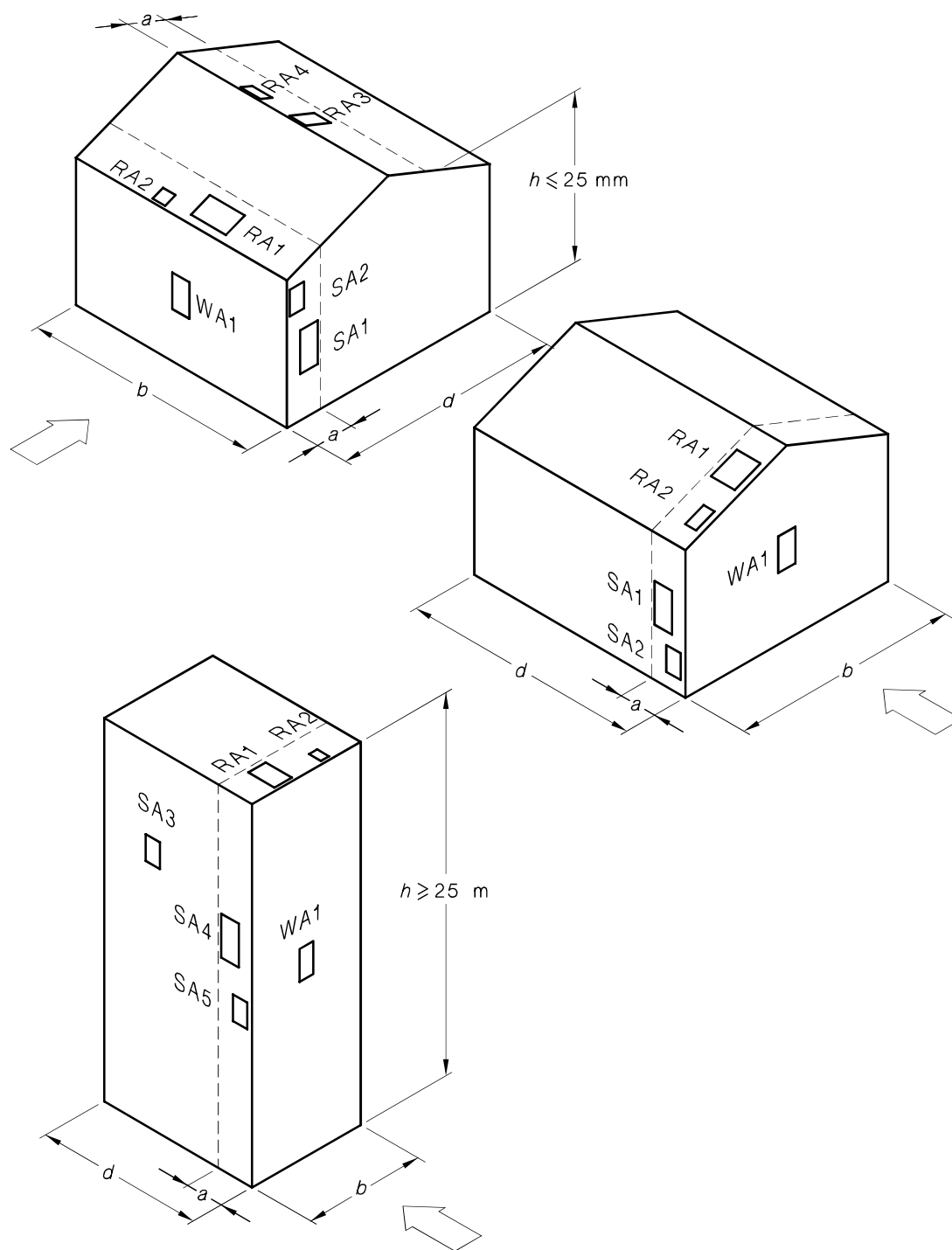
- 1 The dimension a and the Figure reference numbers are defined in Figure 5.3.
- 2 If an area of cladding is covered by more than one case in Table 5.6, use the largest value of K_ℓ obtained for any case.

TABLE 5.7
REDUCTION FACTOR (K_r) DUE TO PARAPETS

h	h_p (see Note)	K_r
≤ 25 m	$\leq 0.07 h$	1.0
	$0.1 h$	0.8
	$\geq 0.2 h$	0.5
> 25 m	$\leq 0.02 w$	1.0
	$0.03 w$	0.8
	$\geq 0.05 w$	0.5

NOTE: h_p = height of parapet above average roof level, w = shortest horizontal dimension of the building. For intermediate values, use linear interpolation.

A1



NOTE: The value of dimension a is the minimum of $0.2b$, $0.2d$ and h .

FIGURE 5.3 LOCAL PRESSURE FACTORS (K_e)

5.4.5 Permeable cladding reduction factor (K_p) for roofs and side walls

The permeable cladding reduction factor (K_p) shall be taken as 1.0 except that where an external surface consists of permeable cladding and the solidity ratio is less than 0.999 and exceeds 0.99, the values given in Table 5.8 may be used for negative pressure. The solidity ratio of the surface is the ratio of solid area to total area of the surface. Figure 5.4 shows dimension d_a .

TABLE 5.8
PERMEABLE CLADDING REDUCTION FACTOR (K_p)

Horizontal distance from windward edge (see Note)	K_p
0 to $0.2d_a$	0.9
$0.2d_a$ to $0.4d_a$	0.8
$0.4d_a$ to $0.8d_a$	0.7
$0.8d_a$ to $1.0d_a$	0.8

NOTE: d_a is the along-wind depth of the surface in metres.

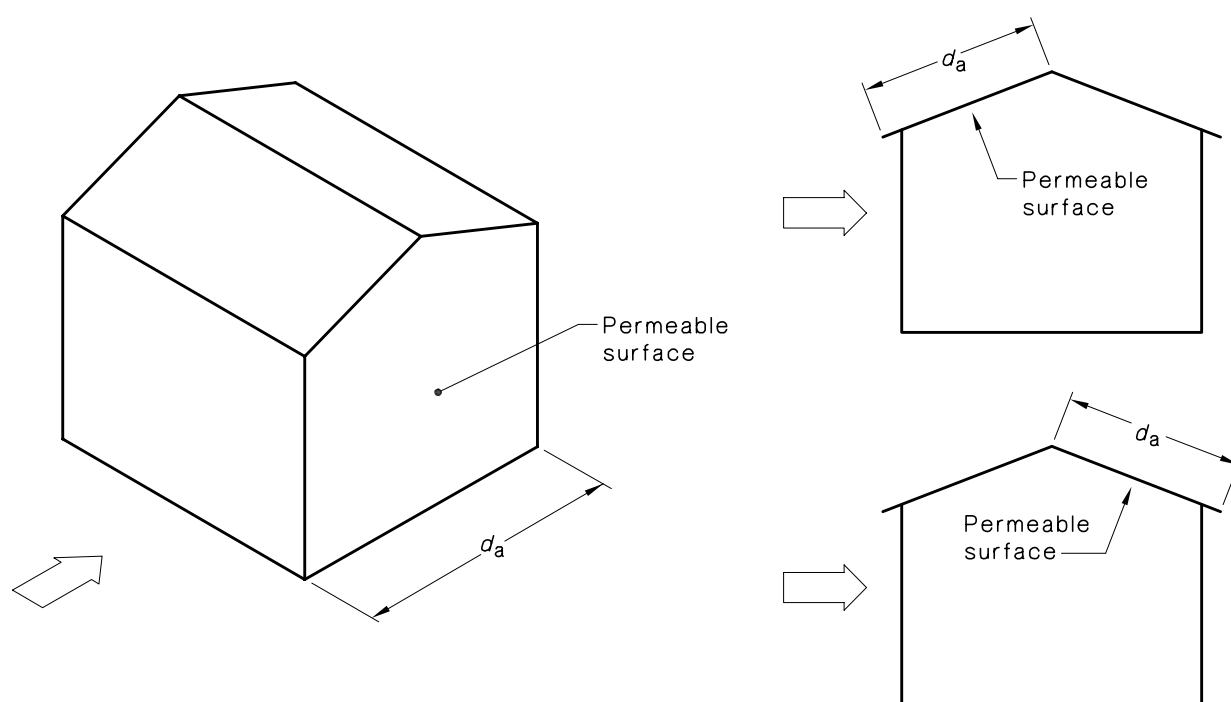


FIGURE 5.4 NOTATION FOR PERMEABLE SURFACES

5.5 FRICTIONAL DRAG FORCES FOR ENCLOSED BUILDINGS

The frictional drag (F) shall be calculated for roofs and side walls of enclosed buildings, in addition to pressures normal to the surface, only where the ratio d/h or d/b is greater than 4. The aerodynamic shape factor (C_{fig}) equals the frictional drag coefficient (C_f) in the direction of the wind as given in Table 5.9.

The effect shall be calculated on the basis of areas as follows:

- (a) For $h \leq b$, area = $(b+2h)(d-4h)$.
- (b) For $h > b$, area = $(b+2h)(d-4b)$.

TABLE 5.9
FRICTIONAL DRAG COEFFICIENT FOR $d/h > 4$ or $d/b > 4$

Distance ' x ' from windward edge	Surface description	C_f
$x \geq$ the lesser of $4h$ and $4b$	Surfaces with ribs across the wind direction	0.04
	Surfaces with corrugations across the wind direction	0.02
	Smooth surfaces without corrugations or ribs or with corrugations or ribs parallel to the wind direction	0.01
$x <$ the lesser of $4h$ and $4b$	All surfaces	0

SECTION 6 DYNAMIC RESPONSE FACTOR

6.1 EVALUATION OF DYNAMIC RESPONSE FACTOR

The dynamic response factor (C_{dyn}) shall be determined for structures or elements of structures with natural first mode fundamental frequencies as follows:

- (a) Greater than 1 Hz, $C_{\text{dyn}} = 1.0$.
- (b) Less than 1 Hz—
 - (i) for tall buildings and towers—
 - (A) less than 0.2 Hz is not covered by this Standard;
 - (B) between 1 Hz and 0.2 Hz, C_{dyn} shall be as defined in Clause 6.2 for along-wind response and Clause 6.3 for crosswind response; and
 - (ii) for cantilever roofs—
 - (A) less than 0.5 Hz is not covered by this Standard;
 - (B) between 1 Hz and 0.5 Hz, C_{dyn} shall be as defined in Paragraph D5, Appendix D.

This Standard does not cover the roofs of podium buildings below tall buildings or where in the walls of tall buildings there are sloping edges or edge discontinuities.

NOTES:

- 1 Information on determining the natural frequency of a structure is given in AS/NZS 1170.2 Supp 1, *Structural design actions—Wind actions—Commentary* (Supplement to AS/NZS 1170.2:2002).
- 2 Appendix G provides information on calculating accelerations for serviceability in tall wind-sensitive structures.

6.2 ALONG-WIND RESPONSE OF TALL BUILDINGS AND TOWERS

6.2.1 General

The dynamic response factor shall be as given in Clause 6.2.2.

NOTE: Information on peak along-wind acceleration for serviceability is given in Appendix G.

6.2.2 Dynamic response factor (C_{dyn})

For calculation of action effects (bending moments, shear forces, member forces) at a height s on the structure (see Figure 6.1), the wind pressures on the structure at a height z shall be multiplied by a dynamic response factor (C_{dyn}). This factor is dependent on both z and s and $s < z < h$. For the calculation of C_{dyn} , the value of $V_{\text{des},\theta}$ is calculated at the reference height, h (top of the building or tower). For calculation of base bending moments, deflections and acceleration at the top of the structure, a single value of C_{dyn} shall be used with s taken as zero.

A1

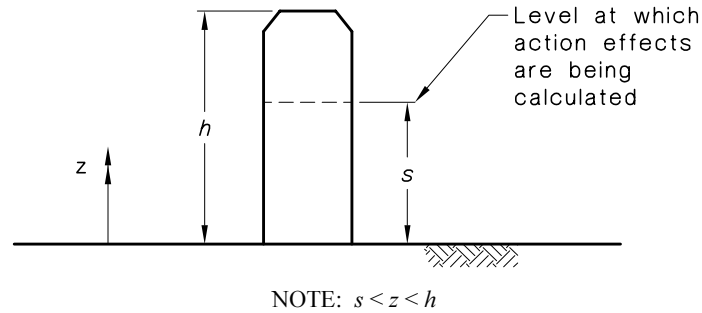


FIGURE 6.1 NOTATION FOR HEIGHTS

The dynamic response factor (C_{dyn}) shall be calculated as follows:

$$C_{\text{dyn}} = \frac{1 + 2I_h \left[g_v^2 B_s + \frac{H_s g_R^2 S E_t}{\zeta} \right]^{0.5}}{(1 + 2g_v I_h)} \quad \dots 6.2(1)$$

where

s = height of the level at which action effects are calculated for a structure

h = average roof height of a structure above the ground

I_h = turbulence intensity, obtained from Table 6.1 by setting z equal to h

g_v = peak factor for the upwind velocity fluctuations, which shall be taken as 3.7

B_s = background factor, which is a measure of the slowly varying background component of the fluctuating response, caused by low frequency wind speed variations, given as follows where b_{sh} is the average breadth of the structure between heights s and h and L_h is a measure of the integral turbulence length scale at height h equal to $85(h/10)^{0.25}$:

$$B_s = \frac{1}{1 + \frac{[0.26(h-s)^2 + 0.46b_{sh}^2]^{0.5}}{L_h}} \quad \dots 6.2(2)$$

H_s = height factor for the resonant response which equals $1 + (s/h)^2$

g_R = peak factor for resonant response (10 min period) given by:

$$= \sqrt{[2 \log_e (600n_a)]} \quad \dots 6.2(3)$$

S = size reduction factor given as follows, where n_a is first mode natural frequency of vibration of a structure in the along-wind direction in Hertz and b_{0h} is the average breadth of the structure between heights 0 and h :

$$= \frac{1}{\left[1 + \frac{3.5n_a h (1 + g_v I_h)}{V_{\text{des}, \theta}} \right] \left[1 + \frac{4n_a b_{0h} (1 + g_v I_h)}{V_{\text{des}, \theta}} \right]} \quad \dots 6.2(4)$$

E_t = $(\pi/4)$ times the spectrum of turbulence in the approaching wind stream, given as follows:

$$= \frac{\pi N}{(1 + 70.8N^2)^{5/6}} \quad \dots 6.2(5)$$

where

N = reduced frequency

$$= n_a L_h [1 + (g_v I_h)] / V_{\text{des},\theta}$$

n_a = first mode natural frequency of vibration of a structure in the along-wind direction in Hertz

$V_{\text{des},\theta}$ = building orthogonal design wind speed determined at the building height, h (see Section 2.3)

ζ = ratio of structural damping to critical damping of a structure (see Table 6.2)

TABLE 6.1
TURBULENCE INTENSITY (I_z)

Height (z) m	Limit state, region and terrain category			
	Serviceability limit states			
	Terrain category 1, all regions	Terrain category 2, all regions	Terrain category 3, all regions	Terrain category 4, all regions
	Ultimate limit states			
	Terrain category 1, Regions A, W and B	Terrain category 2, Regions A, W and B	Terrain category 3, Regions A, W and B Terrain categories 1, 2, and 3: Regions C and D	Terrain category 4, all regions
≤3	0.171	0.207	0.271	0.342
5	0.165	0.196	0.271	0.342
10	0.157	0.183	0.239	0.342
15	0.152	0.176	0.225	0.342
20	0.147	0.171	0.215	0.342
30	0.140	0.162	0.203	0.305
40	0.133	0.156	0.195	0.285
50	0.128	0.151	0.188	0.270
75	0.118	0.140	0.176	0.248
100	0.108	0.131	0.166	0.233
150	0.095	0.117	0.150	0.210
200	0.085	0.107	0.139	0.196
250	0.080	0.098	0.129	0.183
300	0.074	0.092	0.121	0.173
400	0.068	0.082	0.108	0.155
500	0.058	0.074	0.098	0.141

NOTE: For intermediate values of height, z and terrain category, use linear interpolation.

TABLE 6.2
VALUE OF RATIO OF STRUCTURAL DAMPING TO
CRITICAL DAMPING OF THE STRUCTURE (ζ)

Stress levels and type of construction	Ratio of structural damping to critical damping (ζ)
<i>Serviceability limit states</i>	
Steel construction	0.005 to 0.010
Reinforced or prestressed concrete	0.005 to 0.010
<i>Ultimate limit state</i>	
Steel frame welded	0.02
Steel frame bolted	0.05
Reinforced concrete	0.05

6.3 CROSSWIND RESPONSE

6.3.1 General

Clause 6.3.2 gives methods for determining equivalent static forces and base overturning moments and $C_{\text{fig}} C_{\text{dyn}}$ for tall enclosed buildings and towers of rectangular cross-section, and Clause 6.3.3 gives deflections and equivalent static forces for chimneys, masts and poles of circular cross-section. Calculation of crosswind response is not required for porous lattice towers.

NOTES:

- 1 Where calculations of crosswind response yield high values, further guidance is given in AS/NZS 1170.2 Supp 1, *Structural design actions—Wind actions—Commentary* (Supplement to AS/NZS 1170.2:2002).
- 2 Information on peak crosswind acceleration for serviceability is given in Appendix G.

6.3.2 Crosswind response of tall enclosed buildings and towers of rectangular cross-section

6.3.2.1 Equivalent static wind force

The equivalent crosswind static wind force per unit height (w_{eq}) as a function of z , (evaluated using force equals mass times acceleration) in newtons per metre shall be as follows:

$$w_{\text{eq}}(z) = 0.5 \rho_{\text{air}} [V_{\text{des},\theta}]^2 d C_{\text{fig}} C_{\text{dyn}} \quad \dots 6.3(1)$$

where $V_{\text{des},\theta}$ is evaluated at $z = h$, and d is the horizontal depth of the structure parallel to the wind stream and

$$(C_{\text{fig}} C_{\text{dyn}}) = 1.5 g_{\text{R}} \left(\frac{b}{d} \right) \frac{K_{\text{m}}}{(1 + g_{\text{v}} I_{\text{h}})^2} \left(\frac{z}{h} \right)^k \sqrt{\frac{\pi C_{\text{fs}}}{\zeta}} \quad \dots 6.3(2)$$

where

$$\begin{aligned} K_{\text{m}} &= \text{mode shape correction factor for crosswind acceleration, given by:} \\ &= 0.76 + 0.24k \end{aligned}$$

where

$$\begin{aligned} k &= \text{mode shape power exponent for the fundamental mode and values of the exponent } k \text{ should be taken as:} \\ &= 1.5 \text{ for a uniform cantilever} \\ &= 0.5 \text{ for a slender framed structure (moment resisting)} \end{aligned}$$

- = 1.0 for a building with central core and moment resisting façade
- = 2.3 for a tower decreasing in stiffness with height, or with a large mass at the top
- = the value obtained from fitting $\phi_1(z) = (z/h)^k$ to the computed modal shape of the structure
- $\phi_1(z)$ = first mode shape as a function of height z , normalized to unity at $z = h$

C_{fs} = crosswind force spectrum coefficient generalized for a linear mode shape given in Clause 6.3.2.3

6.3.2.2 Crosswind base overturning moment

The crosswind base overturning moment (M_c) in Newton metres (which can be derived by the integration from 0 to h of $w_{eq}(z) z dz$) shall be as follows:

$$M_c = 0.5 g_R b \left[\frac{0.5 \rho_{air} [V_{des,\theta}]^2}{(1 + g_v I_h)^2} \right] h^2 \left(\frac{3}{k+2} \right) K_m \sqrt{\frac{\pi C_{fs}}{\zeta}} \quad \dots 6.3(3)$$

where the value $\left(\frac{3}{k+2} \right) K_m$ is the mode shape correction factor for crosswind base overturning moment.

6.3.2.3 Crosswind force spectrum coefficient (C_{fs})

The reduced velocity (V_n) shall be calculated as follows using $V_{des,\theta}$ calculated at $z = h$, as follows:

$$V_n = \frac{V_{des,\theta}}{n_c b (1 + g_v I_h)} \quad \dots 6.3(4)$$

Values of the crosswind force spectrum coefficient generalized for a linear mode shape (C_{fs}) shall be calculated from the reduced velocity (V_n) as follows (see Figures 6.2 to 6.5):

(a) For a 3:1:1 square section ($h:b:d$), where V_n is in the range 2 to 16:

(i) For turbulence intensity of 0.12 at $2h/3$:

$$\log_{10} C_{fs} = 0.000353 V_n^4 - 0.0134 V_n^3 + 0.15 V_n^2 - 0.345 V_n - 3.109 \quad \dots 6.3(5)$$

(ii) For turbulence intensity of 0.2 at $2h/3$:

$$\log_{10} C_{fs} = 0.00008 V_n^4 - 0.0028 V_n^3 + 0.0199 V_n^2 + 0.13 V_n - 2.985 \quad \dots 6.3(6)$$

(b) For a 6:1:1 square section ($h:b:d$), where V_n is in the range 3 to 16:

(i) For turbulence intensity of 0.12 at $2h/3$:

$$\log_{10} C_{fs} = 0.000406 V_n^4 - 0.0165 V_n^3 + 0.201 V_n^2 - 0.603 V_n - 2.76 \quad \dots 6.3(7)$$

(ii) For turbulence intensity of 0.2 at $2h/3$:

$$\log_{10} C_{fs} = 0.000334 V_n^4 - 0.0125 V_n^3 + 0.141 V_n^2 - 0.384 V_n - 2.36 \quad \dots 6.3(8)$$

(c) For a 6:2:1 rectangular section ($h:b:d$), where V_n is in the range 2 to 18:

(i) For turbulence intensity of 0.12 at $2h/3$:

$$\log_{10} C_{fs} = \frac{-3.2 + 0.0683 V_n^2 - 0.000394 V_n^4}{1 - 0.02 V_n^2 + 0.000123 V_n^4} \quad \dots 6.3(9)$$

(ii) For turbulence intensity of 0.2 at $2h/3$:

$$\log_{10} C_{fs} = \frac{-3 + 0.0637V_n^2 - 0.00037V_n^4}{1 - 0.02V_n^2 + 0.000124V_n^4} \quad \dots 6.3(10)$$

(d) For a 6:1:2 rectangular section ($h:b:d$), where V_n is in the range 2 to 16:

(i) For turbulence intensity of 0.12 at $2h/3$:

$$\log_{10} C_{fs} = 0.000457V_n^3 - 0.0226V_n^2 + 0.396V_n - 4.093 \quad \dots 6.3(11)$$

(ii) For turbulence intensity of 0.2 at $2h/3$:

$$\log_{10} C_{fs} = 0.00038V_n^3 - 0.0197V_n^2 + 0.363V_n - 3.82 \quad \dots 6.3(12)$$

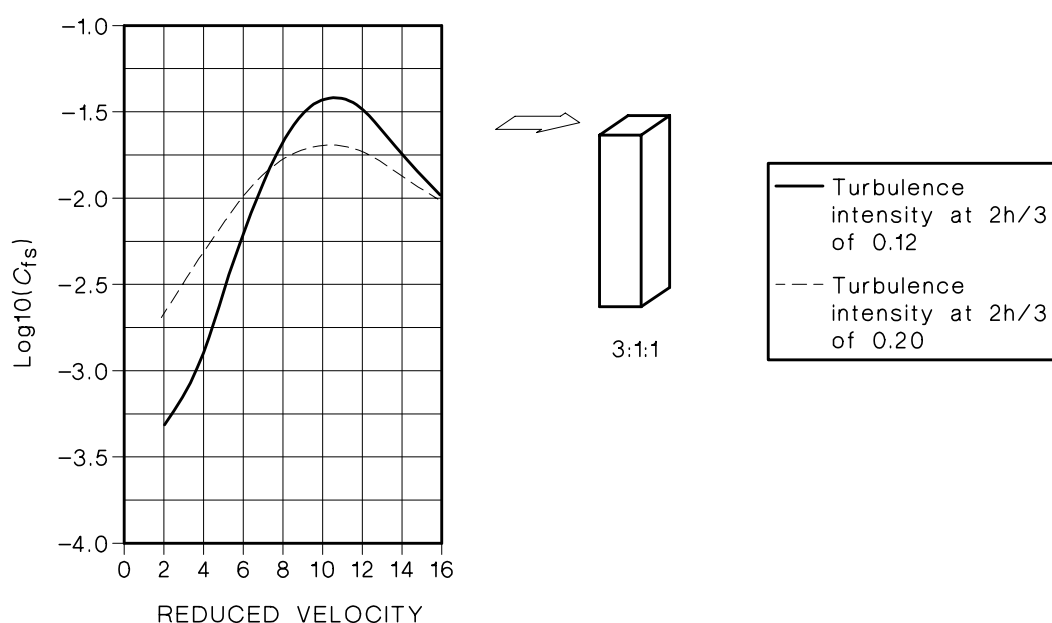


FIGURE 6.2 CROSSWIND FORCE SPECTRUM COEFFICIENT FOR A 3:1:1 SQUARE SECTION

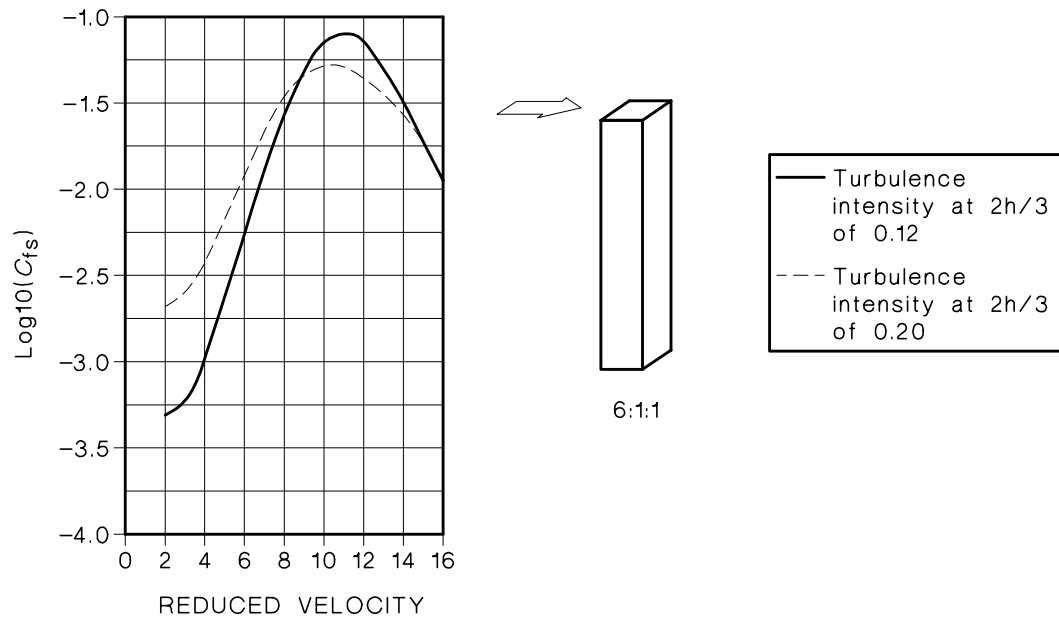


FIGURE 6.3 CROSSWIND FORCE SPECTRUM COEFFICIENT FOR A 6:1:1 SQUARE SECTION

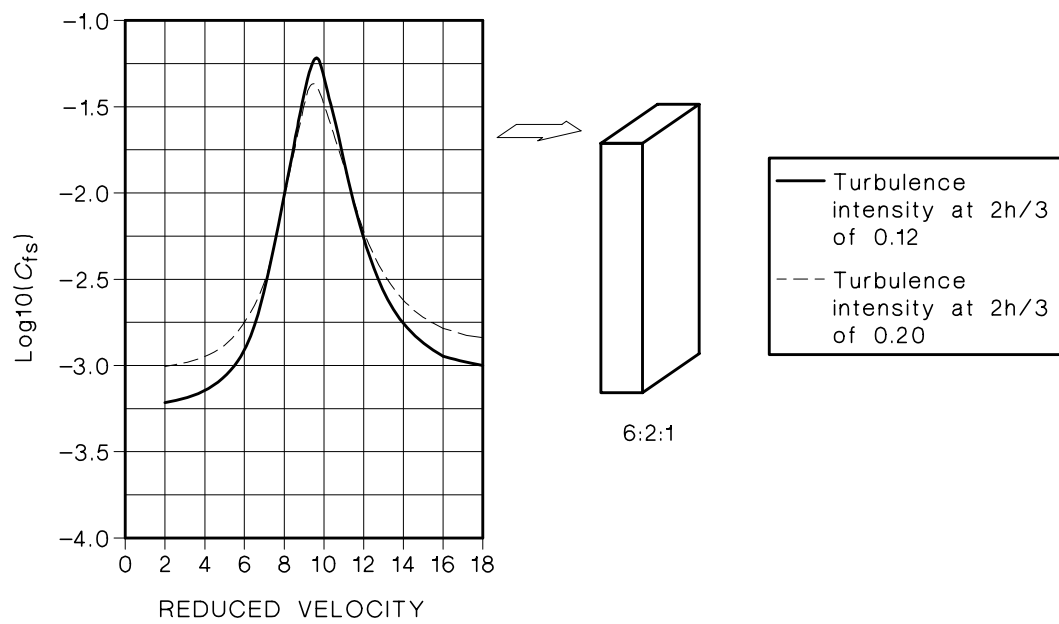


FIGURE 6.4 CROSSWIND FORCE SPECTRUM COEFFICIENT FOR A 6:2:1 RECTANGULAR SECTION

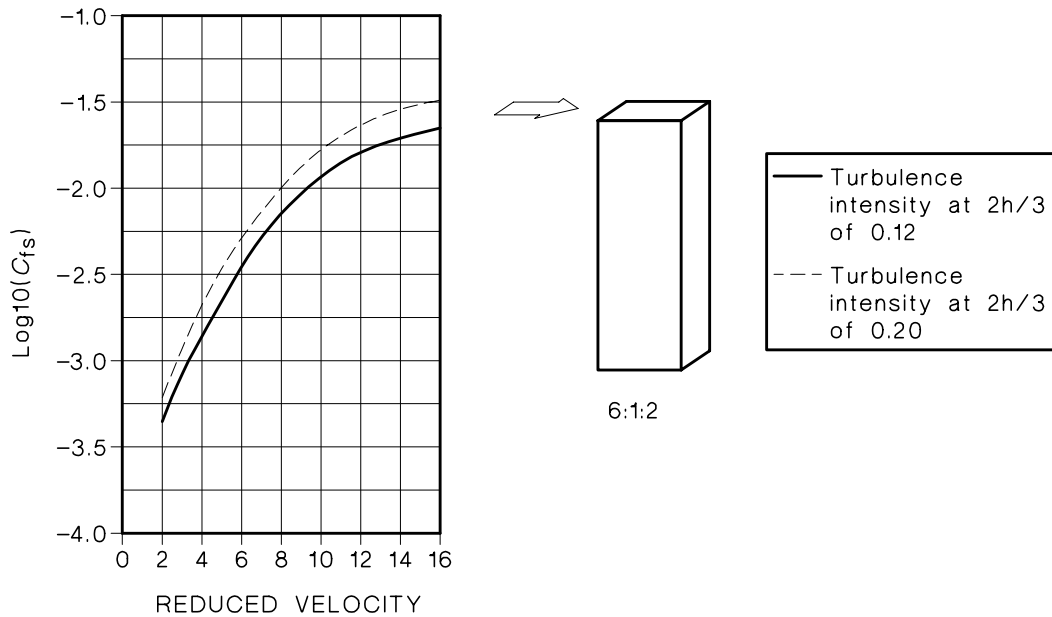


FIGURE 6.5 CROSSWIND FORCE SPECTRUM COEFFICIENT FOR A 6:1:2 RECTANGULAR SECTION

6.3.3 Crosswind response of chimneys, masts and poles of circular cross-section

6.3.3.1 Crosswind tip deflection

The maximum amplitude of tip deflection ($y_{\max}$) in crosswind vibration at the critical wind speed due to vortex shedding for chimneys, masts or poles of circular cross-section (without ladders, strakes or other appendages near the top) shall be calculated as follows:

$$y_{\max} = Kb_t / Sc \quad \dots 6.3(14)$$

where

K = factor for maximum tip deflection, taken as 0.50 for circular cross-sections

b_t = average breadth of the top third of the structure

Sc = Scruton number given by:

$$= 4\pi m_t \zeta / (\rho_{\text{air}} b^2)$$

m_t = average mass per unit height over the top third of the structure

ζ = ratio of structural damping to critical damping of a structure, as given in Table 6.2

6.3.3.2 Equivalent static wind force

The equivalent static wind force per unit height (w_{eq}) for chimneys, masts or poles of circular cross-section (without ladders, strakes or other appendages near the top), as a function of height z , $w_{\text{eq}}(z)$, is given as follows:

$$w_{\text{eq}}(z) = m(z) (2\pi n_1)^2 y_{\max} \phi_1(z) \quad \dots 6.3(15)$$

where

$m(z)$ = mass per unit height as a function of height z

n_1 = first mode natural frequency of vibration of a structure, in Hertz

$\phi_1(z)$ = first mode shape as a function of height z , normalized to unity at $z = h$, which shall be taken as $(z/h)^2$

6.4 COMBINATION OF ALONG-WIND AND CROSSWIND RESPONSE

The total combined peak scalar dynamic action effect (ε_t), such as an axial load in a column, shall be as follows:

$$\varepsilon_t = \varepsilon_{a,m} + [(\varepsilon_{a,p} - \varepsilon_{a,m})^2 + \varepsilon_{c,p}^2]^{0.5} \quad \dots 6.4(1)$$

where

$\varepsilon_{a,m}$ = action effect derived from the mean along-wind response, given as follows, where the values of g_v , I_h and C_{dyn} are defined in Section 6.2.2:

$$= \varepsilon_{a,p} / [C_{dyn} (1 + 2g_v I_h)]$$

$\varepsilon_{a,p}$ = action effect derived from the peak along-wind response

$\varepsilon_{c,p}$ = action effect derived from the peak crosswind response

NOTE: The factor $[C_{dyn} (1 + 2g_v I_h)]$ is a gust factor (G).

APPENDIX A DEFINITIONS

(Normative)

Aerodynamic shape factor

Factor to account for the effects of the geometry of the structure on surface pressure due to wind.

Annual probability of exceedance of the action

The probability that a value will be exceeded in any one year.

NOTE: This is the inverse of the so-called 'return period' better described as the average recurrence interval.

Awning

Roof-like structure, usually of limited extent, projecting from a wall of a building.

Canopy

Roof adjacent to or attached to a building, generally not enclosed by walls.

Cladding

Material that forms the external surface over the framing of a building or structure.

Design wind speed

Wind speed for use in design adjusted for annual probability of exceedance, wind direction, geographic position, surrounding environment and height.

NOTE: For structures over 25 m, design wind speed is expressed as a function of height.

Dominant opening

Opening in the external surface of an enclosed building, which directly influences the average internal pressure in response to external pressures at that particular opening.

NOTE: Dominant openings need not be large.

Drag

Force acting in the direction of the wind stream; *see also* lift.

Dynamic response factor

Factor to account for the effects of fluctuating forces and resonant response on wind-sensitive structures.

Elevated building

Building with a clear, unwallled space underneath the first floor level with a height from ground to underside of the first floor of one-third or more of the total height of the building.

Enclosed building

Building that has a roof and full perimeter walls (nominally sealed) from floor to roof level.

Escarpment

Long, steeply sloping, face between nominally level lower and upper plains where the plains have average slopes of not greater than 5%.

First mode shape

Shape of a structure at its maximum amplitude under first mode natural vibration.

First mode natural frequency

Frequency of free oscillation corresponding to the lowest harmonic of vibration of a structure.

Force coefficient

Coefficient that, when multiplied by the incident wind pressure and an appropriate area (defined in the text), gives the force in a specific direction.

Free roof

Roof (of any type) with no enclosing walls underneath, (e.g., freestanding carport).

Freestanding walls

Walls that are exposed to the wind on both sides, with no roof attached (e.g., fences).

Gable roof

Ridged roof with two sloping surfaces and vertical end walls triangular from lowest points up to the ridge.

Hip roof

A roof with four sloping (pitched) surfaces, pyramidal in shape, and with level eaves all round. A hip roof on a rectangular plan has two triangular sloping roofs at the short sides (hip ends) and two trapezoidal sloping roofs at the long sides.

Hoardings

Freestanding (rectangular) signboards, and the like, supported clear of the ground.

Immediate supports (cladding)

Those supporting members to which cladding is directly fixed (e.g., battens, purlins, girts, studs).

Lag distance

Horizontal distance downwind, required for the effects of a change in terrain roughness on wind speed to reach the height being investigated.

Lattice towers

Three-dimensional frameworks comprising three or more linear boundary members interconnected by linear bracing members joined at common points (nodes), enclosing an open area through which the wind may pass.

Lift

Force acting at 90 degrees to the wind stream; *see also* drag.

Mansard roof

A roof with two slopes on all four sides, the lower slope steeper than the upper slope.

NOTE: A mansard roof with the upper slopes less than 10 degrees may be assumed to be flat topped.

Monoslope roof

Planar roof with a constant slope and without a ridge.

Obstructions

Natural or man-made objects that generate turbulent wind flow, ranging from single trees to forests and from isolated small structures to closely spaced multistorey buildings.

Permeable

Surface with an aggregation of small openings, cracks, and the like, which allows air to pass through under the action of a pressure differential.

Pitched roof

Bi-fold, bi-planar roof (two sloping surfaces) meeting at a central ridge.

Pressure

Air pressure referenced to ambient air pressure. In this Standard, negative values are less than ambient (suction), positive values exceed ambient. Net pressures act normal to a surface in the direction specified within the text.

Pressure coefficient

Ratio of the pressure acting at the point on a surface, to the free stream dynamic pressure of the incident wind.

Rectangular building

For the purposes of Section 5 of this Standard, rectangular buildings include buildings generally made up of rectangular shapes in plan.

Reynolds number

The ratio of the inertial forces to the viscous forces in the airflow.

Ridge (topographic feature)

Long crest or chain of hills that have a nearly linear apex, with sloping faces on either side of the crest.

Roughness length

Theoretical quantification of the turbulence-inducing nature of a particular type of terrain on airflow (wind).

Scruton number

A mass damping parameter.

Solidity (of cladding)

Ratio of the solid area to the total area of the surface.

Structural elements, major

Structural elements whose tributary areas are greater than 10 m².

Structural elements, minor

Structural elements whose tributary areas are less than or equal to 10 m².

Terrain

Surface roughness condition when considering the size and arrangement of obstructions to the wind.

Topography

Major land surface features, comprising hills, valleys and plains, that strongly influence wind flow patterns.

Tornado

Violently rotating column of air, pendant from the base of a convective cloud, and often observable as a funnel cloud attached to the cloud base.

Tributary area

Area of building surface contributing to the force being considered.

Troughed roof

Bi-fold, bi-planar roof with a valley at its lowest point.

APPENDIX B

NOTATION

(Normative)

Unless stated otherwise, the notation used in this Standard shall have the following meanings with respect to a structure, or member, or condition to which a clause is applied. (See Clause 1.5 for units.)

- A = surface area of the element or the tributary area which transmits wind forces to the element, being—
- = the area upon which the pressure acts, which may not always be normal to the wind stream when used in conjunction with the pressure coefficient (C_p);
 - = the projected area normal to the wind stream when used in conjunction with a drag force coefficient (C_d); or
 - = the areas as defined in applicable clauses (see Appendix E) when used in conjunction with a force coefficient, ($C_{F,x}$) or ($C_{F,y}$)
- A_a = reference area of ancillaries on a tower
- A_{ref} = reference area of flag
- $A_{z,s}$ = total projected area of the tower section at height z
- A_z = a reference area, in square metres, at height z , upon which the pressure at that height (p_z) acts
- a = constant for ease of calculation (Paragraph E4.2.3); *or*
dimension used in defining the extent of application of local pressure factors
- B_s = background factor, which is a measure of the slowly varying background component of the fluctuating response, caused by low frequency wind speed variations
- b = breadth of a structure or element, usually normal to the wind stream (see Figures 5.2, C5, C7, D1, E1, E2, E4 and Tables E3, E4 and E5); *or*
average diameter of a circular section
- b_D = diagonal breadth of UHF antennas
- b_i = average diameter or breadth of a section of a tower member
- b_N = normal breadth of UHF antennas
- b_{0h} = average breadth of the structure between heights 0 and h
- b_s = average breadth of shielding buildings, normal to the wind stream
- b_{sh} = average breadth of the structure between heights s and h
- b_t = average breadth of the top third of the structure
- b_z = average breadth of the structure at the section at height z
- b/w = ratio of the average diameter of an ancillary to the average width of a structure
- C_d = drag force coefficient for a structure or member in the direction of the wind stream
- C_{da} = value of drag force coefficient (C_d) on an isolated ancillary on a tower

C_{de}	= effective drag force coefficient for a tower section with ancillaries
C_{dyn}	= dynamic response factor
$C_{F,x}$	= drag force coefficient for a structure or member, in the direction of the x -axis
$C_{F,y}$	= drag force coefficient for a structure or member, in the direction of the y -axis
C_f	= frictional drag force coefficient
C_{fig}	= aerodynamic shape factor
$C_{fig,l}$	= aerodynamic shape factor for the first frame in the upwind direction
C_{fs}	= crosswind force spectrum coefficient generalized for a linear mode shape
$C_{p,b}$	= external pressure coefficient for sides of bins, silos and tanks
$C_{p,e}$	= external pressure coefficient
$C_{p,i}$	= internal pressure coefficient
$C_{p,\ell}$	= net pressure coefficient for the leeward half of a free roof
$C_{p,n}$	= net pressure coefficient acting normal to the surface for canopies, freestanding roofs, walls, and the like
$C_{p,w}$	= net pressure coefficient for the windward half of a free roof
$C_{p1}(\theta_b)$	= external pressure coefficient on walls of bins, silos or tanks of unit aspect ratio ($c/b = 1$) as a function of θ_b
c	= constant for ease of calculation (Paragraph E4.2.3); <i>or</i> = net height of a hoarding, flag, bin, silo or tank (not including roof or lid height); <i>or</i> = height between the highest and lowest points on a hyperbolic paraboloid roof
D	= downwind roof slope
d	= minimum roof plan dimension; <i>or</i> = depth or distance parallel to the wind stream to which the plan or cross-section of a structure or shape extends (e.g., outside diameter); <i>or</i> = length of span of curved roof
d_a	= along-wind depth of a porous wall or roof surface
d_s	= length of span of the first pitched roof in a multi-span building
E	= site elevation above mean sea level, in metres
E_t	= spectrum of turbulence in the approaching wind stream
e	= the base of Napierian logarithms (≈ 2.71828)
e	= horizontal eccentricity of net pressure
F	= force on a building element, in Newtons
F_C	= factor for region C to account for lack of recent analysis of cyclone activity
F_D	= factor for region D to account for lack of recent analysis of cyclone activity
f	= frictional drag force per unit area parallel to a surface, in Newtons per square metre
f_z	= the design frictional distributed force parallel to the surface, calculated in Clause 2.4.2 at height z , in newtons per square metre

g_R	= peak factor for resonant response (10 min period)
g_v	= peak factor for the upwind velocity fluctuations
H	= height of the hill, ridge or escarpment
H_s	= height factor for the resonant response
h	= average roof height of a structure above the ground
h_c	= height from ground to the attached canopy, freestanding roof, wall, or the like
h_i	= developed height of the inner layer, in metres
h_p	= height of parapet above average roof level
h_r	= average height of surface roughness
h_s	= average roof height of shielding buildings
I_h	= turbulence intensity, obtained from Table 6.1 by setting z equal to h
I_z	= turbulence intensity at height z given for various terrain categories in Table 6.1
K	= factor for maximum tip deflection
K_a	= area reduction factor
K_{ar}	= aspect ratio correction factor for individual member forces
K_c	= combination factor
K_i	= factor to account for the angle of inclination of the axis of members to the wind direction
K_{in}	= correction factor for interference
K_ℓ	= local pressure factor
K_m	= mode shape correction factor for crosswind acceleration
K_p	= net porosity factor
K_r	= parapet reduction factor
K_{sh}	= shielding factor for shielded frames in multiple open-framed structures
k	= mode shape power exponent
k_b	= factor for a circular bin
L_h	= measure of integral turbulence length scale at height h
L_u	= horizontal distance upwind from the crest of the hill, ridge or escarpment to a level half the height below the crest
L_1	= length scale, in metres, to determine the vertical variation of M_h , to be taken as the greater of $0.36 L_u$ or $0.4 H$
L_2	= length scale, in metres, to determine the horizontal variation of M_h , to be taken as $4 L_1$ upwind for all types, and downwind for hills and ridges, or $10 L_1$ downwind for escarpments
L	= leeward wall
l	= length of member; <i>or</i> length of a cantilevered roof beam
l_f	= flag length
l_s	= average spacing of shielding buildings

M_c	= crosswind base overturning moment, in Newton metres
M_d	= wind direction multiplier (see Clause 3.3)
M_s	= shielding multiplier
M_t	= topographic multiplier
M_h	= hill shape multiplier
M_{lee}	= lee (effect) multiplier (taken as 1.0, except in New Zealand lee zones, see Clause 4.4.3)
$M_{z,cat}$	= terrain/height multiplier
m_0	= average mass per unit height, in kilograms per metre
m_f	= mass per unit area of flag, in kilograms per square metre
m_t	= average mass per unit height over the top third of the structure, in kilograms per metre
$m(z)$	= mass per unit height as a function of height z , in kilograms per metre
N	= reduced frequency
n	= number of spans of a multi-span roof
n_1	= first mode natural frequency of vibration of a structure, in Hertz
n_a	= first mode natural frequency of vibration of a structure in the along-wind direction, in Hertz
n_c	= first mode natural frequency of vibration of a structure in the crosswind direction, in Hertz
n_s	= number of upwind shielding buildings within a 45° sector of radius $20h$ and with $h_s \geq h$
p	= design wind pressure acting normal to a surface, in pascals = p_e , p_i or p_n where the sign is given by the C_p values used to evaluate C_{fig} NOTE: Pressures are taken as positive, indicating pressures above ambient and negative, indicating pressures below ambient.
p_e	= external wind pressure
p_i	= internal wind pressure
p_n	= net wind pressure
p_z	= design wind pressure, in pascals (normal to the surface), at height z , calculated in Clause 2.4.1 NOTE: The sign convention for pressures leads to forces towards the surface for positive pressures and forces away from the surface for negative pressures.
R	= inverse of the annual probability of exceedance of the wind speed
R	= crosswind roof slope
Re	= Reynolds number
r	= rise of a curved roof; <i>or</i> corner radius of a structural shape
S	= size reduction factor
S	= side wall

	Sc	=	Scruton number
	s	=	shielding parameter; <i>or</i> height of the level at which action effects are calculated for a structure
	T	=	top roof section
	U	=	upwind roof slope
	$V_{des,\theta}$	=	building orthogonal design wind speeds (usually, $\theta = 0^\circ, 90^\circ, 180^\circ$, and 270°), as given in Clause 2.3 NOTE: $V_{des,\theta}$ may be expressed as a function of height z , for some applications, e.g. windward walls of tall buildings ($>25\text{m}$).
	$V_{des,\theta}(z)$	=	building orthogonal design wind speeds as a function of height z
	V_n	=	reduced velocity (non dimensional)
	$V_{sit,\beta}$	=	wind speeds for a site, varying according to compass direction
	V_R	=	regional 3 second gust wind speed, in metres per second, for annual probability of exceedance of $1/R$
A1	W	=	wind actions (see AS/NZS 1170.0)
	W_s	=	wind actions for serviceability limit states (determined using a regional wind speed appropriate to the annual probability of exceedance for serviceability limit states)
	W_u	=	wind actions for ultimate limit states (determined using a regional wind speed appropriate to the annual probability of exceedance specified for ultimate limit states)
	$w_{eq}(z)$	=	equivalent static wind force per unit height as a function of height z , in Newtons per metre
	w	=	width of a tower; <i>or</i> shortest horizontal dimension of the building
	w_c	=	width of canopy, awning carport, or similar, from the face of the building
	x	=	distance from the windward edge of a canopy or cantilevered roof; <i>or</i> horizontal distance upwind or downwind of the structure to the crest of the hill, ridge or escarpment
	x_i	=	distance downwind from the start of a new terrain roughness to the position where the developed height of the inner layer equals z (lag distance)
	$\ddot{x}_{max}$	=	peak acceleration, in metres per second squared, at the top of a structure in the along-wind direction
	$\ddot{y}_{max}$	=	peak acceleration, in metres per second squared, at the top of a structure in the crosswind direction
	y_{max}	=	maximum amplitude of tip deflection in crosswind vibration at the critical wind speed
	z	=	reference height on the structure above the average local ground level
	$z_{0,r}$	=	larger of the two roughness lengths at a boundary between roughnesses
	α	=	angle of slope of a roof, in degrees
	β	=	angle of compass wind direction, in degrees, measured clockwise from North (0°), for determining site wind velocities

	ΔC_d	= additional drag coefficient due to an ancillary attached to one face or located inside the tower section
	Δz	= height of the section of the structure upon which the wind pressure acts
	δ	= solidity ratio of the structure (surface or open frame) which is the ratio of solid area to total area of the structure
	δ_e	= effective solidity ratio for an open frame
	$\varepsilon_{a,m}$	= action effect derived from the mean along-wind response (ε = epsilon)
	$\varepsilon_{a,p}$	= action effect derived from the peak along-wind response
	$\varepsilon_{c,p}$	= action effect derived from the peak crosswind response
	ε_t	= combined peak scalar dynamic action effect
	ζ	= ratio of structural damping to critical damping of a structure (ζ = zeta)
	θ	= angle of the upwind direction to the orthogonal axes of a structure, in degrees
A1	θ_a	= angle of deviation of the wind stream from the line joining the centre of the tower cross-section to the centre of the ancillary, in degrees
	θ_b	= angle from the wind direction to a point on the wall of a circular bin, silo or tank, in degrees
	θ_m	= angle between the wind direction and the longitudinal axis of the member, in degrees
	λ	= spacing ratio for parallel open frames, equal to the frame spacing (centre-to-centre) on the projected frame width normal to the wind direction
	π	= the ratio of the circumference of any circle to its diameter (approx. 3.14159)
A1	ρ_{air}	= density of air, which shall be taken as 1.2 kg/m ³ NOTE: This value is based on 20°C and typical ground level atmospheric pressure and variation may be necessary for very high altitudes or cold environments.
	$\phi_1(z)$	= first mode shape as a function of height z , normalized to unity at $z = h$ (ϕ = phi)

APPENDIX C

ADDITIONAL PRESSURE COEFFICIENTS FOR ENCLOSED BUILDINGS

(Normative)

C1 ADDITIONAL PRESSURE COEFFICIENTS

The external pressure coefficients ($C_{p,e}$) given in this Appendix shall be used to calculate the aerodynamic shape factor for pressures on appropriately shaped enclosed buildings in accordance with Clauses 5.2 and 5.4.

C2 MULTI-SPAN BUILDINGS, ($\alpha < 60^\circ$)

External pressure coefficients ($C_{p,e}$) for the multi-span buildings shown in Figures C1 and C2 for wind directions $\theta = 0^\circ$ and $\theta = 180^\circ$ shall be obtained from Table C1 or Table C2.

Where two values are listed for pressure coefficients in Tables C1 and C2, the roof shall be designed for both values.

All pressure coefficients shall be used with the value of wind speed applying at average roof height (h).

External pressure coefficients for wind directions of $\theta = 90^\circ$ and $\theta = 270^\circ$ shall be obtained from Clause 5.4 but $[-0.05(n - 1)]$ shall be added to the roof pressure coefficients in the region 0 to $1h$ from the leading edge, where n is the total number of spans. For this calculation, take $n = 4$, if n is greater than 4.

TABLE C1
EXTERNAL PRESSURE COEFFICIENTS ($C_{p,e}$) FOR
MULTI-SPAN BUILDINGS—PITCH ROOFS

Surface reference (see Figure C1)				
A	B	C	M	Y
0.7	Use Table 5.2(a), (b) or (c) for same (h/d_s) and α , as appropriate		-0.3 and 0.2 for $\alpha < 10^\circ$ -0.5 and 0.3 for $\alpha \geq 10^\circ$	-0.2

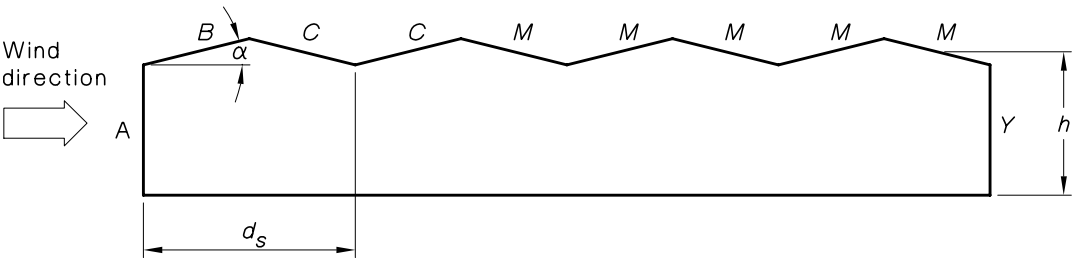


FIGURE C1 EXTERNAL PRESSURE COEFFICIENTS ($C_{p,e}$) FOR MULTI-SPAN BUILDINGS—PITCHED ROOFS

TABLE C2
EXTERNAL PRESSURE COEFFICIENTS ($C_{p,e}$) FOR
MULTI-SPAN BUILDINGS—SAW-TOOTH ROOFS

Wind direction (θ) degrees	Surface reference (see Figure A1.2)								
	A	B	C	D	M	N	W	X	Y
0	0.7	-0.9	-0.9	-0.5, 0.2	-0.5, 0.5	-0.5, 0.3	-0.3, 0.5	-0.4	-0.2
180	-0.2	-0.2, 0.2	-0.3	-0.2, 0.2	-0.4	-0.4	-0.7	-0.3	0.7

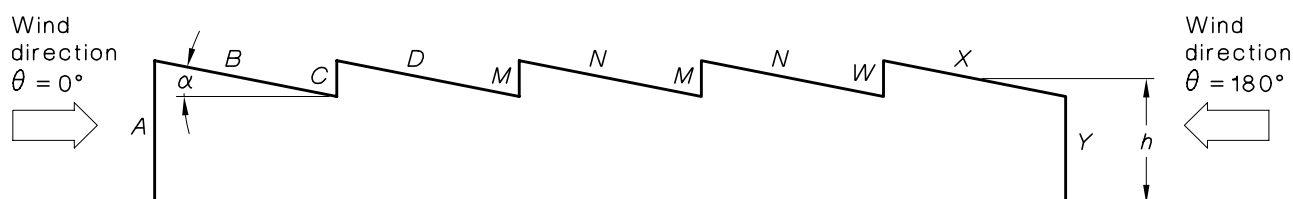


FIGURE C2 EXTERNAL PRESSURE COEFFICIENTS ($C_{p,e}$) FOR MULTI-SPAN BUILDINGS—SAW-TOOTH ROOFS

C3 BUILDINGS WITH CURVED ROOFS

External pressure coefficients ($C_{p,e}$) for curved, arched or domed roofs with profiles approximating a circular arc, for wind directions normal to the axis of the roof and where h/r is not greater than 2 shall be obtained from Table C3.

When two values are listed, the roof shall be designed for both values. In these cases, roof surfaces may be subjected to either positive or negative values due to turbulence. Alternative combinations of external and internal pressures (see Clause 2.5) shall be considered, to obtain the most severe conditions for design.

All pressure coefficients shall be used with the value of wind speed applying at average roof height (h).

External pressure coefficients ($C_{p,e}$) for wind directions parallel to the axis (ridge) of the roof shall be obtained from Table 5.3(A).

The zero values provided for the windward quarter are alternative values for action effects, such as bending, which are sensitive to pressure distribution (turbulence and fluctuations in pressure will produce a range of values occurring at different times during a wind event).

The effect of length to span ratio shall be taken into account by multiplying all the coefficients in Table C3 by a factor of $(b/d)^{0.25}$, where b = breadth normal to the wind and d = span (see Figure C3). If $(b/d)^{0.25}$ is less than 1.0, it shall be taken as 1.0.

Table C3 provides external pressure coefficients for circular arc roofs with no substantial interference to the airflow over the roof. Where a ridge ventilator of a height at least 5% of the total height of the roof is present, the external pressure coefficient on the central half of the roof (T) shall be modified by adding +0.3, that is, the value of a negative coefficient (suction) is reduced by 0.3. Such reductions shall not be made for the wind direction along the axis of the roof, for which the ridge ventilator has little effect on the airflow and resulting external pressures.

All combinations of external pressure coefficients on U, T and D shall be checked.

TABLE C3
EXTERNAL PRESSURE COEFFICIENTS ($C_{p,e}$)—CURVED ROOFS WITH $h/r \leq 2$

Rise-to-span ratio, (r/d)	Windward quarter (U)	Centre half (T)	Leeward quarter (D)
0.05	$-(0.2 + 0.4 h/r)$ or 0.0	$-(0.55 + 0.2 h/r)$ or 0.0	$-(0.4 + 0.2 h/r)$ or 0.0
0.2	$(0.3 - 0.4 h/r)$ or 0.0		$-(0.25 + 0.2 h/r)$ or 0.0
0.5	$(0.5 - 0.4 h/r)$ or 0.0		$-(0.1 + 0.2 h/r)$ or 0.0

- NOTES:
- 1 h is the average roof height and r is the rise of the arch (see Figure C3).
 - 2 For intermediate values of rise to span ratio, use linear interpolation.

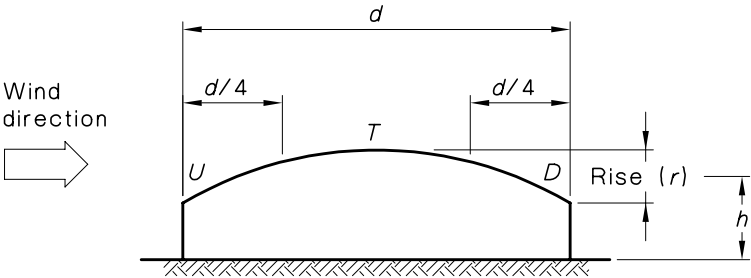


FIGURE C3 EXTERNAL PRESSURE COEFFICIENTS ($C_{p,e}$)—CURVED ROOFS

C4 MANSARD ROOFS

The external pressure coefficients ($C_{p,e}$) for a flat-topped mansard roof (see Figure C4) for the wind direction $\theta = 0^\circ$ shall be determined as follows:

- (a) For upwind slope (U)—using values for upwind slope given in Clause 5.4.1.
- (b) For downwind slope (D)—using values for downwind slope given in Clause 5.4.1, using the same roof pitch α as for the upwind slope.
- (c) For flat top (T)—using the same values as determined for downwind slope.

The external pressure coefficients ($C_{p,e}$) for the wind direction $\theta = 90^\circ$ shall be determined from Clause 5.4.1 assuming R for gable roofs.

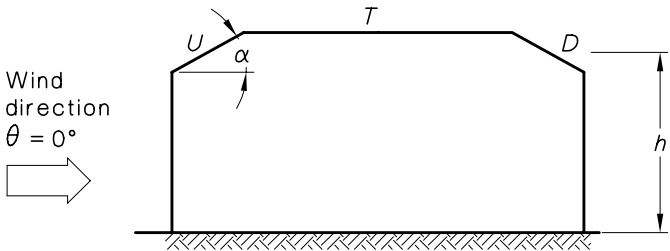


FIGURE C4 EXTERNAL PRESSURE COEFFICIENTS ($C_{p,e}$) FOR MANSARD ROOFS

C5 CIRCULAR BINS, SILOS AND TANKS

C5.1 General

Grouped circular bins, silos and tanks with spacing between walls greater than two diameters shall be treated as isolated silos. Closely spaced groups with spacing less than 0.1 diameters shall be treated as a single structure for wind actions and pressure determined using Tables 5.2 and 5.3. For intermediate spacings use linear interpolation.

C5.2 Isolated circular bins, silos and tanks

C5.2.1 Walls

The aerodynamic shape factor (C_{fig}) for calculating external pressures on the walls of bins, silos and tanks of circular cross-section shall be equal to the external pressure coefficients ($C_{p,b}$) as a function of the angle θ_b (see Figure C5), given as follows for shapes in the ranges indicated:

$$C_{p,b}(\theta_b) = k_b C_{p1}(\theta_b) \quad \dots \text{C5(1)}$$

where

the cylinder is standing on the ground or supported by columns of a height not greater than the height of the cylinder (c),

c/b is in the range 0.25 to 4.0 inclusive

θ_b = angle from the wind direction to a point on the wall of a circular bin, silo or tank, in degrees

k_b = factor (or function) for a circular bin, given as follows:

$$= 1.0 \quad \text{for } C_{p1} \geq -0.15$$

$$= 1.0 - 0.55(C_{p1}(\theta_b) + 0.15) \log_{10}(c/b) \quad \text{for } C_{p1} < -0.15 \quad \dots \text{C5(2)}$$

$$C_{p1}(\theta_b) = -0.5 + 0.4 \cos \theta_b + 0.8 \cos 2 \theta_b + 0.3 \cos 3 \theta_b - 0.1 \cos 4 \theta_b - 0.05 \cos 5 \theta_b \quad \dots \text{C5(3)}$$

For calculating the overall drag force on the wall section of circular bins, silos and tanks (both elevated and on ground) C_{fig} shall be taken as 0.63 (based on an elevation area $b \times c$). This drag force coefficient arises from an integration of the along-wind component of the normal pressures given by Equations C5(2) and C5(3).

External pressure coefficients for the underside of elevated bins, silos and tanks shall be calculated as for elevated enclosed rectangular buildings (see Clause 5.4.1).

Figure C6 is a graphical presentation of the external pressure coefficient (C_{p1}) for circular bins, silos and tanks of unit aspect ratio (i.e., $c/b = 1.0$) at individual locations around the perimeter, and θ_b degrees from the incident wind direction as calculated from Equation C5(1).

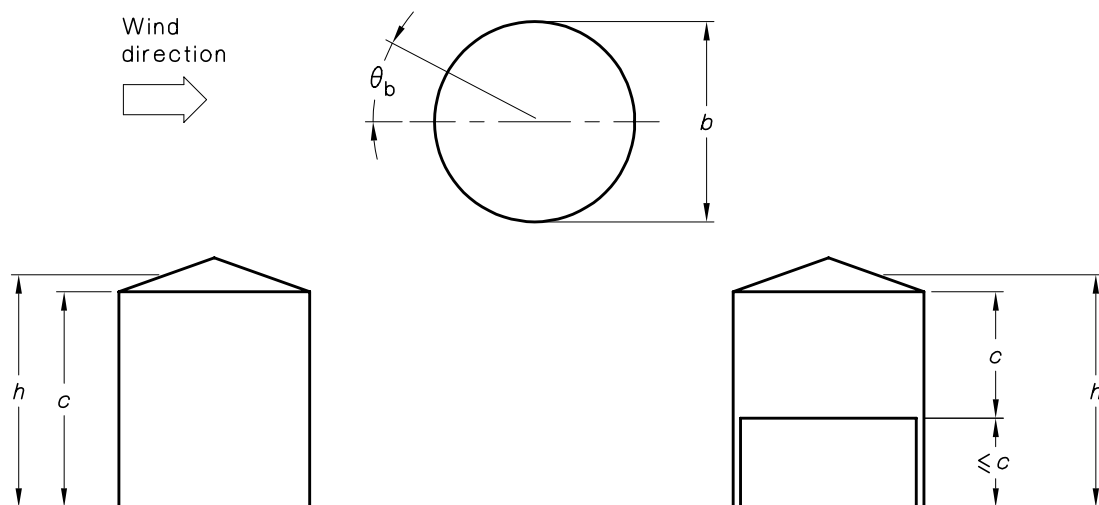


FIGURE C5 EXTERNAL PRESSURE COEFFICIENTS ($C_{p,b}$) ON WALLS OF CIRCULAR BINS, SILOS AND TANKS ($0.25 \leq c/b \leq 4.0$)

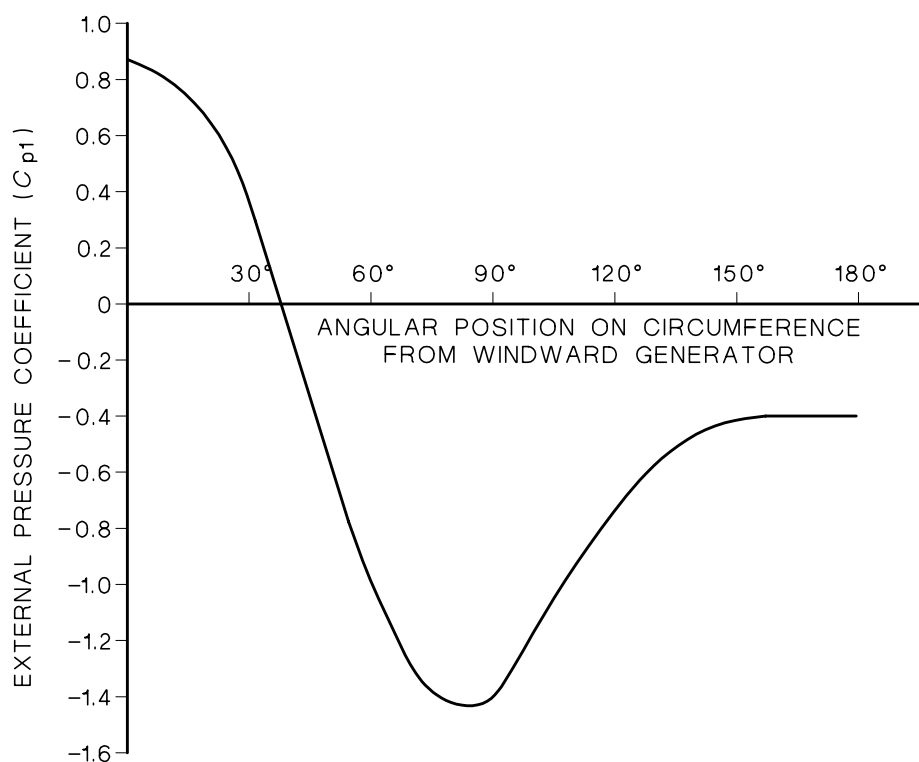


FIGURE C6 PLOT OF EXTERNAL PRESSURE COEFFICIENTS (C_{p1}) ON WALLS OF CIRCULAR BINS, SILOS AND TANKS ($c/b = 1$)

C5.2.2 Roofs and lids

The aerodynamic shape factor (C_{fig}) for calculating external pressures on the roofs or lids of bins, silos or tanks of circular cross-section as shown in Figure C7 shall be as follows:

$$C_{fig} = C_{p,e} K_a K_\ell \quad \dots \text{C4}$$

where $C_{p,e}$ is given in Table C7 for zones A and B as shown in Figure C7, K_a is given in Clause 5.4.2 and K_ℓ is given in Clause 5.4.4.

The local pressure factor (K_ℓ) is applicable to the windward edges of roofs with slope less than or equal to 30° , and to the region near the cone apex for roofs with slope greater than 15° . The applicable areas are shown in Figure C7.

TABLE C7
EXTERNAL PRESSURE COEFFICIENTS
($C_{p,e}$) FOR ROOFS OF CIRCULAR BINS,
SILOS AND TANKS

Zone A	Zone B
-0.8	-0.5

A1

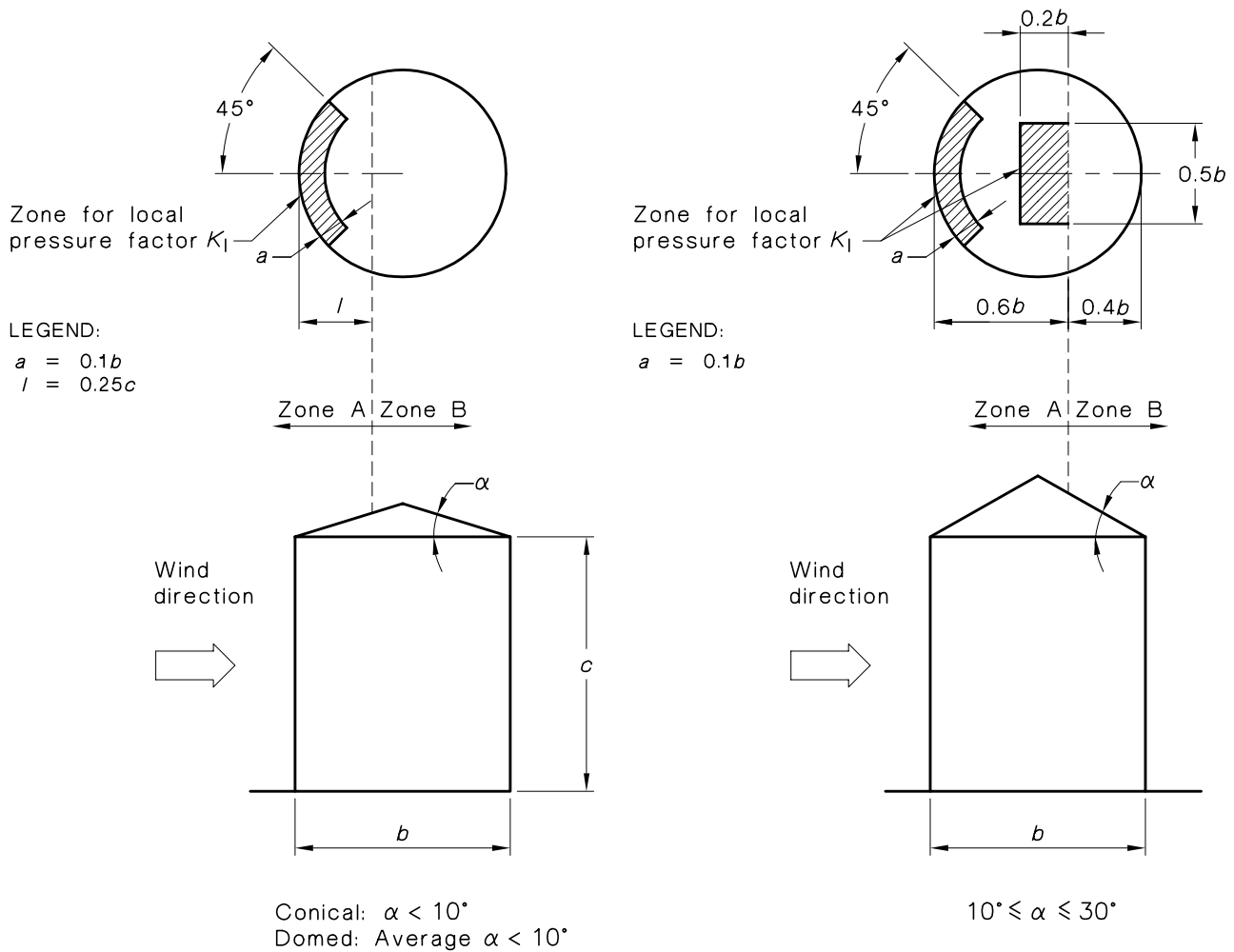


FIGURE C7 EXTERNAL PRESSURE COEFFICIENTS ($C_{p,e}$) FOR ROOFS OF CIRCULAR BINS, SILOS AND TANKS ($0.25 < c/b < 4.0$)

C5.2.3 Internal pressures in bins, silos and tanks

Internal pressures within bins, silos and tanks with vented roofs shall be determined as an area-weighted average of the external pressures at the position of the vents and openings, determined according to Paragraph C5.2.2.

For open-top bins, silos or tanks, the internal pressure shall be determined as follows:

$$C_{fig} = C_{p,i} \quad \dots 6.2(2)$$

$$= -0.9 - 0.35 \log_{10}(c/b)$$

APPENDIX D

FREESTANDING WALLS, HOARDINGS AND CANOPIES

(Normative)

D1 GENERAL

D1.1 Application

This Appendix shall be used to calculate aerodynamic shape factors (C_{fig}) for the following structures and structural elements:

- (a) Free roofs, including hyperbolic paraboloid roofs.
- (b) Canopies, awnings and carports (adjacent to enclosed buildings).
- (c) Cantilevered roofs.
- (d) Hoardings and freestanding walls.

To calculate forces on the structure, use the area of the structure (one side only) as the reference area for normal pressures, and use the area of all the affected sides as the reference area for frictional pressure.

D1.2 Area reduction factor (K_a)

For the design of freestanding roofs and canopies, the area reduction factor (K_a) shall be as defined in Clause 5.4.2. For all other cases in this Appendix, $K_a = 1.0$.

D1.3 Local net pressure factor (K_ℓ)

For the design of cladding elements and elements that offer immediate support to the cladding in free roofs and canopies, the values of local net pressure factor (K_ℓ) given in Table D1 shall be used. For other elements in free roofs and canopies and for all other cases in this Appendix, $K_\ell = 1.0$.

TABLE D1
LOCAL NET PRESSURE FACTORS (K_ℓ) FOR OPEN STRUCTURES

Case	Description	Local net pressure factor (K_ℓ)
1	Pressures on an area between 0 and $1.0a^2$ within a distance $1.0a$ from a roof edge, or a ridge with a pitch of 10° or more	1.5
2	Pressures on an area of $0.25a^2$ or less, within a distance $0.5a$ from a roof edge, or a ridge with a pitch of 10° or more	2.0
3	Upward net pressures on an area of $0.25a^2$ or less, within a distance $0.5a$ from a windward corner of a free roof with a pitch of less than 10°	3.0

NOTES:

- 1 Where a is 20% of the shortest horizontal plan dimension of the free roof or canopy.
- 2 If an area of cladding is covered by more than one case in Table D1, the largest value of K_ℓ shall be used.

D1.4 Net porosity factor (K_p)

For freestanding hoardings and walls, the net porosity factor (K_p) is defined in this Clause. For all other cases in this Appendix, $K_p = 1.0$.

$$K_p = 1 - (1 - \delta)^2 \quad \dots D1$$

where

δ = solidity ratio of the structure (surface or open frame), which is the ratio of solid area to total area of the structure

D2 FREESTANDING HOARDINGS AND WALLS

D2.1 Aerodynamic shape factor for normal net pressure on freestanding hoardings and walls

The aerodynamic shape factor (C_{fig}) for calculating net pressure across freestanding rectangular hoardings or walls (see Figure D1) shall be as follows:

$$C_{fig} = C_{p,n} K_p \quad \dots D2$$

where

$C_{p,n}$ = net pressure coefficient acting normal to the surface, obtained from Table D2 using the dimensions defined in Figure D1

K_p = porous cladding reduction factor, as given in Paragraph D1.4

NOTES:

- 1 The factors K_a and K_ℓ do not appear in this equation as they are taken as 1.0.
- 2 Height for calculation of $V_{des,\theta}$ is the top of the hoarding or wall (i.e., height (h), see Figure D1).

The resultant of the pressure shall be taken to act at half the height of the hoarding, ($h - c/2$), or wall, ($c/2$), with a horizontal eccentricity e .

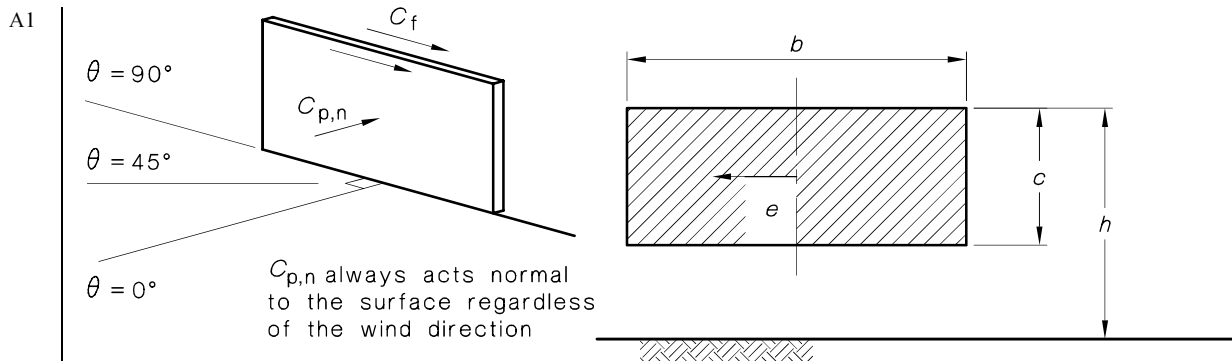


FIGURE D1 FREESTANDING HOARDINGS AND WALLS

TABLE D2(A)

NET PRESSURE COEFFICIENTS FOR HOARDINGS AND FREESTANDING WALLS—WIND NORMAL TO HOARDING OR WALL, $\theta = 0$ degrees

b/c	c/h	$C_{p,n}$	e
0.5 to 5	0.2 to 1	$1.3 + 0.5 (0.3 + \log_{10} (b/c))(0.8 - c/h)$	0
> 5		As above for $b/c = 5$	0
all	<0.2	As above for $c/h = 0.2$	0

TABLE D2(B)

NET PRESSURE COEFFICIENTS FOR HOARDINGS AND FREESTANDING WALLS—WIND AT 45 DEGREES TO HOARDING OR WALL, $\theta = 45$ degrees

b/c	c/h	$C_{p,n}$	e
0.5 to 5 inclusive	0.2 to 1	$1.3 + 0.5 (0.3 + \log_{10} (b/c))(0.8 - c/h)$	$0.2b$
	<0.2	As above for $c/h = 0.2$	$0.2b$

TABLE D2(C)

NET PRESSURE COEFFICIENTS FOR HOARDINGS AND FREESTANDING WALLS—WIND AT 45 DEGREES TO HOARDING OR WALL, $\theta = 45$ degrees

b/c	c/h	Distance from windward free end	$C_{p,n}$ (see Note)
>5	≤ 0.7	0 to $2c$	3.0
		$2c$ to $4c$	1.5
		$>4c$	0.75
	>0.7	0 to $2h$	2.4
		$2h$ to $4h$	1.2
		$>4h$	0.6

NOTE: Where a return wall or hoarding forms a corner extending more than $1c$, the $C_{p,n}$ on 0 to $2c$ for a hoarding shall be 2.2, and 0 to $2h$ for a wall $C_{p,n}$ shall be 1.8.

TABLE D2(D)

NET PRESSURE COEFFICIENTS FOR HOARDINGS AND FREESTANDING WALLS—WIND PARALLEL TO HOARDING OR WALL, $\theta = 90$ degrees

b/c	c/h	Distance from windward free end	$C_{p,n}$ (see Note)
all	≤ 0.7	0 to $2c$	± 1.2
		$2c$ to $4c$	± 0.6
		$>4c$	± 0.3
	>0.7	0 to $2h$	± 1.0
		$2h$ to $4h$	± 0.25
		$>4h$	± 0.25

NOTE: Take values of $C_{p,n}$ of the same sign.

D2.2 Aerodynamic shape factor for frictional drag

The aerodynamic shape factor (C_{fig}) for calculating frictional drag effects on freestanding hoardings and walls, where the wind is parallel to the hoarding or wall, shall be equal to C_f , which shall be determined as given in Table D3. The frictional drag on both surfaces shall be calculated and summed and added to the force on any exposed members calculated in accordance with Appendix E.

TABLE D3
FRICTIONAL DRAG COEFFICIENT

Surface description	C_f
Surfaces with ribs across the wind direction	0.04
Surfaces with corrugations across the wind direction	0.02
Smooth surfaces without corrugations or ribs or with corrugations or ribs parallel to the wind direction	0.01

D3 FREE ROOFS AND CANOPIES

D3.1 Aerodynamic shape factor for local net pressure on free roofs

The aerodynamic shape factor (C_{fig}) for calculating local net pressures normal to free roofs of monoslope, pitched or troughed configuration shall be as follows:

$$C_{fig} = C_{p,n} K_a K_\ell \quad \dots D3$$

where

$C_{p,n}$ = net pressure coefficient acting normal to the surface, obtained for the windward half of a free roof ($C_{p,w}$) or net pressure coefficient for the leeward half of a free roof ($C_{p,\ell}$) as given in Tables D4 to D7 for roofs within the geometrical limits given (positive indicates net downward pressure)

K_a = area reduction factor, as given in Paragraph D1.2

K_ℓ = local pressure factor, as given in Paragraph D1.3

NOTE: The factor K_p does not appear in this equation as it is taken as 1.0.

For free roofs of low pitch with fascia panels, the fascia panel shall be treated as the wall of an elevated building, and the C_{fig} found from Clause 5.4.

In Tables D4, D5, D6 and D7, 'empty under' implies that any goods or materials stored under the roof, block less than 50 percent of the cross-section exposed to the wind. 'Blocked under' implies that goods or materials stored under the roof block more than 75 percent of the cross-section exposed to the wind.

To obtain intermediate values of blockage and roof slopes other than those shown, use linear interpolation. Interpolation shall be carried out only between values of the same sign. Where no value of the same sign is given, for interpolation purposes 0.0 shall be assumed.

A1 | Where alternative pressure coefficient values are listed in Tables D4(A), D4(B), D5 and D6, for the appropriate roof slope, blockage and wind direction, all combinations of values of $C_{p,w}$ and $C_{p,\ell}$ shall be considered.

For $\theta = 90$ degrees, and all configurations, the roof pitch is effectively zero, and Table D4(A) shall be used to determine $C_{p,n}$.

TABLE D4(A)
NET PRESSURE COEFFICIENTS ($C_{p,n}$) FOR
MONOSLOPE FREE ROOFS— $0.25 \leq h/d \leq 1$ (see Figure D2)

Roof pitch (α) degrees	$\theta = 0$ degrees				$\theta = 180$ degrees			
	$C_{p,w}$		$C_{p,\ell}$		$C_{p,w}$		$C_{p,\ell}$	
	Empty under	Blocked under	Empty under	Blocked under	Empty under	Blocked under	Empty under	Blocked under
0	-0.3, 0.4	-1.0, 0.4	-0.4, 0.0	-0.8, 0.4	-0.3, 0.4	-1.0, 0.4	-0.4, 0.0	-0.8, 0.4
15	-1.0	-1.5	-0.6, 0.0	-1.0, 0.2	0.8	0.8	0.4	-0.2
30	-2.2	-2.7	-1.1, -0.2	-1.3, 0.0	1.6	1.6	0.8	0.0

TABLE D4(B)
NET PRESSURE COEFFICIENTS ($C_{p,n}$) FOR
MONOSLOPE FREE ROOFS— $0.05 \leq h/d < 0.25$ (see Figure D2)

Conditions	h/d	Horizontal distance (x) from windward edge	Net pressure coefficients ($C_{p,n}$)
For $\alpha \leq 5^\circ$, or For all α with $\theta = 90^\circ$	$0.05 \leq h/d < 0.25$	$x \leq 1h$	Values given for $C_{p,w}$ in Table D4(A), for $\alpha = 0^\circ$
		$1h < x \leq 2h$	Values given for $C_{p,\ell}$ in Table D4(A), for $\alpha = 0^\circ$
		$x > 2h$	-0.2, 0.2 for empty under -0.4, 0.2 for blocked under

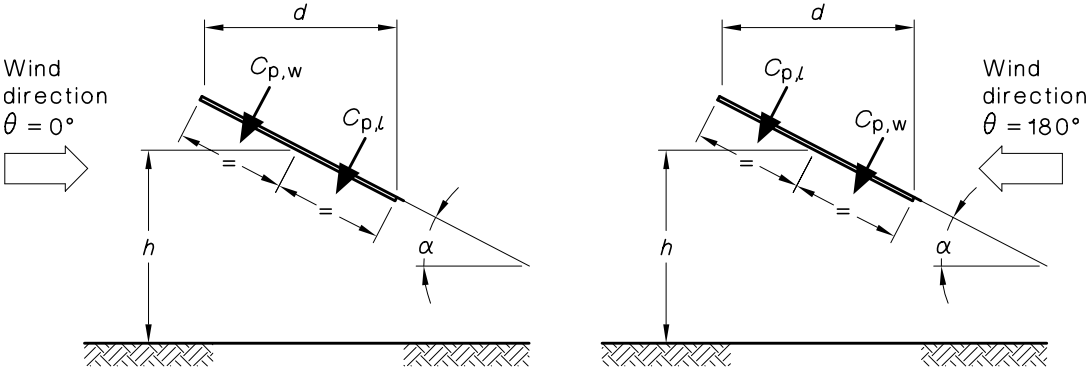


FIGURE D2 MONOSLOPE FREE ROOFS

TABLE D5
NET PRESSURE COEFFICIENTS ($C_{p,n}$) FOR
PITCHED FREE ROOFS— $0.25 \leq h/d \leq 1$ (see Figure D3)

Roof pitch (α) degrees	$\theta = 0^\circ$			
	$C_{p,w}$		$C_{p,l}$	
	Empty under	Blocked under	Empty under	Blocked under
≤ 15	-0.3, 0.4	-1.2	-0.4, 0.0	-0.9
22.5	-0.3, 0.6	-0.9	-0.6, 0.0	-1.1
30	-0.3, 0.8	-0.5	-0.7, 0.0	-1.3

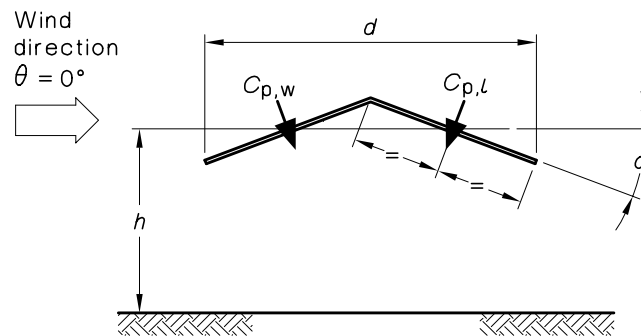


FIGURE D3 PITCHED FREE ROOFS

TABLE D6
NET PRESSURE COEFFICIENTS ($C_{p,n}$) FOR
TROUGHED FREE ROOFS— $0.25 \leq h/d \leq 1$ (see Figure D4)

Roof pitch (α) degrees	$\theta = 0^\circ$			
	$C_{p,w}$		$C_{p,l}$	
	Empty under	Blocked under	Empty under	Blocked under
7.5	-0.6, 0.4	-0.7	0.3	-0.3
15	-0.6, 0.4	-0.8	0.5	-0.2
22.5	-0.7, 0.3	-1.0	0.7	-0.2

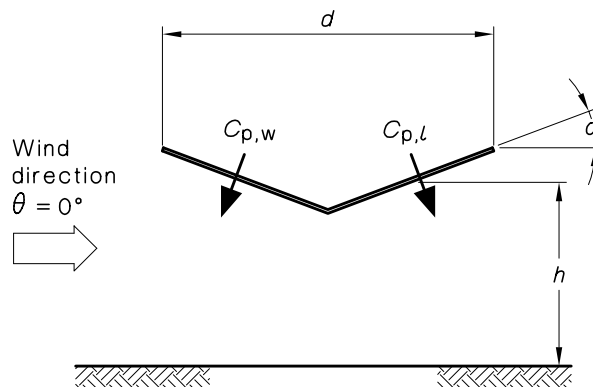


FIGURE D4 TROUGHED FREE ROOFS

TABLE D7
NET PRESSURE COEFFICIENTS ($C_{p,n}$) FOR
HYPAR FREE ROOFS—EMPTY UNDER (see Figure D5)

Conditions	θ , degrees	$C_{p,w}$	$C_{p,\ell}$
Empty under, $0.25 < h/d < 0.5$, $0.1 < c/d < 0.3$, and $0.75 < b/d < 1.25$	0	+0.45	+0.25
		−0.45	−0.25
	90	+0.45	+0.25
		−0.45	−0.25

NOTE: $C_{p,n}$ is defined as positive downwards, and only combinations of values of the same sign need to be considered.

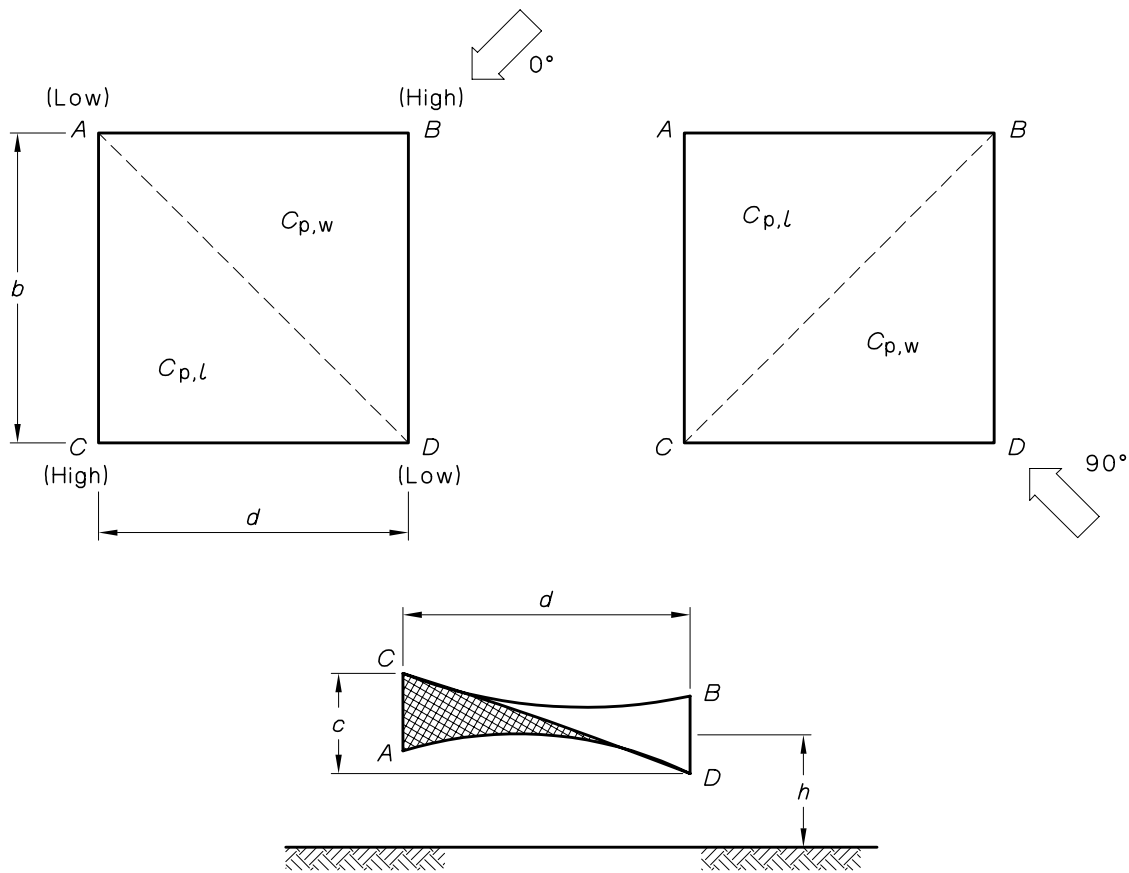


FIGURE D5 HYPERBOLIC PARABOLOID (HYPAR) ROOFS

D3.2 Aerodynamic shape factor for frictional drag and drag on exposed members for free roofs

The aerodynamic shape factor (C_{fig}) for calculating frictional drag effects and drag on exposed members of free roofs of monoslope, pitched or troughed configuration shall be equal to C_f calculated as given in Table D3. For free roofs, the frictional drag on both upper and lower surfaces shall be calculated and summed and added to the drag on any exposed members calculated in accordance with Appendix E (see Clause 2.5).

Calculation of frictional drag pressure is not required for wind directions of 0 degrees or 180 degrees as shown in Figures D2, D3 and D4 for free roofs with pitches of 10 degrees or more.

D4 ATTACHED CANOPIES, AWNINGS AND CARPORTS (ROOFS)

D4.1 Aerodynamic shape factor for normal net pressure on attached canopies

The aerodynamic shape factor (C_{fig}) for calculating net pressures normal to the roof on canopies, awnings or carports adjacent to enclosed buildings and with a roof slope of 10 degrees or less shall be calculated as follows:

$$C_{fig} = C_{p,n} K_a K_\ell \quad \dots D4$$

where

$C_{p,n}$ = net pressure coefficient acting normal to the surface, as given in Tables D8 and D9

K_a = area reduction factor, as given in Paragraph D1.2

K_ℓ = local pressure factor, as given in Paragraph D1.3

NOTES:

- 1 The values given for $C_{p,n}$ assume that any goods and materials stored under the canopy do not represent more than a 75 percent blockage.
- 2 The factor K_p does not appear in this equation as it is taken as 1.0.

Where indicated, attached canopies, awnings or carports shall be designed for both downward (positive) and upward (negative) net wind pressures.

For wind directions normal to the attached wall ($\theta = 0$ degrees), for canopies and awnings, $C_{p,n}$ shall be taken from Tables D8 or D9 with reference to Figure D6. All pressure coefficients shall be used with the value of wind speed applying at average roof height (h) and h_c is the average height of the canopy above ground.

For wind directions parallel to the wall of the attached building ($\theta = 90$ degrees or 270 degrees), the canopy or awning shall be considered as a free roof and the net pressure coefficients ($C_{p,n}$) shall be obtained in accordance with Table D4(A) or D4(B), or where the canopy is partially enclosed, from Table D9.

TABLE D8
NET PRESSURE COEFFICIENTS ($C_{p,n}$) FOR CANOPIES AND AWNINGS
ATTACHED TO BUILDINGS (see Figure D6(a)) FOR $\theta = 0$ DEGREES

Design case	h_c/h (see Note 1)	Net pressure coefficients ($C_{p,n}$)
$h_c/h < 0.5$	0.1	1.2, -0.2
	0.2	0.7, -0.2
	0.5	0.4, -0.2
$h_c/h \geq 0.5$	0.5	0.5, -0.3
	0.75	0.4, $[-0.3 - 0.2(h_c/w_c)]$ or -1.5 (see Note 2)
	1.0	0.2, $[-0.3 - 0.6(h_c/w_c)]$ or -1.5 (see Note 2)

NOTES:

- 1 For intermediate values of h_c/h , use linear interpolation.
- 2 Whichever is the lower magnitude.

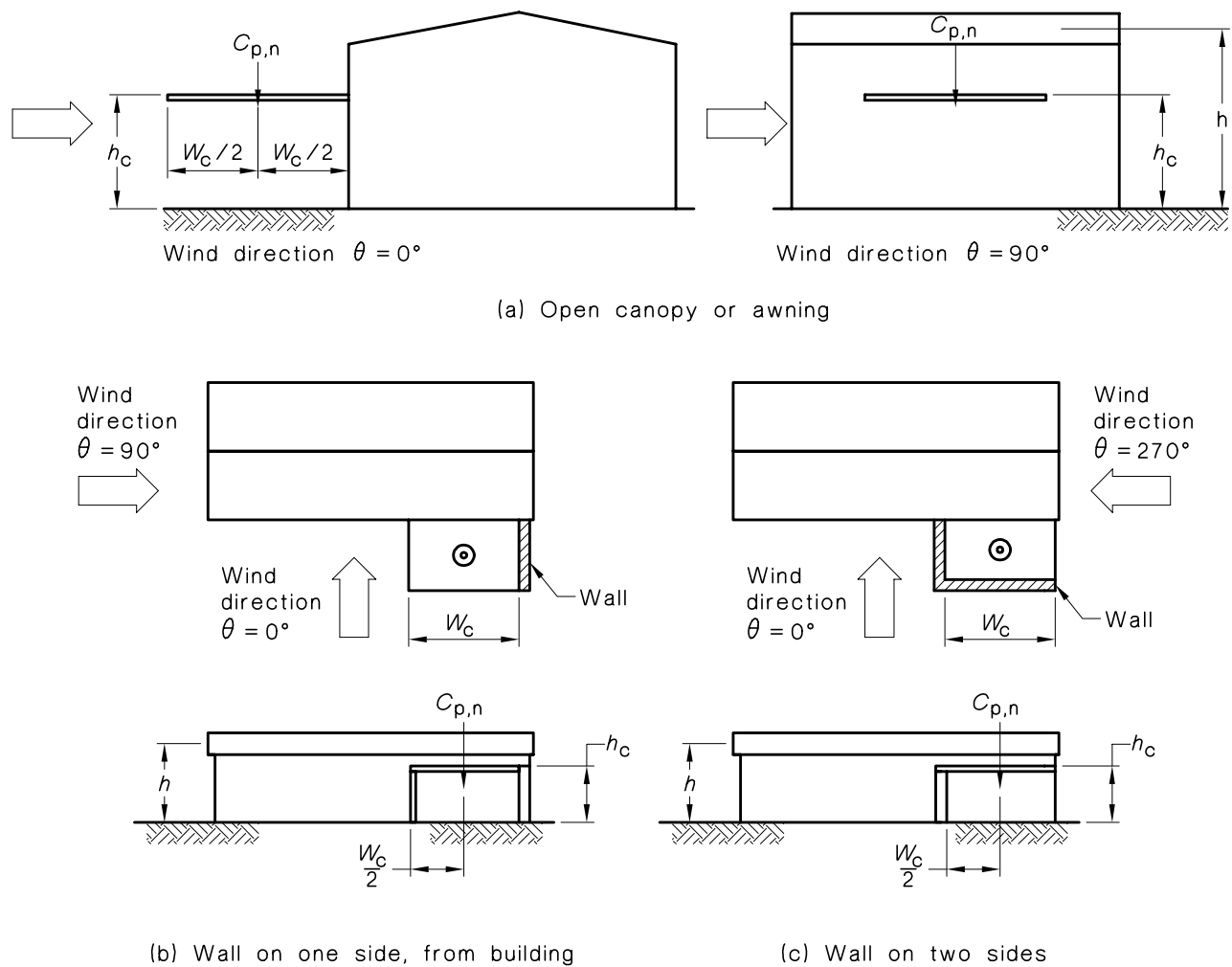


FIGURE D6 NET PRESSURE COEFFICIENTS ($C_{p,n}$) FOR CANOPIES, AWNINGS AND CARPORTS ATTACHED TO BUILDINGS

TABLE D9
NET PRESSURE COEFFICIENTS ($C_{p,n}$) FOR
PARTIALLY ENCLOSED CARPORTS (see Figures D6(b) and D6(c))

Conditions	Partially enclosed	Wind direction (θ), degrees	Net pressure coefficients ($C_{p,n}$)
$h_c/w_c \leq 0.5$ and $h_c/h < 0.8$	Wall on one side attached to building, see Figure D6(b)	0	-0.7
		90	-1.0
	Wall on two sides, see Figure D6(c)	0	-0.6
		270	-1.2

D4.2 Aerodynamic shape factor for frictional drag and drag on exposed members of attached canopies

The aerodynamic shape factor (C_{fi}) for calculating frictional drag effects on attached canopies, awnings or carport roofs, where the wind is parallel to the attached wall, shall be equal to C_f as given in Table D3. For canopies, the frictional drag on both upper and lower surfaces shall be calculated and summed and added to the drag on any exposed members calculated in accordance with Appendix E (see Clause 2.5).

D5 CANTILEVERED ROOFS AND CANOPIES

For long cantilevered roof beams, both C_{fig} and C_{dyn} shall be determined.

- A1 | The aerodynamic shape factor (C_{fig}) shall be taken as $1.5(1 - x/l)$ applied as shown in Figure D7, where l is the length of the cantilevered roof beam and x is the distance from the leading edge.

Dynamic response shall be taken into account by determining the dynamic response factor (C_{dyn}) as follows:

- (a) For cases where beams are greater than 15 m long, $\left(\frac{V_{des,\theta}}{1 + g_v I_h} \right) \left(\frac{1}{n_1 l} \right) > 0.4$ and $n_1 < 1$ Hz;

$$C_{dyn} = \left[1.0 + 0.5 \left[\left(\frac{V_{des,\theta}}{1 + g_v I_h} \right) \left(\frac{1}{n_1 l} \right) - 0.4 \right] \right] \quad \dots D5$$

where

n_1 = first mode frequency of vibration of the cantilevered roof in the vertical bending mode

- (b) For all other cases, $C_{dyn} = 1.0$.

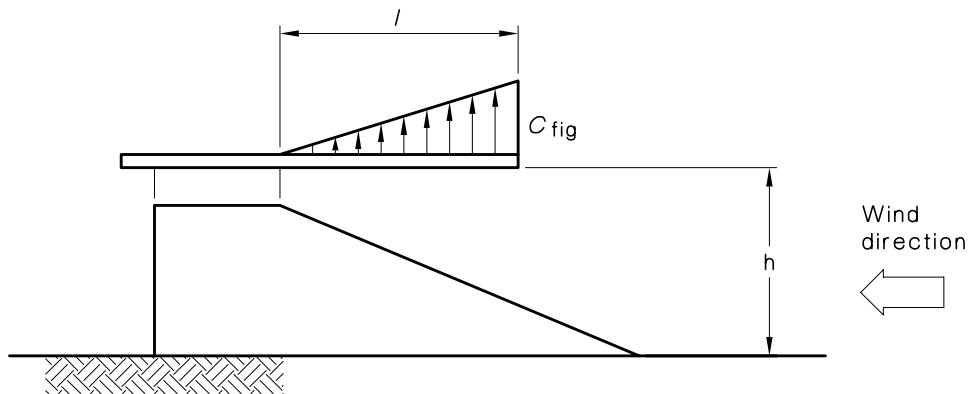


FIGURE D7 CANTILEVERED ROOF AND CANOPY

APPENDIX E

AERODYNAMIC SHAPE FACTORS FOR EXPOSED STRUCTURAL MEMBERS, FRAMES AND LATTICE TOWERS

(Normative)

E1 GENERAL

This Appendix shall be used to calculate aerodynamic shape factors (C_{fig}) for structures and components consisting of exposed members, such as lattice frames, trusses and towers.

All pressure coefficients shall be used with the value of wind speed applying at the height of the component being considered.

E2 AERODYNAMIC SHAPE FACTORS FOR INDIVIDUAL MEMBERS AND FRAMES

E2.1 Simple shapes and individual members

The aerodynamic shape factor (C_{fig}) for individual exposed structural members, whose aspect ratio (l/b) is greater than 8, shall be calculated as follows:

(a) For wind axes:

$$C_{fig} = K_{ar} K_i C_d \quad \dots E2(1)$$

(b) For body axes:

$$C_{fig} = K_{ar} K_i C_{F,x} \quad \text{along member's x-axis (major axis)} \quad \dots E2(2)$$

$$C_{fig} = K_{ar} K_i C_{F,y} \quad \text{along member's y-axis (minor axis)} \quad \dots E2(3)$$

where

l = length of member

b = breadth of element, normal to the wind stream

K_{ar} = aspect ratio correction factor for individual member forces, as given in Table E1

K_i = factor to account for the angle of inclination of the axis of members to the wind direction, determined as follows:

= 1.0, when the wind is normal to the member

= $\sin^2 \theta_m$ for rounded cylindrical shapes

= $\sin \theta_m$ for sharp-edged prisms, (sharp-edged prisms are those with b/r less than 16)

θ_m = angle between the wind direction and the longitudinal axis of the member, in degrees

r = corner radius of a structural shape

C_d = drag force coefficient for a structure or member in the direction of the wind stream, as given in Paragraph E3

$C_{F,x}$ and $C_{F,y}$ = drag force coefficients for a structure or member, in the direction of the x - and y -axes respectively, as given in Paragraph E3

TABLE E1
ASPECT RATIO CORRECTION FACTORS (K_{ar})

Aspect ratio, l/b (see Note)	Correction factor K_{ar}
8	0.7
14	0.8
30	0.9
40 or more	1.0

NOTE: For intermediate values of l/b , use linear interpolation.

E2.2 Open frames in a single plane

The aerodynamic shape factor (C_{fig}) for a structure of open frame type, comprising a number of members where the members are sharp-edged rectangular or structural sections, lying in a single plane normal to the wind direction (see Figure E1), shall be taken as follows:

- A1 (a) For $0.2 < \delta_e < 0.8$ and $1/3 < (b/l) < 3$ (where b/l is the aspect ratio of the whole frame).

$$C_{fig} = 1.2 + 0.26 (1 - \delta_e) \quad \dots E2(4)$$

- (b) For all other cases, wind action shall be the sum of the effects calculated on individual members and attachments determined in accordance with Clause 2.5.3.3 and Paragraph E2.1. For calculating $C_{fig,1}$, for the purposes of Paragraph E2.3, sum the $C_{fig}bl$ for each of the individual members and divide the result by the overall frame area (i.e., bl as shown in Figure E1).

where

- δ_e = effective solidity ratio for an open frame, given as follows:
 = δ for flat-sided members
 = $1.2\delta^{1.75}$ for circular cross-section members

where

δ = solidity ratio of the structure (surface or open frame), which is the ratio of solid area to total area of the structure

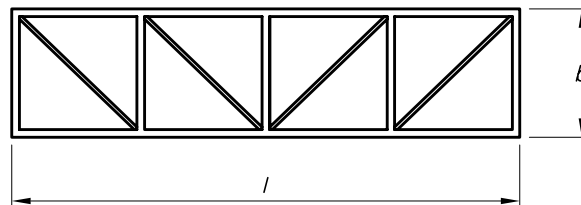


FIGURE E1 NOTATION FOR FRAME DIMENSIONS

E2.3 Multiple open frames

For structures comprising a series of similar open frames in parallel, the aerodynamic shape factors for the second and subsequent frames shall be taken as the aerodynamic shape factors on the windward frame calculated as in Paragraph E2.2, multiplied by the shielding factor (K_{sh}) obtained from Table E2. The aerodynamic shape factor (C_{fig}) for the structure shall be as follows:

$$C_{fig} = C_{fig,1} + \Sigma K_{sh} C_{fig,1} \quad \dots E2(5)$$

where

- $C_{\text{fig},1}$ = aerodynamic shape factor for the first frame in the upwind direction, as given in Paragraph E2.2
- K_{sh} = shielding factor for shielded frames in multiple open-framed structures, as given in Table E2
- λ = spacing ratio for parallel open frames, equal to the frame spacing (centre-to-centre) on the projected frame width normal to the wind direction

TABLE E2
SHIELDING FACTORS (K_{sh}) FOR MULTIPLE FRAMES

Angle of wind to frames (θ), degrees	Frame spacing ratio (λ)	Shielding factors (K_{sh})							
		Effective solidity (δ_e)							
		0	0.1	0.2	0.3	0.4	0.5	0.7	1.0
0 (wind normal to frames)	≤ 0.2	1.0	0.8	0.5	0.3	0.2	0.2	0.2	0.2
	0.5	1.0	1.0	0.8	0.6	0.4	0.2	0.2	0.2
	1.0	1.0	1.0	0.8	0.7	0.5	0.3	0.2	0.2
	2.0	1.0	1.0	0.9	0.7	0.6	0.4	0.2	0.2
	4.0	1.0	1.0	1.0	0.8	0.7	0.6	0.4	0.2
	≥ 8.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
45	≤ 0.5	1.0	0.9	0.8	0.7	0.6	0.5	0.3	0.3
	1.0	1.0	1.0	0.9	0.8	0.7	0.6	0.6	0.6
	2.0	1.0	1.0	1.0	1.0	1.0	0.9	0.8	0.6
	4.0	1.0	1.0	0.9	0.7	0.6	0.4	0.2	0.2
	≥ 8.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0

NOTES:

- 1 The notation λ = spacing ratio for parallel open frames, equal to the frame spacing (centre-to-centre) on the projected frame width normal to the wind direction.
- 2 For intermediate values of δ_e and λ , use linear interpolation.

E3 DRAG FACTORS FOR STRUCTURAL MEMBERS AND SIMPLE SECTIONS

E3.1 Rounded cylindrical shapes, sharp-edged prisms and structural sections

Values of drag force coefficients (C_d , $C_{F,x}$ and $C_{F,y}$) for rounded cylindrical shapes, sharp-edged prisms and some structural sections shall be as given in Tables E3, E4 and E5 respectively.

Table E4 gives values for the most common polygonal sharp-edged cross-sections except for rectangular prisms that are covered separately in Paragraph E3.2.

NOTES:

- 1 Drag force coefficients of sharp-edged section are independent of the Reynolds number.
- 2 Note that in Table E5, the dimension b used in the definition of the force coefficients is not always normal to the flow direction and d is not always parallel.

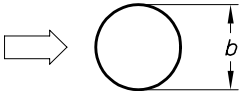
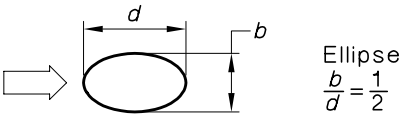
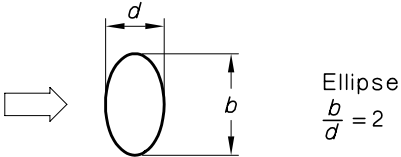
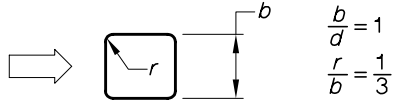
In the absence of experimental information for particular cables, C_d for helically wound, unwrapped cables shall be as follows:

- (a) 1.2 for $bV_{\text{des},\theta} < 0.5$.
- (b) 1.0 for $bV_{\text{des},\theta} > 5.0$.

For values of $bV_{\text{des},\theta}$ between 0.5 and 5.0, use interpolation.

Where icing of cables is considered, the increased cross-sectional area and changed shape shall be taken into account.

TABLE E3
DRAG FORCE COEFFICIENTS (C_d) FOR ROUNDED CYLINDRICAL SHAPES

Cross-sectional shape	Description	Drag force coefficient (C_d) (see Note 1)	
		$bV_{des,0} < 4 \text{ m}^2/\text{s}$	$bV_{des,0} > 10 \text{ m}^2/\text{s}$
	Cylindrical	1.2	(see Note 2)
	Ellipse narrow side to wind	0.7	0.3
	Ellipse broad side to wind	1.7	1.5
	Square with rounded corners	1.2	0.6

NOTES:

- For intermediate values of $bV_{des,0}$, use linear interpolation.
- For smooth circular cross-sections and polygonal sections with more than 16 sides, where $bV_{des,0} > 10 \text{ m}^2/\text{s}$, C_d shall be as follows:

$$C_d = 0.5 \quad \text{for } h_r/d \leq 0.00002$$

$$C_d = 1.6 + 0.105 \log_e (h_r/d) \quad \text{for } h_r/d > 0.00002$$

where

h_r = average height of surface roughness

d = outside diameter

Some typical values for h_r in millimetres are as follows:

Glass, plastic: 0.0015

Steel: galvanized 0.15; light rust 2.5; heavy rust 15

Concrete, new smooth 0.06; new rough: 1.0

Metal, painted: 0.003 to 0.03

TABLE E4
DRAG FORCE COEFFICIENT (C_d) FOR SHARP-EDGED PRISMS

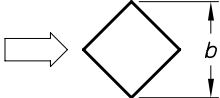
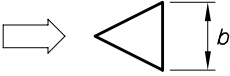
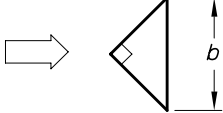
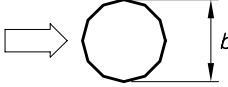
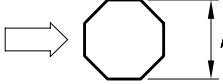
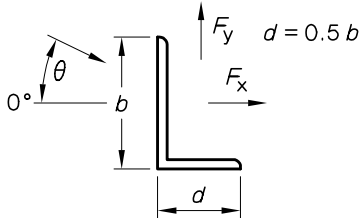
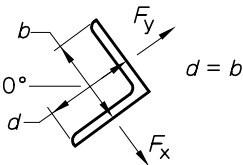
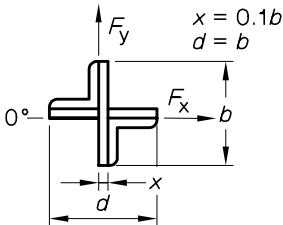
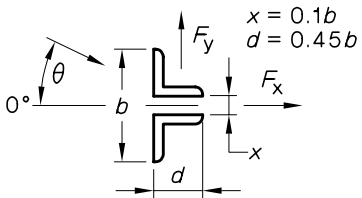
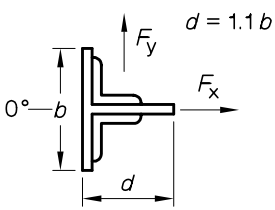
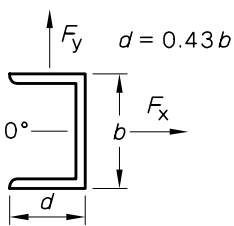
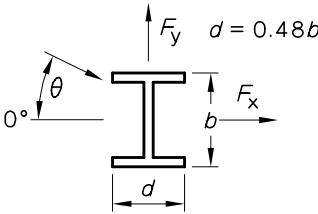
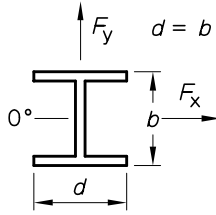
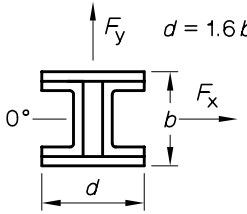
Cross-sectional shape	Drag force coefficient (C_d)
 Square with face to wind	2.2
 Square with corner to wind	1.5
 Equilateral triangle—apex to wind	1.2
 Equilateral triangle—face to wind	2.0
 Right angled triangle	1.55
 12-sided polygon	1.3
 Octagon	1.4

TABLE E5
FORCE COEFFICIENTS ($C_{F,x}$) AND ($C_{F,y}$) FOR STRUCTURAL SECTIONS

Section shape		Wind direction measured clockwise (θ)				
		0	45	90	135	180
	$C_{F,x}$	2.0	1.8	-2.0	-1.8	-1.9
	$C_{F,y}$	-0.1	0.1	-1.7	-0.8	-0.95
	$C_{F,x}$	1.8	1.8	-1.0	0.3	-1.4
	$C_{F,y}$	1.8	2.1	-1.9	-2.0	-1.4
	$C_{F,x}$	1.75	0.75	-0.1	-0.85	-1.75
	$C_{F,y}$	0.1	-0.75	-1.75	-0.85	-0.1
	$C_{F,x}$	1.6	1.5	-0.95	-0.5	-1.5
	$C_{F,y}$	0	0.1	-0.7	-1.05	0
	$C_{F,x}$	2.0	1.2	-1.6	-1.1	-1.7
	$C_{F,y}$	0	-0.9	-2.15	-2.4	± 2.1
	$C_{F,x}$	2.05	1.85	0	-1.6	-1.8
	$C_{F,y}$	0	-0.6	-0.6	-0.4	0

(continued)

TABLE E5 (continued)

Section shape		Wind direction measured clockwise (θ)				
		0	45	90	135	180
	$C_{F,x}$	2.05	1.95	± 0.5	—	—
	$C_{F,y}$	0	-0.6	-0.9	—	—
	$C_{F,x}$	1.6	1.5	0	—	—
	$C_{F,y}$	0	-1.5	-1.9	—	—
	$C_{F,x}$	1.4	1.2	0	—	—
	$C_{F,y}$	0	-1.6	-2.2	—	—

NOTE: Note that the direction of θ has been changed to clockwise and the values transposed accordingly, to align with the clockwise direction used elsewhere in the Standard. Also, dimension b , used in the definitions of the force coefficients, is not always normal to the wind direction.

E3.2 Rectangular prismatic sections

Values of drag and lift force coefficients ($C_{F,x}$ and $C_{F,y}$) for rectangular prismatic cross-sections are given in Figures E2(A) and E2(B). This Paragraph does not cover the case where the wind direction angle θ is greater than 20 degrees. For intermediate values of d/b , use linear interpolation.

NOTE: Figure E2(B) contains maximum values of $C_{F,y}$ for angles within 20 degrees of the directions parallel to the faces of the rectangle. Fluctuations in wind direction of up to 20 degrees may occur in turbulent flow nominally parallel to one face.

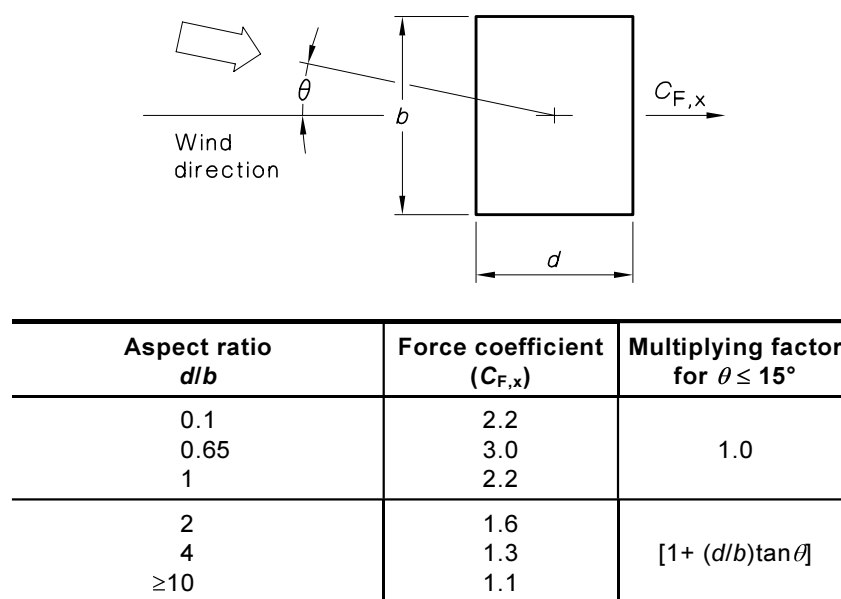


FIGURE E2(A) ALONG-WIND COEFFICIENTS FOR RECTANGULAR PRISMS

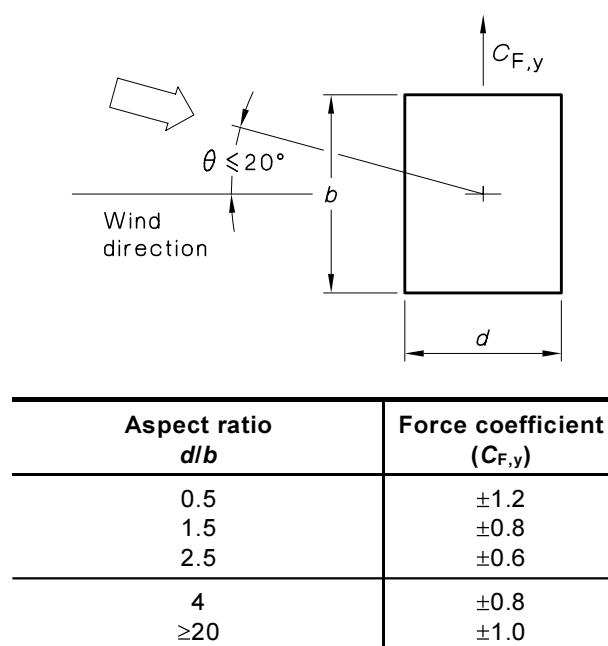


FIGURE E2(B) CROSSWIND COEFFICIENTS FOR RECTANGULAR PRISMS

E4 LATTICE TOWERS

E4.1 General

Lattice towers shall be divided vertically into a series of sections (levels) and the aerodynamic shape factors (C_{fig}) shall be calculated for each section.

NOTE: A minimum of 10 sections should be used where possible.

They shall be designed for winds in 8 directions with $V_{des,\theta}$ being the value of $V_{sit,\beta}$ in a sector ± 22.5 degrees from the 45 degrees direction being considered.

The aerodynamic shape factor (C_{fig}) shall be equal to the values calculated, as follows:

- (a) C_d , for a tower section without ancillaries, as given in Paragraph E4.2.1.

- (b) C_{de} , for a tower section with ancillaries, as given in Paragraph E4.2.2.
- (c) $1.2 \sin^2 \theta_m$, for guy cables, using the wind speed calculated for 2/3 of the height of the cable

where

C_{de} = effective drag force coefficient for a tower section with ancillaries

θ_m = angle between the wind direction and the longitudinal axis of the member, in degrees

E4.2 Drag force coefficient

E4.2.1 Tower sections without ancillaries

The drag force coefficients (C_d) for complete lattice tower sections shall be taken from Tables E6(A) to E6(C) using linear interpolation for values of $b_i V_{des, \theta}$ between 3 and 6, and using the following notation:

δ = solidity ratio of the structure (surface or open frame), which is the ratio of solid area to total area of the structure, which for the front face of a tower section shall be taken as the ratio of the total projected area of members (A_z) to the projected area enclosed over the section height by the boundaries of the frame

b_i = average diameter or breadth of a section of a tower member

For equilateral-triangle lattice towers with flat-sided members, the drag force coefficient (C_d) shall be assumed to be constant for any inclination of the wind to a face.

For complete clad tower sections, C_d shall be taken as the value given in Tables E3 and E4, and Figure E2 for the appropriate tower section shapes.

For UHF antenna sections, C_d shall be obtained from Table E7 and Figure E3. To calculate the area for the application of the pressure, take breadth as b_D or b_N , as appropriate to the wind direction.

Where used, the reduction for aspect ratio shall be carried out by multiplying by the correction factor (K_{ar}), given in Table E1, taking l equal to two times the height of the end-mounted antennas.

TABLE E6(A)

DRAG FORCE COEFFICIENTS (C_d) FOR LATTICE TOWERS—SQUARE AND EQUILATERAL TRIANGLE IN PLAN WITH FLAT-SIDED MEMBERS

Solidity of front face (δ)	Square towers		Equilateral-triangle towers
	Onto face	Onto corner	
≤ 0.1	3.5	3.9	3.1
0.2	2.8	3.2	2.7
0.3	2.5	2.9	2.3
0.4	2.1	2.6	2.1
≥ 0.5	1.8	2.3	1.9

TABLE E6(B)
DRAG FORCE COEFFICIENTS (C_d) FOR LATTICE TOWERS—
SQUARE PLAN WITH CIRCULAR MEMBERS

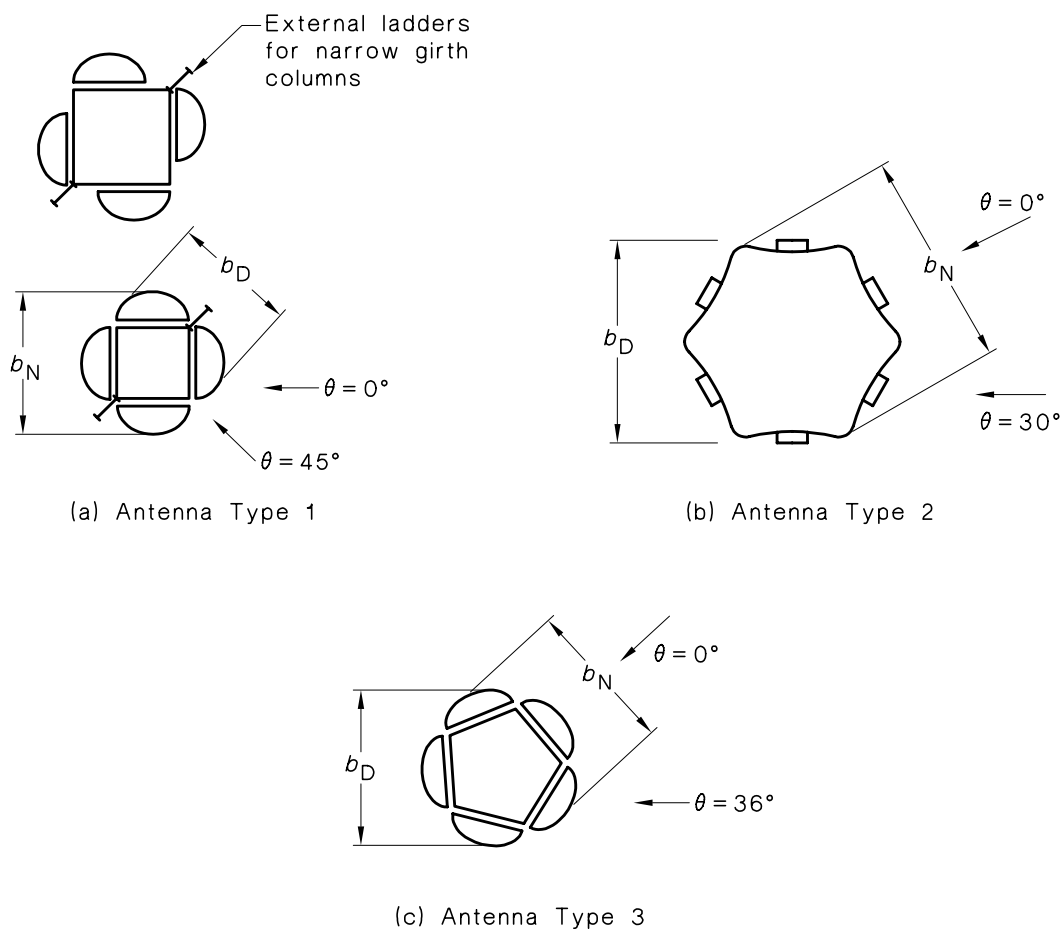
Solidity of front face (δ)	Parts of tower in sub-critical flow $b_i V_{des,\theta} < 3 \text{ m}^2/\text{s}$		Parts of tower in super-critical flow $b_i V_{des,\theta} \geq 6 \text{ m}^2/\text{s}$	
	Onto face	Onto corner	Onto face	Onto corner
≤ 0.05	2.2	2.5	1.4	1.2
0.1	2.0	2.3	1.4	1.3
0.2	1.8	2.1	1.4	1.6
0.3	1.6	1.9	1.4	1.6
0.4	1.5	1.9	1.4	1.6
≥ 0.5	1.4	1.9	1.4	1.6

TABLE E6(C)
DRAG FORCE COEFFICIENTS (C_d) FOR LATTICE TOWERS—
EQUILATERAL TRIANGLE PLAN WITH CIRCULAR MEMBERS

Solidity of front face (δ)	Parts of tower in sub-critical flow $b_i V_{des,\theta} < 3 \text{ m}^2/\text{s}$ (all wind directions)	Parts of tower in super-critical flow $b_i V_{des,\theta} \geq 6 \text{ m}^2/\text{s}$ (all wind directions)
≤ 0.05	1.8	1.1
0.1	1.7	1.1
0.2	1.6	1.1
0.3	1.5	1.1
0.4	1.5	1.1
≥ 0.5	1.4	1.2

TABLE E7
DRAG FORCE COEFFICIENT (C_d) FOR UHF-ANTENNA SECTIONS
(see Figure E2)

Antenna type	Wind direction (θ) degrees	Drag force coefficient (C_d)
1	0, 45	1.5
2	0, 30	1.9
3	0, 36	1.6

FIGURE E3 DRAG FORCE COEFFICIENTS (C_d) FOR SECTIONS OF UHF ANTENNAS**E4.2.2 Tower sections with ancillaries**

The effective drag force coefficient for a tower section with ancillaries (C_{de}) shall be calculated as follows:

- Where ancillaries are attached symmetrically to all faces, their projected area shall be added to the projected area of the tower members (A_z).
- When ancillaries are not symmetrically placed, the total effective drag force coefficient (C_{de}) for a tower section shall be taken as follows:

$$C_{de} = C_d + \Sigma \Delta C_d \quad \dots E4(1)$$

where

ΔC_d = additional drag coefficient due to an ancillary attached to one face or located inside the tower section, given by:

$$= C_{da} K_{ar} K_{in} (A_a/A_z) \quad \dots E4(2)$$

where

C_{da} = value of drag force coefficient (C_d) on an isolated ancillary on a tower, as given in Tables E3 and E4 and Figure E2

K_{ar} = aspect ratio correction factor for individual member forces

- = as given in Table E1, for linear ancillaries with aspect ratios less than 40
- = 1.0, for all other cases
- K_{in} = correction factor for interference, as given in Paragraph E4.2.3
- A_a = reference area of ancillaries on a tower, given as
 - = lb , where l is the length of the linear ancillary and b is defined in Figure E4 and Tables E3 and E4
- $A_{z,s}$ = total projected area of the tower section at height z

E4.2.3 Correction factor for interference

The correction factor for interference (K_{in}) shall be calculated as follows:

- (a) For ancillaries attached to the face of the tower:
 - (i) To the face of a square tower (see Figure E4(a))—

$$K_{in} = [1.5 + 0.5 \cos 2(\theta_a - 90^\circ)] \exp [-1.2(C_d \delta)^2] \quad \dots \text{E4(3)}$$
 - (ii) To the face of a triangular tower (see Figure E4(b))—

$$K_{in} = [1.5 + 0.5 \cos 2(\theta_a - 90^\circ)] \exp [-1.8(C_d \delta)^2] \quad \dots \text{E4(4)}$$
- (b) For lattice-like ancillaries inside the tower, K_{in} shall be taken either as 1.0 or shall be determined as follows:
 - (i) Inside a square tower (see Figure E4(c))—

$$K_{in} = \exp [-1.4 (C_d \delta)^{1.5}] \quad \dots \text{E4(5)}$$
 - (ii) Inside a triangular tower (see Figure E4(d))—

$$K_{in} = \exp [-1.8(C_d \delta)^{1.5}] \quad \dots \text{E4(6)}$$
- (c) For cylindrical ancillaries inside the tower, K_{in} shall be taken either as 1.0 or shall be determined as follows:
 - (i) Inside a square tower (see Figure E4(e))—

$$K_{in} = \exp [-a(C_d \delta)^{1.5}] \quad \dots \text{E4(7)}$$

$$a = 2.7 - 1.3 \exp [-3(b/w)^2] \quad \dots \text{E4(8)}$$
 - (ii) Inside a triangular tower (see Figure E4(f))—

$$K_{in} = 1.5 \exp [-c(C_d \delta)^{1.5}] \quad \dots \text{E4(9)}$$

$$c = 6.8 - 5 \exp [-40(b/w)^3] \quad \dots \text{E4(10)}$$

where

- θ_a = angle of deviation of the wind stream from the line joining the centre of the tower cross-section to the centre of the ancillary, in degrees
- δ = solidity ratio of the structure, as given in Clause E4.2.1
- a, c = constants for ease of calculation
- b/w = ratio of the average diameter of an ancillary to the average width of a structure

A1

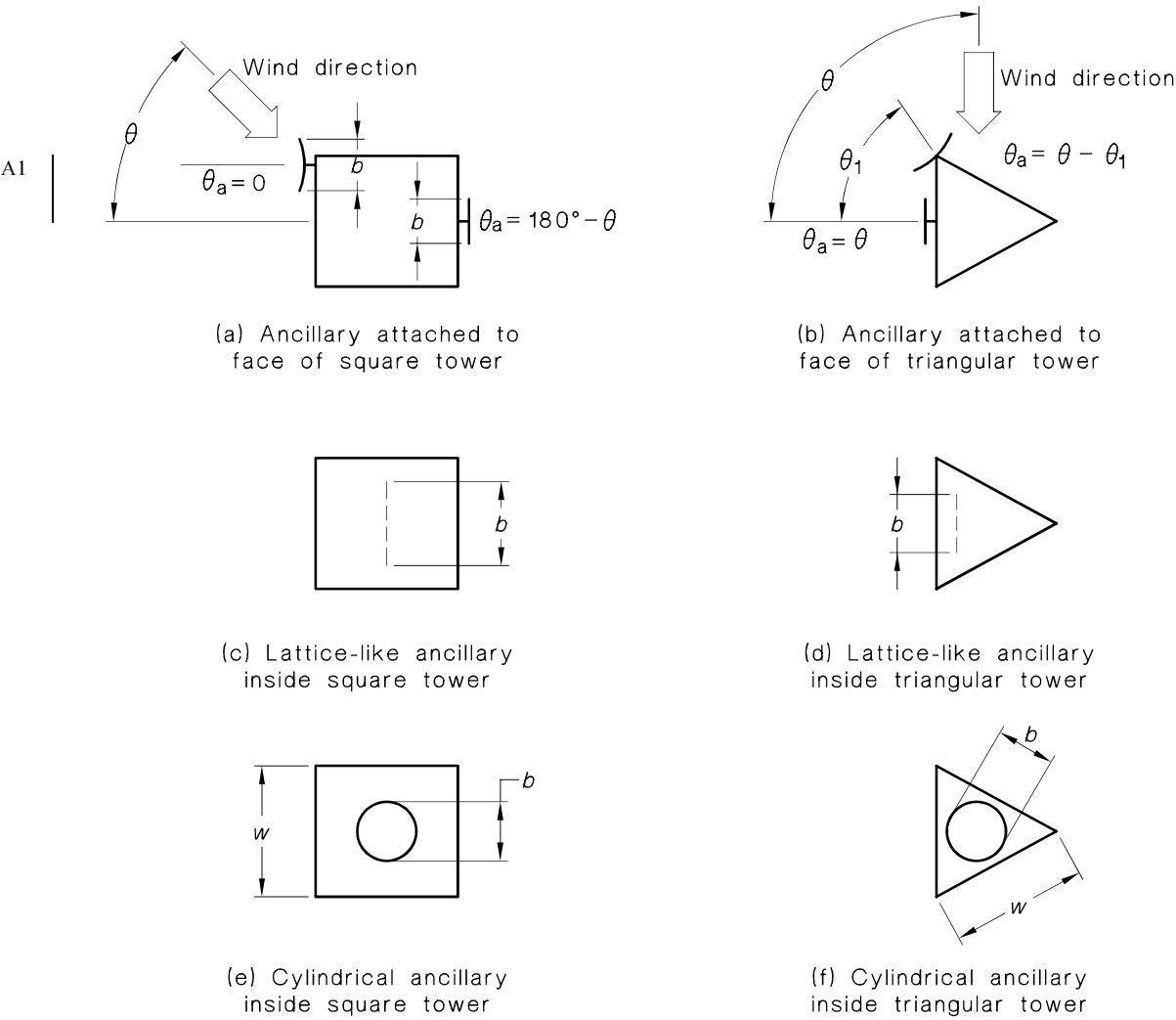


FIGURE E4 TOWER SECTIONS WITH ANCILLARIES

APPENDIX F

FLAGS AND CIRCULAR SHAPES

(Normative)

F1 GENERAL

This Appendix shall be used to calculate aerodynamic shape factors (C_{fig}) for flags, discs and spherical shapes.

All pressure coefficients shall be used with the value of wind speed applying at the height of the component being considered.

F2 FLAGS

The aerodynamic shape factor (C_{fig}) for flags is as follows:

- (a) *Fixed flags*, treat as elevated hoarding (see Appendix D).
- (b) *Free flags (including dynamic effects from flutter)*;

A1

$$C_{fig} = 0.05 + 0.7 \frac{m_f}{\rho_{air} c} \left(\frac{A_{ref}}{c^2} \right)^{-1.25} \quad \text{but not greater than } 0.76 \quad \dots F1$$

where

m_f = mass per unit area of flag, in kilograms per square metre

ρ_{air} = density of air which shall be taken as 1.2 kg/m^3

c = net height of flag (see Figure F1)

l_f = flag length (see Figure F1)

A_{ref} = reference area of flag, as given in Figure F1 (area of flag perpendicular to the wind direction)

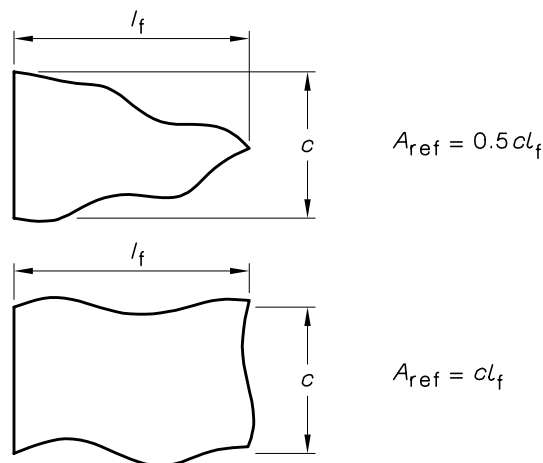
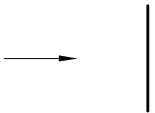
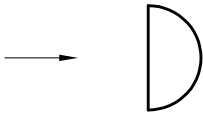
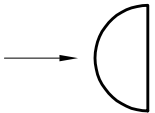
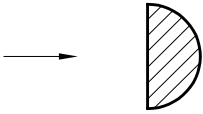
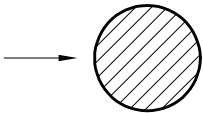


FIGURE F1 REFERENCE AREA FOR FLAGS

F3 CIRCULAR SHAPES

The aerodynamic shape factor (C_{fig}) for calculating drag pressures on circular shapes shall be as given in Table F1.

TABLE F1
AERODYNAMIC SHAPE FACTOR FOR CIRCULAR SHAPES

Cross-sectional shape	Description of shape	Aerodynamic shape factor (C_{fig})
	Circular disc	1.3
	Hemispherical bowl (cup to wind)	1.4
	Hemispherical bowl	0.4
	Hemispherical solid (flat to wind)	1.2
	Spherical solid	0.5 for $bV_{des,\theta} < 7$ 0.2 for $bV_{des,\theta} \geq 7$

APPENDIX G

ACCELERATIONS FOR WIND SENSITIVE STRUCTURES

(Informative)

G1 ACCELERATION FOR SERVICEABILITY

To provide some indication of motion serviceability it is noted that for wind sensitive buildings, mostly exposed to free stream flow, acceptable crosswind acceleration levels may be exceeded if—

$$h^{1.3}/m_0 > 0.0016 \quad \dots \text{G1}$$

where

h = average roof height of a structure above the ground

m_0 = average mass per unit height, in kg/m

In conditions of high turbulence, due to interference from other buildings, a more conservative approach to the use of this indicator should be taken. Should the inequality indicate likely high acceleration levels then the designer should undertake more detailed analysis or wind tunnel model studies.

G2 PEAK ALONG-WIND ACCELERATION FOR SERVICEABILITY

The peak acceleration at the top of a structure in the along-wind direction ($\ddot{x}_{\max}$) in metres per second squared, is as follows:

$\ddot{x}_{\max} = m_0 h^2 \times \text{resonant component of peak base bending moment}$

A1

$$= \frac{3}{m_0 h^2} \frac{\rho_{\text{air}} g_R I_h \sqrt{\frac{SE_t}{\zeta}}}{(1 + 2g_v I_h)} \left\{ C_{\text{fig, windward}} \sum_{z=0}^h [V_{\text{des},\theta}(z)]^2 b_z z \Delta z - C_{\text{fig, leeward}} [V_{\text{des},\theta}(h)]^2 \sum_{z=0}^h b_z z \Delta z \right\} \quad \dots \text{G2}$$

where

m_0 = average mass per unit height, in kg/m

ρ_{air} = density of air which shall be taken as 1.2 kg/m³

$V_{\text{des},\theta}(z)$ = building orthogonal design wind speeds as a function of height z

$V_{\text{des},\theta}(h)$ = building orthogonal design wind speeds evaluated at height h

b_z = average breadth of the structure at the section at height z

Δ_z = height of the section of the structure upon which the wind pressure acts

G3 CROSSWIND ACCELERATION FOR SERVICEABILITY OF TALL BUILDINGS AND TOWERS OF RECTANGULAR CROSS-SECTION

G3.1 General

This Paragraph gives methods for determining peak accelerations at the top of tall enclosed buildings and towers of rectangular cross-section. Calculation of crosswind response is not required for porous lattice towers.

NOTE: AS/NZS 1170.2 Supp 1, *Structural design actions—Wind actions—Commentary* (Supplement to AS/NZS 1170.2:2002) gives further guidance where calculations of crosswind response yield high values.

G3.2 Peak crosswind acceleration for serviceability

The peak acceleration in the crosswind direction ($\ddot{y}_{\max}$) in metres/second squared, at the top of a structure with constant mass per unit height (m_0) should be determined as follows:

$$\ddot{y}_{\max} = \frac{1.5bg_R}{m_0} \left[\frac{0.5\rho_{\text{air}} [V_{\text{des},\theta}]^2}{(1 + g_v I_h)^2} \right] K_m \sqrt{\frac{\pi C_{fs}}{\zeta}} \quad \dots \text{G3(1)}$$

where

b = breadth of a structure, normal to the wind stream

g_R = peak factor for resonant response (10 min period) given by:

$$g_R = \sqrt{[2 \times \log_e (600n_c)]} \quad \dots \text{G3(2)}$$

n_c = first mode natural frequency of vibration of a structure in the crosswind direction, in Hertz

m_0 = average mass per unit height, in kg/m

g_v = peak factor for the upwind velocity fluctuations, which may be taken as 3.7

I_h = turbulence intensity, obtained from Table 6.1 by setting z equal to h

K_m = mode shape correction factor for crosswind acceleration, given by:

$$= 0.76 + 0.24k$$

where

k = mode shape power exponent for the fundamental mode and values of the exponent k should be taken as:

= 1.5 for a uniform cantilever

= 0.5 for a slender framed structure (moment resisting)

= 1.0 for a building with central core and moment resisting façade

= 2.3. for a tower decreasing in stiffness with height, or with a large mass at the top

= the value obtained from fitting $\phi_1(z) = (z/h)^k$ to the computed modal shape of the structure

$\phi_1(z)$ = first mode shape as a function of height z , normalized to unity at $z = h$

C_{fs} = crosswind force spectrum coefficient generalized for a linear mode shape given in Clause 6.3.2.3

ζ = ratio of structural damping to critical damping of a structure (see Table 6.2)

AMENDMENT CONTROL SHEET**AS/NZS 1170.2:2002**

Amendment No. 1 (2005)

CORRECTION

SUMMARY: This Amendment applies to the Preface, Clauses 1.1, 1.2, 1.4, 2.5.1, 3.2, 4.3.2, 4.3.3, 5.4.1, 6.2.2, 6.4, Figures 3.1, 4.4, 5.1 (in part), 5.2, 5.3, Tables 3.1, 4.2(A), 5.2(B), 5.6, Equations 6.2(2), 6.2(5), 6.3(6), 6.4(1) and Appendices B, C, D, E, F and G.

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NOTES

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